

RESIDUAL GEOMETRY AND SINGULARITIES OF MULTIPLICATION FAILURE SCHEMES

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ABSTRACT. For symmetric multiplication systems, polar residual spaces measure the tangent excess of annihilator incidences. We identify joint polar relations at every corank and construct exact residual matrix equations retaining all nonlinear terms. Transversality gives generic determinantal neighbourhoods; at corank one, the polar kernel counts the equations of a minimal presentation and determines their quadratic obstruction. Residual minors describe restricted cotangent Fitting ideals and the expected-codimension singular support. For quadratic binary subseries we determine the failure components in all parameter dimensions, with maximal-dimensional components classified in the excess regime, and construct a relative residual determinant divisor with a dense ordinary-double-crossing locus. For hyperplane subseries in every symmetric degree, we determine the entire failure support and every corank. In degree at least three the associated subvarieties are precisely the secant threefold and an embedded evaluation curve; the nilpotency index grows unboundedly with the degree. The arguments also give exceptional orientation descent, same-annihilator signed intersections, rank-two fibre primary decomposition, and algebraic dihedral descent for higher-product two-point sectors. Full information-recovery, native-realization and statistical applications are retained in the accompanying appendices.

1. INTRODUCTION

Let $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$ and $k = n + 1$. Multiplication of sections defines, on the product of Grassmannians $X_L = \text{Gr}(c, V_n)^L$, the bundle map

$$\mu : \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{U}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}. \quad (1.1)$$

There are no products between different factors. We study the scheme $\mathcal{D}_L$ of maximal minors of this map, rather than only the rank at a general tuple. Put

$$q = \binom{c+1}{2}, \quad e = Lq, \quad p = 2k - 1, \quad M = Lc(k - c), \quad a = e - p + 1, \quad b = Lc - 3,$$

and assume $e \geq p$. The competition between a and b reflects two geometrically different kinds of annihilator. A nondegenerate Hankel form leads to the expected determinantal codimension a ; a rank-two form leads to signed secant families of codimension b . This competition is measured by multiplication on a smaller residual space and governs the singularities of the failure scheme.

The reduced support $\Delta^{(L)} = |\mathcal{D}_L|$ is the set of tuples for which $\sum_\nu U_\nu^2 \neq V_{2n}$. The basic codimension formula holds in the entire range $4 \leq c < k$.

Theorem 1.1 (Reduced failure locus and independent contacts). *Suppose $4 \leq c < k$, $L \geq 1$, and $Lq \geq p$. Then*

$$\text{codim}_{X_L} \Delta^{(L)} = \min\{Lq - p + 1, Lc - 3\}. \quad (1.2)$$

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There is an irreducible common-secant subvariety $\Sigma_L \subset \Delta^{(L)}$ of codimension $Lc - 3$. If $Lc - 3 \leq Lq - p + 1$, it is a maximal-dimensional irreducible component of $\Delta^{(L)}$. Consequently a general L -tuple is informative if and only if $Lq \geq p$.

Theorem 7.1 refines this formula to a component classification in all these dimensions. For $a \leq b$ it determines all components and proves reducedness and Cohen–Macaulayness; for $b < a$ it determines all maximal-dimensional components. Exactly five additional equality incidences occur outside the two principal rank families. In each, the plane contains the full radical and is maximal isotropic in the quotient. The orientation covers in Section 6 determine their irreducibility.

The principal deformation result is independent of this binary classification. Given any map $\beta : \text{Sym}^m V \rightarrow F$, the plane derivative at an annihilator defines a polar residual space Q and a map $\tau : Q \rightarrow \ker \ell$. Theorem 2.1 expresses the tangent excess of the incidence, and of its corank-one image, as $\dim \ker \tau$. In degree two,

$$Q = \bigoplus_{\nu} \text{Sym}^2(U_{\nu} \cap \text{rad } H_{\ell}),$$

so the obstruction is a smaller multiplication problem. The formulation also applies to products of sections on arbitrary projective varieties and to higher symmetric powers.

This tangent calculation has a scheme-theoretic global form. Theorem 2.2 determines the full singular support in expected codimension: all points of original corank at least two are singular, and a corank-one point is singular exactly when τ is not injective. On every constant-rank polar stratum, the Fitting ideals of the cotangent sheaf are the indicated minors of τ . These identities include the higher tangent-excess strata and glue with exact Schur–Jacobian presentations at larger corank. Corollary 2.3 consequently handles all annihilator ranks of the quadratic binary problem, including support collisions. Proposition 2.5 gives the corresponding local criterion in excess codimension.

The distinction between restricted ideals and neighbourhoods is addressed by Theorem 3.2. At arbitrary original corank, the joint polar relations are the obstruction to a transverse generic matrix chart. Even when that obstruction is nonzero, the theorem gives an exact matrix of residual convergent equations, retaining all higher terms and all nested rank ideals. In corank one, Theorem 3.5 reduces these to $\delta = \dim \ker \tau$ equations in $M - a + \delta$ parameters and identifies their quadratic obstruction. In expected codimension they form a regular sequence. Proposition 3.7 constructs transverse examples in every symmetric degree; the cubic example in Section 3 also exhibits a nontransverse cuspidal failure germ. These are exact local statements, not a claim that the binary higher-product component problem follows from the two-point calculation.

At a rank-two annihilator with distinct support, its radical is gV_{n-2} for the vanishing quadratic g . Writing $U_{\nu} \cap \text{rad } H_{\ell} = gK_{\nu}$, the wall $a = b$ is exactly

$$L \binom{c}{2} = 2k - 5,$$

the square threshold for the residual multiplication of the $(c-1)$ -planes K_{ν} . Its entire determinantal rank stratification is the pullback of the corresponding cotangent Fitting stratification on the original corank-one locus (Corollary 2.4). On a nonempty dense open subset of its integral divisor, the analytic equation is $uv = 0$ (Theorem 8.9). The normal-bundle factorization, relative determinant line and simultaneous nonemptiness needed for this equation are proved separately in Section 8.3. We also give all same-annihilator signed intersections over the split rank-two base and an explicit primary decomposition of the fixed-annihilator isotropy fibre. These describe different objects: fibre embedded structure is not being asserted for the global scheme.

For higher symmetric powers the general polar theorem remains applicable. Section 9 additionally computes the two-point annihilator sectors by algebraic dihedral descent. This explicit

family is not identified with all components of the higher-product failure scheme. The arbitrary- β theorem, rather than such an identification, is the general extension of the quadratic residual mechanism.

A complete higher-product family is provided by Theorem 10.1. For all hyperplanes $U \subset V_n$, $n \geq 4$, and every $m \geq 2$, the entire reduced failure locus of $\text{Sym}^m U \rightarrow V_{mn}$ is the same secant threefold. We determine every corank, including the jump to m on the evaluation curve. In degree at least three we also determine all associated points of the projective failure scheme: the secant generic point and the embedded evaluation-curve generic point. A lower bound $\lceil m/2 \rceil$ for the nilpotency index shows that the high-degree scheme structure is not recovered from this fixed reduced support. This supplies a global excess-codimension consequence in every higher degree, beyond merely inserting a multiplication map into the general polar formalism.

The geometry arises naturally in recovery of calibrated information metrics: second-order distance contacts give the moment functional on the within-contact products U_ν^2 . A short application is given in Section 12. The appendices give the exact-contact, native-realization, boundary-experiment and finite-precision arguments, with their distinct hypotheses. None is used as an assumption in the algebraic proofs. Classical inputs and the precise extent of the primary-source comparison are recorded in Section 11.

2. POLAR RESIDUAL SPACES AND GLOBAL SINGULARITY STRATA

The deformation mechanism does not require binary forms. Let V, F be finite-dimensional complex vector spaces, $\dim F = p$, and fix a linear map

$$\beta : \text{Sym}^m V \longrightarrow F, \quad m \geq 2.$$

Choose integers $0 < c_\nu < \dim V$, put $X = \prod_\nu \text{Gr}(c_\nu, V)$, and write $M = \dim X$. The tautological multiplication map, its source rank and its expected failure codimension are

$$\mu : \mathcal{E} = \bigoplus_\nu \text{Sym}^m \mathcal{U}_\nu \longrightarrow F \otimes \mathcal{O}_X, \quad e = \text{rank } \mathcal{E}, \quad a = e - p + 1.$$

Assume $e \geq p$, and let $D = V(I_p(\mu))$. No assumption on the codimension of D is made until explicitly stated. The incidence $\mathcal{I} \subset \mathbb{P}(F^*) \times X$ is the zero scheme of $\ell\mu = 0$. Write $D^{(1)}$ for the locus where μ has corank one.

At a point $z = ([\ell], U)$ of this incidence, choose a representative ℓ and define the plane derivative

$$\begin{aligned} B_z : T_U X &\longrightarrow \mathcal{E}_U^*, \\ B_z(A)(u_1 \cdots u_m) &= \sum_{i=1}^m \ell \beta(u_1 \cdots A_\nu u_i \cdots u_m), \quad u_i \in U_\nu. \end{aligned} \quad (2.1)$$

This is well defined for $A_\nu : U_\nu \rightarrow V/U_\nu$: changing a lift adds a product annihilated by ℓ . Define the *polar residual space* and its multiplication map by

$$Q_z = \ker(B_z^* : \mathcal{E}_U \longrightarrow T_U^* X), \quad \tau_z = \mu_U|_{Q_z} : Q_z \longrightarrow \ker \ell \subset F. \quad (2.2)$$

The kernel Q_z is independent of the chosen nonzero representative of ℓ . On a constant-rank stratum of B , these spaces are vector bundles and τ is a bundle map. Equivalently, before trivializing the tautological line $\mathcal{L} = \mathcal{O}_{\mathbb{P}(F^*)}(-1)$, the plane derivative has target $\mathcal{E}^* \otimes \mathcal{L}^*$.

Theorem 2.1 (Polar residual tangent sequence). *At every point $z = ([\ell], U) \in \mathcal{I}$ there is an exact sequence*

$$0 \longrightarrow \ker B_z \longrightarrow T_z \mathcal{I} \longrightarrow (\mu_U Q_z)^\perp / \mathbb{C} \ell \longrightarrow 0. \quad (2.3)$$

In particular

$$\dim T_z \mathcal{I} = M - a + \dim \ker \tau_z. \quad (2.4)$$

The projection $\mathcal{I} \rightarrow D$ is a scheme isomorphism over $D^{(1)}$, so the same formula computes $\dim T_U D$ there. For $m = 2$, set $H_\ell(v, w) = \ell\beta(vw)$ and $W_\nu = U_\nu \cap \text{rad } H_\ell$. Then

$$Q_z = \bigoplus_\nu \text{Sym}^2 W_\nu, \quad \tau_z((q_\nu)_\nu) = \sum_\nu \beta(q_\nu). \quad (2.5)$$

Thus the quadratic obstruction is an actual smaller multiplication problem for every symmetric bilinear system, not only for binary forms.

Proof. Differentiating the incidence equations gives $\dot{\ell}\mu_U + B_z(\dot{U}) = 0$. A vector in $\mathcal{E}_U^*$ belongs to $\text{im } B_z$ exactly when it annihilates Q_z . The equation is therefore soluble in $\dot{U}$ exactly when $\dot{\ell}$ annihilates $\mu_U Q_z$. Its set of solutions is a torsor under $\ker B_z$. Taking $\dot{\ell}$ modulo $\mathbb{C}\ell$ proves (2.3). If $h = \dim Q_z$, then $\text{rank } B_z = e - h$, so its middle dimension is

$$M - (e - h) + p - 1 - \text{rank } \tau_z = M - a + \dim \ker \tau_z.$$

On a chart with an invertible $(p - 1)$ -minor, invertible row and column operations put μ in the form $\text{diag}(I_{p-1}, S)$, where S is a single row. The incidence equations first solve the normalized left kernel vector uniquely and then impose $S = 0$. This proves the scheme isomorphism, not just a bijection on points.

For the quadratic assertion, the map $V/U_\nu \rightarrow U_\nu^*$ induced by H_ℓ has image $(U_\nu/W_\nu)^*$. Its transpose has kernel W_ν , and it descends through U_ν because U_ν is isotropic. Choose $U_\nu = W_\nu \oplus Z_\nu$. Varying vectors of W_ν in lifts dual to Z_ν prescribes every mixed symmetric entry; varying the vectors of Z_ν prescribes their symmetric block, using division by two. Consequently $\text{im } B_z$ is exactly the space of forms vanishing on every $\text{Sym}^2 W_\nu$. Taking annihilators gives (2.5). $\square$

We use $I_s(A)$ for the ideal of s -minors of a matrix, with $I_s = (1)$ for $s \leq 0$ and $I_s = (0)$ if the requested minors are too large. For a finite module with presentation $R^q \rightarrow R^N \rightarrow C \rightarrow 0$, $\text{Fitt}_i C$ is generated by the $(N - i)$ -minors of the presentation matrix. This definition is independent of the presentation. All statements about restrictions of these ideals below mean pullback of the ideals of D ; they do not mean the intrinsic singular scheme of a locally closed stratum.

Theorem 2.2 (Global singularities and residual Fitting strata). *Suppose D is nonempty and of pure expected codimension $a > 0$ in X , and put $d = M - a$. Then its full singular support is*

$$\text{Sing}(D)_{\text{red}} = \{U : \text{corank } \mu_U \geq 2\}_{\text{red}} \cup \{U \in D^{(1)} : \ker \tau_{([U], U)} \neq 0\}_{\text{red}}. \quad (2.6)$$

The union denotes the indicated subsets of D , with their reduced closed union; the second set is locally closed in D . On any locally closed constant-rank stratum $T \subset D^{(1)}$ of B , with $h = \text{rank } Q$, one has, for every integer $j \geq 1$,

$$\text{Fitt}_{d+j-1}(\Omega_{D/\mathbb{C}})\mathcal{O}_T = I_{h-j+1}(\tau). \quad (2.7)$$

In particular the restriction of the singular scheme $V(\text{Fitt}_d \Omega_{D/\mathbb{C}})$ is the residual noninjectivity scheme $V(I_h(\tau))$, including its determinantal scheme structure. Formula (2.7) detects all tangent-excess strata, including points outside a generic node locus and points whose annihilator has colliding support.

At a point of original corank $\rho \geq 1$, write $S = D_0 - CA^{-1}B_0$ for the Schur complement of an invertible $(p - \rho)$ -block of μ . Let f_α be all its ρ -minors and choose smooth local coordinates $x_1, \dots, x_M$ on X . On this chart the singular scheme has the exact ideal

$$I_\rho(S) + I_a((\partial f_\alpha / \partial x_i)_{\alpha, i}) \quad \text{in } \mathcal{O}_X. \quad (2.8)$$

The Schur–Jacobian presentations hold on open charts for every $\rho \geq 1$, so they cover the whole failure scheme and are compatible on overlaps. The residual-minor simplification is a pullback to the stated strata, not an assertion of a product neighbourhood transverse to those strata.

Proof. Over $D^{(1)}$ use the incidence isomorphism of Theorem 2.1. In local coordinates on $\mathbb{P}(F^*) \times X$, the differential of its e equations is the map from a free module of rank $M + p - 1$ to one of rank e obtained by adjoining $\dot{\ell}\mu$ to B . Its transpose presents the cotangent module of the incidence. Restrict to T and split off an invertible $(e - h)$ -minor of B . Constant rank means that its remaining Schur block is identically zero on T , not just at a chosen point. After elementary operations the remaining nonzero block is the map from $F^*/\mathbb{C}\ell$ to Q^* induced by $\dot{\ell}\mu$; it is τ^* , with the tautological-line twist if no representative of ℓ was chosen. Fitting ideals commute with base change. The minors defining Fitt_{d+j-1} have size

$$(M + p - 1) - (d + j - 1) = e - j + 1.$$

After removing the identity block their size is $h - j + 1$. This proves (2.7). Change of frames or of a representative of ℓ multiplies the matrices by invertible matrices and hence leaves these ideals unchanged.

Since the local dimension on D is d , the tangent formula gives regularity at a corank-one point exactly when τ is injective. Over $\mathbb{C}$, regularity is smoothness. At a point of corank $\rho \geq 2$, Schur elimination identifies the defining ideal with $I_\rho(S)$, where every entry of S vanishes at the point. Thus this ideal is contained in the square of the maximal ideal. All its first differentials vanish, and $T_U D = T_U X$ has dimension $M > d$. This proves (2.6). Finally the usual exact cotangent presentation for the quotient by $(f_\alpha)_\alpha$ gives (2.8) directly from the definition of Fitt_d . The presentation independence of Fitting ideals proves compatibility with every other Schur chart and with (2.7). $\square$

Corollary 2.3 (Every annihilator rank for quadratic binary subseries). *In the notation of Theorem 1.1, suppose $4 \leq c < k$, $e \geq p$, and $a \leq b$. Theorem 2.2 applies to $\mathcal{D}_L$. On its corank-one locus, put $W_\nu = U_\nu \cap \text{rad } H_{\ell_U}$ and $h = \sum_\nu \binom{\dim W_\nu + 1}{2}$. Then*

$$U \text{ is smooth} \iff \bigoplus_\nu \text{Sym}^2 W_\nu \longrightarrow V_{2n} \text{ is injective.} \quad (2.9)$$

On each fixed-radical-intersection-dimension stratum, the singular ideal and all higher tangent-excess ideals are the minors in (2.7) of this smaller multiplication map. No assumption on distinct support, maximal annihilator rank, or membership in just one signed component is required.

Proof. Theorem 1.1 and Proposition 5.6 give pure expected codimension. Equation (2.5) identifies the residual map. Constant dimensions of the intersections give vector bundles after stratification. The preceding theorem applies. $\square$

Corollary 2.4 (The entire residual wall stratification). *At $a = b$, on a signed rank-two stratum with original multiplication of corank one and $U_\nu \not\subset \text{rad } H_\ell$, the pullback of the singular scheme is exactly the residual determinant scheme of μ'_K . More generally its tangent-excess-at-least- j ideal is $I_{h-j+1}(\mu'_K)$, where $h = L\binom{c}{2} = 2k - 5$. This includes every residual corank, not only the open ordinary-double-crossing locus.*

Proof. Here $Q = \bigoplus \text{Sym}^2(gK_\nu)$, and multiplication factors as μ'_K followed by the injective multiplication by g^2 . The source and residual target both have dimension h . Apply (2.7); multiplication by g^2 is a subbundle inclusion on the split rank-two base. $\square$

Proposition 2.5 (Excess-codimension form of the residual criterion). *Without the expected-codimension hypothesis, let $T \subset D^{(1)}$ be a constant-rank stratum of B . For any integer t , locally*

$$\text{Fitt}_{M-t}(\Omega_{D/\mathbb{C}})\mathcal{O}_T = I_{h-a+t}(\tau). \quad (2.10)$$

At a point where $\dim_U D = M - t$, smoothness is equivalent to $\dim \ker \tau = a - t$. For quadratic binary subseries in the excess regime $b < a$, this gives $t = b$ at every corank-one point lying on a

maximal-dimensional component. It is a statement about the full local failure scheme there, not only its selected component.

Proof. The same cotangent presentation has $M + p - 1$ generators, so the requested minors have size $p - 1 + t = e - a + t$. Remove the $(e - h)$ -identity block. This gives (2.10). Compare (2.4) with the stated local dimension. In the last assertion the global dimension bound is $M - b$, and the component through the point attains it. $\square$

These results separate a full singular-locus calculation from a classification of local analytic types or of associated primes. Equation (2.8) retains nonlinear Schur entries: replacing it by minors of their linear parts would lose information. Neither the global criterion nor the excess formula asserts that every germ is a normal crossing, or that every excess scheme is unmixed.

The generality in β has concrete instances. For a projective variety Y and a line bundle A , one may take $V = H^0(Y, A)$, $F = H^0(Y, A^m)$ and β to be multiplication of sections; the same construction applies to multivariate polynomial spaces and their linear quotients. Theorems 2.1 and 2.2 use precisely the hypotheses stated above in each case. They do not require a secant parameterization or a classification of catalecticant ranks. For $m > 2$ the polar residual Q need not be a symmetric power of a radical, which is why the general statement uses the kernel of the plane derivative instead of imposing a quadratic description.

3. INTRINSIC RESIDUAL EQUATIONS AT EVERY MULTIPLICATION CORANK

The residual Fitting identities describe restrictions of intrinsic ideals. To pass from these restrictions to a neighbourhood, one must retain the nonlinear equations. We give such a presentation at every point. The linear algebra in this section applies to a map of vector bundles on any smooth complex variety; the polar formula identifies its intrinsic obstruction for the multiplication systems of Section 2.

The matrix normal derivative and the transverse determinantal model used below are classical [10, Remark 1.18; Section 1.3.2]. The point here is their explicit expression in multiplication polars, including the joint relations between distinct annihilators, and the nonlinear equations left when transversality fails.

3.1. The joint polar obstruction. Fix $U \in D$, and write

$$K_U = \ker \mu_U, \quad C_U = \operatorname{coker} \mu_U, \quad \rho = \dim C_U, \quad s = \dim K_U = e - p + \rho.$$

Here $C_U^* \subset F^*$ is the entire space of annihilators, not a choice of one projective annihilator. Differentiating μ and then restricting to its kernel and projecting to its cokernel defines

$$\partial_U : T_U X \longrightarrow \operatorname{Hom}(K_U, C_U), \quad A \longmapsto [\dot{\mu}_U(A)|_{K_U}]. \quad (3.1)$$

This map is intrinsic. Indeed a change of source or target frame changes $\dot{\mu}$ by terms of the form $P\mu_U + \mu_U Q$; these vanish after restriction to K_U and projection to C_U .

Proposition 3.1 (Joint polar relations). *For a symmetric multiplication system, the transpose of (3.1) is*

$$\begin{aligned} \partial_U^* : K_U \otimes C_U^* &\longrightarrow T_U^* X, \\ \sum_i q_i \otimes \ell_i &\longmapsto \sum_i B_{([\ell_i], U)}^*(q_i), \end{aligned} \quad (3.2)$$

where the representative ℓ_i is used in the definition of B ; the expression is linear in that representative and extends to $\ell_i = 0$. Put

$$\mathcal{R}_U = \ker \partial_U^*, \quad t = \operatorname{rank} \partial_U, \quad b_U = \dim \mathcal{R}_U = \rho s - t.$$

If $\rho = 1$, a choice of nonzero annihilator identifies $\mathcal{R}_U$ with $\ker \tau_{([\ell], U)}$. If $\rho > 1$, the kernel in (3.2) includes relations between distinct annihilators; it is not in general the direct sum of the single-annihilator kernels.

Proof. For $q \in K_U$, $\ell \in C_U^*$ and $A \in T_U X$, the natural trace pairing gives

$$\langle q \otimes \ell, \partial_U(A) \rangle = \ell \dot{\mu}_U(A)(q) = B_{([\ell], U)}(A)(q).$$

Summing proves the formula. If $C_U^* = \mathbb{C}\ell$, its kernel is $K_U \cap \ker B^* = K_U \cap Q = \ker(\mu_U|_Q)$. All identifications are compatible with changes of frame by the intrinsic construction of ∂_U . $\square$

3.2. An exact residual presentation. Let $H_U = \text{Hom}(K_U, C_U)$ and $H_0 = \text{im } \partial_U$. Choose a vector-space complement H_1 so that $H_U = H_0 \oplus H_1$, and choose an isomorphism $L : \mathbb{C}^t \rightarrow H_0$. Such choices specify coordinates, not additional hypotheses. In particular H_1 identifies with $\text{coker } \partial_U$ and has dimension b_U .

Theorem 3.2 (Residual matrix germ). *At every $U \in D$ there are analytic coordinates $(y_1, \dots, y_t, x_1, \dots, x_{M-t})$ centred at U and an H_1 -valued convergent power series $\Psi(y, x)$ with $\Psi(0) = d\Psi(0) = 0$ such that*

$$\widehat{\mathcal{O}}_{D, U} \simeq \mathbb{C}[[y_1, \dots, y_t, x_1, \dots, x_{M-t}]]/I_\rho(L(y) + \Psi(y, x)). \quad (3.3)$$

The same presentation holds with convergent power series for the analytic local ring. Moreover, for $1 \leq j \leq \rho$, the ideal of the locus where the original multiplication has rank at most $p - \rho + j - 1$ is, in this presentation,

$$I_j(L(y) + \Psi(y, x)). \quad (3.4)$$

These are equalities of ideals after analytic change of coordinates, not merely tangent-space or set-theoretic assertions. No expected-codimension assumption is required.

Proof. Put $r = p - \rho$. Trivialize source and target and choose an invertible r -minor at U . After constant changes of frame the matrix is

$$\mu = \begin{pmatrix} A & B_0 \\ C & D_0 \end{pmatrix}, \quad A(U) = I_r, \quad B_0(U) = C(U) = D_0(U) = 0.$$

Invertible analytic row and column operations reduce it to $\text{diag}(I_r, S)$, where $S = D_0 - CA^{-1}B_0$ is a $\rho \times s$ matrix vanishing at U . If $r = 0$, this means $S = \mu$, with no inverse block. The differential of S at U is precisely (3.1); the correction $CA^{-1}B_0$ starts in degree two.

Let $P_0 : H_U \rightarrow H_0$ and $P_1 : H_U \rightarrow H_1$ be the projections. The map $L^{-1}P_0S$ has differential of rank t . Complete its t coordinate functions to analytic local coordinates (y, x) on X using the inverse function theorem. Then $P_0S = L(y)$ exactly, whereas $\Psi = P_1S$ has no constant or linear term. This proves the claimed matrix expression. For a block diagonal matrix with an invertible r -block,

$$I_{r+j}(\text{diag}(I_r, S)) = I_j(S) \quad (1 \leq j \leq \rho).$$

To see both inclusions, a j -minor of S extends by the identity block to an $(r+j)$ -minor; all other terms are minors of S of size at least j and belong to $I_j(S)$. This proves (3.4) and, with $j = \rho$, the analytic presentation of D . Completion gives (3.3). $\square$

Remark 3.3. The power series Ψ is not a canonical function: complements, frames and coordinates enter its construction. The completed local ring and the nested determinantal ideals are canonical. Their compatibility on overlaps follows because each presentation is obtained from the original ideals $I_{r+j}(\mu)$ by invertible operations. The theorem therefore supplies local equations also at support collisions and at points with several annihilators. It does not replace Ψ by its quadratic term or assert that a constant polar-rank stratum has a product neighbourhood.

The first nonzero residual terms can be computed without solving all the equations. For example, along a curve with parameter h , write

$$\mu(h) = \begin{pmatrix} A_0 + hA_1 + O(h^2) & hB_1 + h^2B_2 + O(h^3) \\ hC_1 + h^2C_2 + O(h^3) & hD_1 + h^2D_2 + O(h^3) \end{pmatrix}, \quad A_0 \text{ invertible.}$$

Expansion of the inverse gives

$$S(h) = hD_1 + h^2(D_2 - C_1A_0^{-1}B_1) + O(h^3). \quad (3.5)$$

For a velocity in $\ker \partial_U$, the class of the second coefficient in H_U/H_0 is independent of the acceleration of the curve. An acceleration changes that coefficient by an element of $\text{im } \partial_U$. The same argument, using the chain rule and $D_1 = 0$, proves independence of coordinates and frames. Thus (3.5), paired with $\mathcal{R}_U = (H_U/H_0)^*$, is the intrinsic quadratic obstruction on $\ker \partial_U$. It describes the quadratic part of $\Psi(0, x)$; it is not by itself the initial ideal of all the maximal minors.

Corollary 3.4 (Transverse determinantal neighbourhoods). *If $\mathcal{R}_U = 0$, then $M \geq \rho s$ and*

$$\hat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[z_1, \dots, z_{M-\rho s}, (y_{ij})_{1 \leq i \leq \rho, 1 \leq j \leq s}]]/I_\rho((y_{ij})). \quad (3.6)$$

This is also an analytic product decomposition. The germ is reduced, Cohen–Macaulay and of dimension $M - a$. If $\rho = 1$ it is smooth; if $\rho \geq 2$ its singular support is the product of the smooth factor with the rank-at-most- $(\rho - 2)$ generic matrix locus. This last assertion concerns reduced support, not equality of the singular-scheme ideal with $I_{\rho-1}((y_{ij}))$.

Proof. Here $t = \rho s$, $H_1 = 0$, and (3.3) has no nonlinear residual part. Choosing the entries of S as coordinates proves (3.6). For completeness, the generic rank-at-most- $(\rho - 1)$ matrix variety is irreducible: it is the closure of products of a $\rho \times (\rho - 1)$ matrix and a $(\rho - 1) \times s$ matrix. Its rank- $(\rho - 1)$ stratum has dimension $(\rho - 1)(s + 1)$, as one sees by choosing its image and a surjection onto it. Its codimension is $s - \rho + 1 = a$. The expected-grade maximal-minor resolution [8] consequently gives a Cohen–Macaulay quotient. At rank $\rho - 1$, Schur elimination leaves a independent entries, so the quotient is generically smooth and hence generically reduced. Cohen–Macaulayness gives S_1 ; $R_0 + S_1$ implies reducedness [24]. At rank at most $\rho - 2$, every first derivative of every maximal minor vanishes. The tangent dimension is the ambient dimension, strictly larger than the local dimension. These and only these points are singular. For $\rho = 1$ the ideal consists of independent coordinates. The dimension and all product assertions follow. $\square$

3.3. Corank-one equations and their quadratic obstruction. Corank one admits a smaller presentation than the general matrix normal form: all independent linear equations may be eliminated.

Theorem 3.5 (Minimal residual equations in expected codimension). *Suppose μ_U has corank one, put $z = ([\ell], U)$ and $\delta = \dim \ker \tau_z$, and let $d = M - a$. There is an exact presentation*

$$\hat{\mathcal{O}}_{D,U} \simeq \mathbb{C}[[x_1, \dots, x_{d+\delta}]]/(f_1, \dots, f_\delta), \quad f_i \in (x_1, \dots, x_{d+\delta})^2, \quad (3.7)$$

and an analogous convergent presentation. If D has codimension a at U , these δ equations form a regular sequence. In that case the displayed embedding dimension and number of equations are minimal. Without that codimension hypothesis the exact presentation still holds, but regularity of the sequence is not asserted.

The quadratic obstruction takes values in $(\ker \tau_z)^*$. For a tangent vector $(\lambda, A) \in T_z \mathcal{I}$, choose graph lifts of the maps $A_\nu : U_\nu \rightarrow V/U_\nu$. On a monomial $u_1 \cdots u_m \in \text{Sym}^m U_\nu$ define

$$\begin{aligned} \mathcal{H}_{\lambda, A}(u_1 \cdots u_m) &= \sum_{i < j} \ell \beta(u_1 \cdots A_\nu u_i \cdots A_\nu u_j \cdots u_m) \\ &\quad + \sum_i \lambda \beta(u_1 \cdots A_\nu u_i \cdots u_m). \end{aligned} \tag{3.8}$$

Extend linearly to $\mathcal{E}_U$. Its restriction to $\ker \tau_z$ is the intrinsic quadratic obstruction and equals the degree-two part of the residual equations, after identifying their coordinate space with $(\ker \tau_z)^*$.

Proof. The incidence is the zero scheme of a rank- e section on the smooth $(M+p-1)$ -dimensional variety $\mathbb{P}(F^*) \times X$. Its differential is

$$T_{[\ell]} \mathbb{P}(F^*) \oplus T_U X \longrightarrow \mathcal{E}_U^*, \quad (\lambda, A) \longmapsto \lambda \mu_U + B_z(A).$$

The annihilator of its image is $\ker \mu_U \cap \ker B_z^* = \ker \tau_z$; thus its rank is $e - \delta$. Choose that many independent equations and eliminate them, together with the corresponding coordinates, by the analytic implicit function theorem. The resulting smooth space has dimension $M+p-1-(e-\delta) = d + \delta$. Subtracting appropriate linear combinations of the eliminated equations leaves δ equations with zero differential. Projection from the incidence is a scheme isomorphism here by Theorem 2.1; this proves (3.7).

If the local codimension is a , this local ring has dimension d . In the regular local ring with $d + \delta$ parameters its ideal, generated by δ elements, has height δ . Such a sequence is regular: a regular local ring is Cohen–Macaulay, and a height-equal-to-length generating sequence is a partial system of parameters, hence a regular sequence. Since all f_i are quadratic or higher, no further smooth coordinate can be eliminated, and a regular sequence of this length minimally generates the ideal.

To compute the quadratic terms, trivialize the projective line near $[\ell]$ and use the graph path $(\ell + h\lambda, u_i + hA_\nu u_i)$. Expanding the incidence section, its coefficient of h^2 is exactly (3.8). The order- h coefficient vanishes because (λ, A) is tangent. A second-order change of path changes the coefficient of h^2 by the image of the section's differential, which is annihilated by $\ker \tau_z$. Changes of graph lifts, projective trivialization or frames have precisely the same effect, by the chain rule and the vanishing first-order section. Thus the restriction is well defined. The nonlinear corrections used in eliminating the independent equations are again second-order changes of path and have zero pairing with this kernel. This proves the asserted identification with the degree-two residual equations. $\square$

Remark 3.6. Theorem 3.5 identifies an actual complete-intersection germ in expected codimension, not just its tangent dimension. Its quadratic forms need not generate the full initial ideal. Higher-order terms in the f_i or in Ψ must be retained unless an additional normal-form argument removes them. The simple-normal-determinant argument below supplies such an argument on the residual wall. In the excess regime the same exact equations remain available without silently imposing expected grade.

3.4. Realization in every symmetric degree. The transverse class is not confined to two-point annihilators or to quadratic binary multiplication. The following explicit model realizes it for every symmetric degree and every possible multiplication corank.

Proposition 3.7 (Realization of transverse matrix germs). *Let $m \geq 2$, $e \geq p \geq 1$ and $1 \leq \rho \leq p$ be integers. There exist a vector space V , a surjection $\beta : \text{Sym}^m V \rightarrow \mathbb{C}^p$, and e one-dimensional contacts in $X = \text{Gr}(1, V)^e$ for which the multiplication has corank ρ and $\mathcal{R}_U = 0$. Consequently its failure germ is the full generic $\rho \times (e - p + \rho)$ maximal-minor germ, times a smooth factor, for every m .*

Proof. Put $r = p - \rho$ and take V with basis $x_1, \dots, x_e, w_1, \dots, w_\rho$. Let $f_1, \dots, f_p$ be the target basis. Define β on monomials by

$$\begin{aligned} \beta(x_i^m) &= f_i & (1 \leq i \leq r), \\ \beta(x_j^{m-1}w_\alpha) &= \frac{1}{m}f_{r+\alpha} & (r < j \leq e, 1 \leq \alpha \leq \rho), \end{aligned} \tag{3.9}$$

and set it equal to zero on all remaining basis monomials. The listed monomials are distinct, including when $m = 2$, so this defines a linear map. It is surjective because $e - r = e - p + \rho \geq \rho > 0$. At $U_j = \mathbb{C}x_j$, the source of the contact multiplication has basis $(x_j^m)_j$, its first r columns are $f_1, \dots, f_r$, and its remaining columns are zero. Thus K_U is spanned by the last $s = e - r$ source vectors and C_U by $f_{r+1}, \dots, f_p$. Varying only the j th line from x_j toward w_α has derivative

$$\left. \frac{d}{dh} \right|_{h=0} \beta((x_j + hw_\alpha)^m) = f_{r+\alpha}, \quad j > r.$$

These ρs independent tangent directions map under ∂_U to every matrix unit in $\text{Hom}(K_U, C_U)$. The map is surjective, and Corollary 3.4 applies. $\square$

This proposition concerns arbitrary symmetric linear systems. It exhibits genuine higher-degree failure germs and supplies concrete models for the intrinsic theorem. It is separate from, and does not presuppose, a global classification for $\text{Sym}^m U \rightarrow V_{mn}$ on the binary Grassmannian. The latter system's two-point sectors remain the subject of Section 9.

3.5. Two explicit nontransverse residual germs. These familiar singularities provide direct tests of the distinction between a residual space and the nonlinear residual equations.

A simultaneous-annihilator relation. For the bundle matrix on $\mathbb{A}^3$

$$\mu = \begin{pmatrix} x & y \\ y & z \end{pmatrix},$$

the origin has $K = C = \mathbb{C}^2$. The intrinsic derivative is inclusion of the symmetric matrices into all 2×2 matrices. Its transpose has the one-dimensional kernel generated by $e_1 \otimes e_2^* - e_2 \otimes e_1^*$. For each fixed nonzero $\ell \in C^*$, however, the map $K \rightarrow T_0^* \mathbb{A}^3$, $q \mapsto \partial^*(q \otimes \ell)$, is injective: its coefficients are those of the product of two nonzero linear binary forms. Thus checking one annihilator at a time would miss the joint relation. In Theorem 3.2, H_0 is the symmetric subspace and $\Psi = 0$; the exact failure germ is $xz - y^2 = 0$. This example is a bundle-matrix example for the intrinsic theorem, not an additional claim about binary subseries.

A cubic multiplication cusp. Let $V = \mathbb{C}X \oplus \mathbb{C}Y \oplus \mathbb{C}Z$, $F = \mathbb{C}$, and define $\beta : \text{Sym}^3 V \rightarrow F$ by $\beta(Y^2Z) = 1/3$, $\beta(X^3) = -1$, and zero on the other basis monomials. Take one line contact, and use the chart $U = \mathbb{C}(Z + xX + yY)$ near $\mathbb{C}Z$. Its multiplication matrix is the one-entry matrix

$$\beta((Z + xX + yY)^3) = y^2 - x^3.$$

At the origin $\delta = 1$ and the quadratic obstruction is y^2 , whereas the exact residual equation is $y^2 - x^3$. The map from its completed local ring to $\mathbb{C}[[t]]$ given by $x = t^2$, $y = t^3$ is injective: reducing a power series modulo $y^2 - x^3$ leaves $A(x) + yB(x)$, and the even powers in $A(t^2)$ cannot cancel the odd powers in $t^3B(t^2)$. Thus the cusp has one branch, unlike the double crossing. This explicit higher-product system shows why the full residual equation, rather than only its quadratic part, occurs in the theorem.

4. THE GLOBAL MULTIPLICATION FAILURE LOCUS

Throughout this section the field is $\mathbb{C}$, $k = n + 1$, $4 \leq c < k$, $q = \binom{c+1}{2}$, and $p = 2k - 1$. On $X_L = \text{Gr}(c, V_n)^L$ let $\mathcal{S}_\nu$ denote the tautological bundles. The failure scheme is defined by the maximal minors of

$$\mu : \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu \longrightarrow V_{2n} \otimes \mathcal{O}_{X_L}.$$

Its underlying set is $\Delta^{(L)}$. We shall prove Theorem 1.1 without using a maximal-rank theorem for general curves.

4.1. Annihilating Hankel forms. For $\ell \in V_{2n}^*$, let $H_\ell(f, g) = \ell(fg)$ on V_n . The following classical input concerns the whole rank locus, including confluent and limiting forms; it is not an assertion only about sums of distinct point evaluations.

Lemma 4.1 (Classical Hankel rank strata). *In $\mathbb{P}(V_{2n}^*)$, the rank-exactly- r stratum of H_ℓ is nonempty and has dimension*

$$h_r = \begin{cases} 2r - 1, & 1 \leq r < k, \\ 2k - 2, & r = k. \end{cases}$$

For $r < k$, its closure is the variety of sums of at most r point evaluations, with Zariski closure understood.

Identification with the classical statement. The rank- $\leq r$ equations are the $(r + 1)$ -minors of the $k \times k$ Hankel matrix in $2k - 1$ independent coordinates. The equality of Hankel minor ideals under changing the rectangular shape, recalled in [7, Section 1], identifies their coordinate ring with that of the maximal minors of an $(r + 1) \times (2k - r - 1)$ Hankel matrix. That ring has affine dimension $2r$. The secant identification is [7, Section 2, equation (2.0.1)]; thus the projective dimension is $2r - 1$. The rank- $\leq r - 1$ subvariety has smaller dimension, so the exact-rank stratum is nonempty and dense. The full-rank stratum is open in $\mathbb{P}^{2k-2}$. These are classical Hankel determinantal facts; no claim about their rational singularities is needed here. $\square$

Lemma 4.2 (Isotropic incidence dimensions). *Let a symmetric form on a k -space have rank r and radical E of dimension $k - r$. For a totally isotropic c -plane U , put*

$$s = \dim((U + E)/E), \quad a = c - s.$$

Such planes exist precisely for

$$\max\{0, c - k + r\} \leq s \leq \min\{c, \lfloor r/2 \rfloor\}. \quad (4.1)$$

The stratum with this value of s has codimension

$$D_{r,s} = (c - s)(r - s) + \frac{s(s + 1)}{2} \quad (4.2)$$

in $\text{Gr}(c, k)$.

Proof. Choose $U \cap E$, an a -plane in E , with dimension $a(k - r - a)$. The image of U in the nondegenerate r -dimensional quotient is an isotropic s -plane, whose Grassmannian has dimension $s(r - s) - s(s + 1)/2$. This follows by differentiating the symmetric restriction equations at an isotropic frame: pairing with a complementary dual frame gives all symmetric variations, so the equations are independent. Existence is equivalent to $2s \leq r$. Finally the lifts are an affine space of dimension $s(k - r - a)$. Subtracting the sum from $c(k - c)$ gives (4.2). Maximal orthogonal Grassmannians may have two components; both have this same dimension, which is all the count uses. $\square$

Lemma 4.3 (Rank optimization). *Assume $Lq \geq 2k - 1$. For all r, s satisfying (4.1),*

$$LD_{r,s} - h_r \geq \min\{Lq - 2k + 2, Lc - 3\}. \quad (4.3)$$

Proof. Write $A_0 = Lq - 2k + 2$ and $B_0 = Lc - 3$. If $r = k$, feasibility forces $s = c$, giving exactly A_0 . Suppose $r < k$. For $s = 0$, the left side is $(Lc - 2)r + 1 \geq Lc - 1 > B_0$.

For $1 \leq s \leq c - 2$, the coefficient of r in $LD_{r,s} - 2r + 1$ is $L(c - s) - 2 \geq 0$. Since $r \geq 2s$, the expression is at least

$$g(s) = 1 + s(Lc - 4) - \frac{Ls(s - 1)}{2}.$$

This is concave as a function of s , and

$$g(1) = B_0, \quad g(c - 2) - B_0 = (c - 3) \left(\frac{L(c + 2)}{2} - 4 \right).$$

The latter is nonnegative except when $(L, c) = (1, 4)$ or $(1, 5)$. In the first exception, saturation forces $k = 5$ and feasibility forces $s \leq 1$. In the second, $k \leq 8$; the values $g(1), g(2), g(3)$ are 2, 2, 1. The last value is feasible only when $k = 8$, in which case $A_0 = 1$. Both exceptions therefore satisfy (4.3).

For $s = c - 1$ and $L = 1$, the expression decreases with r , hence is at least its value at $r = k - 1$, namely $A_0 + k - 2c + 1$. Feasibility implies $k \geq 2c - 1$. For $s = c - 1$ and $L \geq 2$, it is minimized at $r = 2c - 2$ and exceeds B_0 by

$$(c - 2) \left(\frac{L(c + 1)}{2} - 4 \right) \geq 0.$$

Finally, if $s = c$ and $r < k$, it is $Lq - 2r + 1 \geq A_0 + 1$. These cases exhaust (4.1). $\square$

4.2. The common-secant family. Let $C_n \subset \mathbb{P}(V_n^*)$ be the evaluation rational normal curve, and let S_n be the closure of its secant lines. Since $n \geq 4$, S_n is irreducible of dimension three. Define

$$\mathfrak{S}_L = \{([h], U_1, \dots, U_L) : [h] \in S_n, U_\nu \subset \ker h\}, \quad \Sigma_L = \text{pr}_{X_L} \mathfrak{S}_L. \quad (4.4)$$

Proposition 4.4 (A global secant obstruction). *The closed set Σ_L is irreducible, has codimension $Lc - 3$ in X_L , and lies in $\Delta^{(L)}$.*

Proof. The incidence is a relative product of Grassmannians over S_n , so it is irreducible of dimension $3 + Lc(k - 1 - c)$ and its projection is closed. For a general $h = \alpha \text{ev}_x + \beta \text{ev}_y$, with $x \neq y$ and $\alpha\beta \neq 0$, every $f \in \ker h$ satisfies $\alpha f(x) = -\beta f(y)$. Consequently

$$\ell = \alpha^2 \text{ev}_x^{(2n)} - \beta^2 \text{ev}_y^{(2n)}$$

is nonzero and annihilates all products from each U_ν . Closedness extends this conclusion to the whole incidence.

It remains to check the dimension of the image, rather than merely that of the incidence. Fix a general h . If $c = k - 1$, the hyperplane U_1 determines h . Otherwise, the c -planes in $\ker h$ that are also annihilated by some $h' \in S_n \setminus \{h\}$ form a constructible set of dimension at most

$$3 + c(k - 2 - c) < c(k - 1 - c),$$

because $c \geq 4$. Here, for each h' , the plane lies in the intersection of two distinct hyperplanes. Thus a general $U_1 \subset \ker h$ admits no such h' , and the incidence projection is generically one-to-one. Subtraction from $\dim X_L = Lc(k - c)$ gives $Lc - 3$. $\square$

Proof of Theorem 1.1. Failure is equivalent to the existence of a nonzero ℓ annihilating every U_ν^2 . Consider the closed projective incidence of such pairs $((U_\nu), [\ell])$. On the rank- r stratum of H_ℓ , stratify each U_ν by s_ν . Lemmas 4.1 and 4.2 bound the incidence dimension by

$$\dim X_L + h_r - \sum_{\nu} D_{r,s_\nu}.$$

Its image cannot have larger dimension. For fixed r , each term is at least the smallest feasible $D_{r,s}$, so Lemma 4.3 gives

$$\text{codim } \Delta^{(L)} \geq \min\{Lq - p + 1, Lc - 3\} \geq 1.$$

This proves, in particular, generic surjectivity without assuming it.

Proposition 4.4 gives the upper bound $Lc - 3$ and shows that the failure set is nonempty. On a Grassmannian chart the map μ is a $p \times Lq$ matrix over a regular ring. The classical maximal-minor height bound gives codimension at most $Lq - p + 1$ for its nonempty failure locus; see [9, Introduction], with generic rank p and minor size p . The two upper bounds match the lower bound. When $Lc - 3$ is the minimum, the irreducible closed subvariety Σ_L has the maximal dimension of $\Delta^{(L)}$, so it is an irreducible component of its underlying reduced set. If $Lq < p$, the dimension of the domain of μ makes surjectivity impossible. This proves the final equivalence. $\square$

Corollary 4.5 (The base-point locus and the contact count). *For $L = 1$, the locus $\mathcal{B}$ of spaces with a common projective zero is irreducible of codimension $c - 1$, has multiplication corank at least two, and is properly contained in Σ_1 . It is not an irreducible component of $\Delta^{(1)}$. For arbitrary L , the least number of independently chosen c -planes with full product span is*

$$L_{\min} = \left\lceil \frac{2k - 1}{\binom{c+1}{2}} \right\rceil.$$

Proof. For a fixed linear factor l , common-base-point spaces form $\text{Gr}(c, lV_{n-1})$. Varying $[l] \in \mathbb{P}^1$ gives an irreducible incidence of dimension $1 + c(k - 1 - c)$, generically one-to-one because a general residual space has no further common factor. The codimension is $c - 1$. Products are divisible by l^2 , whose multiples have dimension $p - 2$. The annihilating hyperplane is an evaluation point of $C_n \subset S_n$, proving inclusion in Σ_1 . The codimensions $c - 1$ and $c - 3$ make that inclusion proper. An irreducible proper subset of the irreducible closed set $\Sigma_1 \subset \Delta^{(1)}$ cannot itself be a component of $\Delta^{(1)}$. The contact count follows from Theorem 1.1 and the dimension obstruction. $\square$

The theorem determines the exact dimension of the entire bad set and one explicit maximal-dimensional component in the stated regime. It does not assert that every component is a secant component, nor that the determinantal scheme is reduced, equidimensional, or everywhere transverse. In particular, the equality case of the two codimension bounds does not exclude additional components of the same dimension.

5. RESIDUAL CONORMALS AND EXACT LOCAL EQUATIONS

We work over $\mathbb{C}$. Put $E_\mu = \bigoplus_{\nu} \text{Sym}^2 U_\nu$, $M = Lc(k - c)$, $e = L\binom{c+1}{2}$, $p = 2k - 1$, and

$$a = e - p + 1, \quad b = Lc - 3.$$

When $e \geq p$, write $\mathcal{D}_L$ for the scheme of maximal minors of $\mu : E_\mu \rightarrow V_{2n}$, and $\mathfrak{I}$ for the incidence $\ell\mu = 0$ in $\mathbb{P}(V_{2n}^*) \times X_L$. The following result is not restricted to a general point or to a stable range of dimensions.

Theorem 5.1 (Residual tangent sequence). *Let $([\ell], U) \in \mathfrak{I}$, let $E = \text{rad } H_\ell$, and put*

$$W_\nu = U_\nu \cap E, \quad d_\nu = \dim W_\nu, \quad R_W = \sum_{\nu} W_\nu^2 \subset V_{2n}, \quad q = \binom{c+1}{2}.$$

For each ν define

$$B_\nu : \text{Hom}(U_\nu, V_n/U_\nu) \longrightarrow \text{Sym}^2 U_\nu^*, \quad B_\nu(A)(u, v) = H_\ell(Au, v) + H_\ell(u, Av).$$

Then $\text{im } B_\nu$ consists exactly of the symmetric forms whose restriction to $W_\nu \times W_\nu$ is zero. There is an exact sequence

$$0 \longrightarrow \bigoplus_\nu \ker B_\nu \longrightarrow T_{([\ell], U)} \mathfrak{J} \longrightarrow R_W^\perp / \mathbb{C}\ell \longrightarrow 0. \quad (5.1)$$

In particular

$$\dim T\mathfrak{J} = M + p - 1 - \sum_\nu \left(q - \binom{d_\nu + 1}{2} \right) - \dim R_W. \quad (5.2)$$

If μ_U has corank one, these are also the tangent space and tangent dimension of $\mathcal{D}_L$ at U ; its conormal rank in X_L is

$$\sum_\nu \left(q - \binom{d_\nu + 1}{2} \right) - p + 1 + \dim R_W. \quad (5.3)$$

Proof. The map $V_n/U_\nu \longrightarrow U_\nu^*$ induced by H_ℓ has image $(U_\nu/W_\nu)^*$. Indeed its transpose has kernel W_ν ; quotienting by U_ν is permitted because U_ν is isotropic. Choose a complement Z_ν of W_ν in U_ν and lifts of a dual basis of Z_ν^* . The coordinates of $A|_{W_\nu}$ in these lifts prescribe every $W_\nu \times Z_\nu$ entry. The coordinates of $A|_{Z_\nu}$ prescribe every symmetric $Z_\nu \times Z_\nu$ entry, by taking one half of the desired symmetric matrix. This proves the assertion about B_ν , including its rank.

Differentiating $\ell(uv) = 0$ gives

$$\dot{\ell}(uv) + B_\nu(\dot{U}_\nu)(u, v) = 0.$$

It is solvable for the $\dot{U}_\nu$ precisely when $\dot{\ell}$ annihilates all W_ν^2 . Solutions for different contacts are independent. Taking $\dot{\ell}$ modulo $\mathbb{C}\ell$ proves the exact sequence and its dimension. If μ_U has corank one, invert a $(p-1)$ -minor. Row and column operations leave an identity block and one row of a entries. The normalized left kernel vector is uniquely solved by the identity block; the remaining incidence equations are exactly these a entries. Hence the incidence projects isomorphically, as a scheme, to $\mathcal{D}_L$ on this chart. Subtracting the tangent dimension from M gives the conormal rank. $\square$

Proposition 5.2 (Local presentation at every corank). *At a point where μ has corank $\rho \geq 1$, invert a $(p-\rho)$ -minor and write its matrix in blocks as $\begin{pmatrix} A & B \\ C & D \end{pmatrix}$. On this open chart, and also after completion of its local ring,*

$$I_p(\mu) = I_\rho(D - CA^{-1}B). \quad (5.4)$$

The matrix on the right has size $\rho \times (e - p + \rho)$ and vanishes at the given point.

Proof. Multiply by the invertible block matrices eliminating C and B , and then by A^{-1} on the first block. The resulting matrix is $\text{diag}(I_{p-\rho}, D - CA^{-1}B)$, in rectangular block notation. Its maximal minors generate the indicated ideal. Invertible row and column operations preserve determinantal ideals, and localization and completion preserve this identity. $\square$

This is a scheme presentation, not a replacement of its tangent cone by minors of the linear part. Initial ideals can contain additional initial forms obtained from cancellations. Nor does the presentation by itself classify the associated primes of every excess-dimensional germ.

5.1. The rank-two residual morphism. Let $[\ell]$ have rank two and two distinct support points x, y . Choose the two isotropic hyperplanes H_+, H_- of H_ℓ and their intersection $E = gV_{n-2}$, where g vanishes at x, y . Fix a sign pattern ε and consider the open incidence on which $U_\nu \subset H_{\varepsilon_\nu}$ and $U_\nu \not\subset E$.

Proposition 5.3 (Dominance with affine fibres). *For fixed $[\ell]$ and choices of its hyperplanes, the morphism*

$$(U_\nu)_\nu \longmapsto (g^{-1}(U_\nu \cap E))_\nu \text{ to } \mathrm{Gr}(c-1, V_{n-2})^L \quad (5.5)$$

is smooth and surjective. It is locally an affine-space bundle of relative dimension $L(k-c-1)$. The same assertion holds relatively as the distinct support points and their two hyperplanes vary, after trivializing the indicated bundles.

Proof. For a prescribed $W_\nu = gK_\nu$, an admissible U_ν is the inverse image of a line in $H_{\varepsilon_\nu}/W_\nu$ projecting isomorphically to H_{ε_ν}/E . Its choices form a torsor under

$$\mathrm{Hom}(H_{\varepsilon_\nu}/E, E/W_\nu),$$

an affine space of dimension $k-c-1$. Local choices of a splitting trivialize this torsor. Intersections with E have constant dimension $c-1$, so the assignment is a morphism. Products give the result. $\square$

Corollary 5.4 (Exact singularity recursion on a signed stratum). *At a rank-two point as above, suppose that μ_U has corank one. Let μ'_K be multiplication from $\bigoplus_\nu \mathrm{Sym}^2 K_\nu$ to V_{2n-4} and put $\delta = \dim \mathrm{coker} \mu'_K$. Then*

$$\dim T_U \mathcal{D}_L = M - b + \delta. \quad (5.6)$$

If $4 \leq c < k$, $e \geq p$ and $b \leq a$, this point is smooth in $\mathcal{D}_L$ if and only if μ'_K is onto. In this range the singular set on the stated corank-one open stratum is exactly the inverse image of the residual multiplication failure locus.

Proof. Here $d_\nu = c-1$ and $R_W = g^2 \mathrm{im} \mu'_K$, of dimension $p-4-\delta$. Equation (5.2) gives the displayed formula. The signed incidence has image of dimension $M-b$: a general first c -plane in its secant hyperplane determines that hyperplane uniquely, since the locus also lying in a second secant hyperplane has dimension at most $3+c(k-2-c) < c(k-1-c)$. On the corank-one open set the local signed incidence is embedded by the incidence isomorphism. Thus its local dimension is $M-b$. Theorem 1.1 bounds the dimension of the entire failure locus by $M-b$ when $b \leq a$. Consequently its local dimension here is $M-b$, and equality of tangent and local dimensions is equivalent to $\delta = 0$. $\square$

5.2. Projection and commutative algebra.

Proposition 5.5 (From rank incidences to components). *Stratify the annihilator incidence by the rank r of H_ℓ and by $s_\nu = \dim(U_\nu/(U_\nu \cap \mathrm{rad} H_\ell))$. Every irreducible component C of the support of $\mathcal{D}_L$ is dominated by the closure of an irreducible piece of one of these finitely many strata. In particular a stratum of dimension less than $\dim C$ cannot dominate C . If a dominating incidence has dimension $\dim C$, and its generic fibre contains a nonempty open subset of the projective annihilator space, then that annihilator space is a point and its projection is birational in characteristic zero.*

Proof. The projective incidence maps onto the failure locus. Intersect the images of its finitely many locally closed strata and their irreducible pieces with C . These constructible sets cover C ; at least one is dense. The dimension theorem gives the first assertion and implies that the generic fibre of an equal-dimensional dominating piece is zero-dimensional. The projective annihilator space is the projectivization of a vector space, not an arbitrary finite collection. A nonempty open subset of a positive-dimensional projective space has positive dimension. Thus the open-fibre

hypothesis forces the projective space to be a point. On a corank-one chart the incidence is the graph of its uniquely solved annihilator. This proves birationality, including the scheme-theoretic generic-point assertion. $\square$

Proposition 5.6 (Expected grade and absence of embedded primes). *Suppose $4 \leq c < k$, $e \geq p$, and $a \leq b$. On every nonempty affine Grassmannian chart meeting $\mathcal{D}_L$, its maximal-minor ideal has height and grade a . The Eagon–Northcott complex resolves its quotient, which is Cohen–Macaulay of pure codimension a and has no embedded associated primes.*

Proof. The chart ring is regular. The global codimension theorem gives height at least a , whereas the maximal-minor height bound gives height at most $e - p + 1 = a$. In a regular ring grade equals height. The Eagon–Northcott grade criterion gives a free resolution of length a and the Cohen–Macaulay conclusion [8, 19]. Localization gives the same conclusion at every prime over the ideal. A Cohen–Macaulay quotient of this expected dimension is unmixed and has no embedded associated primes. No independence assumption on the matrix entries is used. $\square$

Proposition 5.7 (The reducedness step). *In the situation of Proposition 5.6, if $\mathcal{D}_L$ is generically reduced on each of its irreducible components, then it is reduced. If its support is also irreducible, it is integral.*

Proof. A Noetherian ring without embedded associated primes satisfies S_1 . Generic reducedness is R_0 . The criterion $R_0 + S_1$ for reducedness therefore applies [24, Tag 031R]. Alternatively a nonzero nilradical would have an associated prime; without embedded primes this would be a minimal prime, contradicting generic reducedness. The last assertion is the definition of an integral scheme. $\square$

6. ORIENTATION COVERS AND DESCENT OF ISOTROPIC FAMILIES

We give the descent statements used in the component theorem explicitly. They distinguish connectedness of a cover from geometric irreducibility of a fibre, and also distinguish one maximal-isotropic factor from a product of such factors.

Proposition 6.1 (Relative orthogonal descent). *Let S be a smooth integral complex variety and let $(\mathcal{V}, q)$ be a nondegenerate quadratic bundle of rank $2s$, with values in a line bundle. Its orientation cover $\tilde{S} \rightarrow S$ is the degree-two étale cover whose local equation, after choosing frames, is $z^2 = (-1)^s \det q$. Changes of frames glue these local covers. Suppose this cover is connected. The relative maximal orthogonal Grassmannian $\mathrm{OGr}_S(s, \mathcal{V})$ is integral. More generally, the product of L such Grassmannians for the same quadratic bundle has 2^{L-1} irreducible components, indexed by $\{+, -\}^L$ modulo simultaneous sign change. Each component is smooth over S . Adding relative nonmaximal orthogonal Grassmannian factors does not change this count.*

Proof. If a frame of $\mathcal{V}$ changes by A , its determinant changes by $(\det A)^2$. If the frame of the value line changes by u , the determinant changes by u^{-2s} . Thus the square-root equations are compatible with the transition functions of $\det \mathcal{V}^* \otimes (\text{value line})^s$. Their zero-free right-hand side makes the cover étale. After an étale base change the form has a hyperbolic basis $e_1, \dots, e_s, f_1, \dots, f_s$.

Over an algebraically closed field a maximal isotropic plane belongs to one of two families, distinguished by the parity of its intersection dimension with $\mathrm{span}(e_1, \dots, e_s)$. A Witt basis sends any such plane to the reference plane by an orthogonal transformation. The transformations of determinant one preserve this parity and act transitively on each family; interchanging e_1 and f_1 reverses it. The special orthogonal group is connected: in a hyperbolic basis its elementary orthogonal root subgroups are affine lines, its diagonal torus is a product of multiplicative groups, and these generate the group by the usual elimination of a Witt basis. Therefore each of its two orbits is irreducible. Equivalently, the graph charts within a family are alternating matrices

and are translated under this connected group. The same Witt-basis elimination gives a single irreducible orbit for isotropic planes of dimension strictly less than half the rank.

The two families are consequently locally constant étale labels, with the determinant character describing their interchange. The orientation cover kills this character. Over $\tilde{S}$ each family is a smooth proper relative variety with geometrically irreducible fibres. It is integral: smoothness gives reducedness, every component dominates the integral base, and the geometric generic fibre has only one component. Here domination follows also from local smoothness and properness: each component is open and closed in the smooth total space, so its nonempty image is open and closed in the connected base. The nontrivial deck transformation exchanges the two families. Their images in the descended space are therefore the same irreducible image, which is the whole space. This proves integrality for one factor.

For L factors the geometric components over $\tilde{S}$ are the 2^L products of these integral family bundles. The deck action sends ε to $-\varepsilon$. The orbit unions descend to the irreducible components: the image of any one integral product is irreducible and equals the image of its entire orbit, and distinct orbits have distinct geometric generic fibres. There are 2^{L-1} orbits. Smoothness descends étale locally. Nonmaximal factors have geometrically irreducible fibres and add no labels. $\square$

Proposition 6.2 (The five exceptional orientation bases). *For each exceptional tuple of Theorem 7.1, the incidence containing the full radical and a maximal isotropic quotient is irreducible. For the full-rank case $(c, k, L) = (7, 14, 1)$ the nondegenerate isotropic incidence is irreducible as well.*

Proof. For an exceptional rank $r = 2s < k$, work first over the dense ordered support chart

$$T_r = \{(x_1, \dots, x_r, \alpha_1, \dots, \alpha_{r-1}) : x_i \neq x_j, \alpha_i \neq 0\}, \quad \alpha_r = 1.$$

This is a smooth integral open subset of $\mathbb{A}^r \times \mathbb{G}_m^{r-1}$. It parameterizes the projective functionals $[\ell] = [\sum_i \alpha_i \text{ev}_{x_i}]$. Evaluation identifies the quotient by the radical with $\mathbb{C}^r$, and the induced quadratic form is $\text{diag}(\alpha_1, \dots, \alpha_r)$. In the function field $\mathbb{C}(x_1, \dots, x_r, \alpha_1, \dots, \alpha_{r-1})$, the product $\prod_{i < r} \alpha_i$ is not a square: its valuation at $\alpha_1 = 0$ is one. This valuation is a function-field valuation; the boundary $\alpha_1 = 0$ need not belong to T_r . The orientation cover obtained by adjoining its square root is therefore integral and connected. Proposition 6.1 applies.

The radical is gV_{k-r-1} with $g = \prod_i (t - x_i)$. Containing this radical identifies the exceptional plane incidence with the maximal orthogonal Grassmannian of the quotient. There is one such factor in each of the five cases. The ordered chart maps dominantly to the corresponding exact-rank base. Its generic fibres merely relabel the support: the radical determines g , its distinct roots recover the unordered points, and a Vandermonde system recovers the weights up to common scale. Taking the unordered quotient cannot split the irreducible image of this incidence. Its closure is the exceptional incidence claimed in the theorem. Colliding-support points do not create another component of this closure.

In the remaining full-rank case the universal quadratic form has even rank fourteen. The determinant of the universal Hankel matrix has the irreducible corank-one Hankel hypersurface as its zero divisor, with multiplicity one. The irreducibility follows from the Hankel maximal-minor description in Section 4; simplicity can also be checked at a corank-one form represented by thirteen distinct evaluations. Its kernel is generated by a nonzero polynomial v ; the derivative of the determinant in a Hankel direction ℓ is a nonzero scalar multiple of $\ell(v^2)$, which is not identically zero. On an affine projective chart meeting its generic point this simple prime divisor proves nonsquareness in the function field. Scaling the representative of a rank-fourteen form changes the determinant by a fourteenth power, hence by a square, so this is the projective orientation cover. It is connected and Proposition 6.1 gives the assertion. $\square$

7. COMPONENTS IN ALL DIMENSIONS

We retain the entire range $4 \leq c < k$ and $Lq \geq p$ of Theorem 1.1. Let Φ_L be the closure of the image of the incidence with nonsingular H_ℓ . Let Σ_ε denote the signed secant image: for distinct x, y and $t \neq 0$ its points satisfy

$$U_\nu \subset \ker(\text{ev}_x + \varepsilon_\nu t \text{ev}_y), \quad \varepsilon \in \{+1, -1\}^L / \{\pm 1\}.$$

Only in the five triples in Table 1 do we need an additional notation. For those triples, $\mathcal{T}_{c,k}$ is the closure of the image of the rank- r incidence indicated in the table. Equivalently, on its dense distinct-support open set,

$$E = \text{rad } H_\ell \subset U, \quad U/E \text{ is a maximal isotropic subspace of } V_n/E.$$

Thus these exceptions have one uniform geometric description; they are not five unrelated numerical accidents.

c	k	L	(a, b)	r	$\dim E$	$\dim(U/E)$
5	7	1	(3, 2)	4	3	2
5	8	1	(1, 2)	6	2	3
6	10	1	(3, 3)	8	2	4
6	11	1	(1, 3)	10	1	5
7	13	1	(4, 4)	12	1	6

TABLE 1. All nonendpoint equality incidences.

Theorem 7.1 (All-dimension component theorem). *Suppose $4 \leq c < k$ and $Lq \geq p$.*

- (1) *If $a < b$, the scheme $\mathcal{D}_L$ is integral and Cohen–Macaulay of codimension a . Its support is Φ_L , except at $(c, k, L) = (5, 8, 1), (6, 11, 1)$, where its support is $\mathcal{T}_{5,8}, \mathcal{T}_{6,11}$, respectively.*
- (2) *If $a = b$, the scheme is reduced and Cohen–Macaulay of pure codimension a . Its components are Φ_L and the 2^{L-1} varieties Σ_ε , except at $(6, 10, 1), (7, 13, 1)$, where Φ_L is replaced by $\mathcal{T}_{6,10}, \mathcal{T}_{7,13}$.*
- (3) *If $b < a$, the maximal-dimensional components are exactly the 2^{L-1} signed secant varieties, except at $(5, 7, 1)$, where $\mathcal{T}_{5,7}$ is one additional maximal-dimensional component. Their codimension is b .*

Every component listed above has general multiplication corank one and is generically smooth in the maximal-minor scheme. Each of its indicated annihilator incidences is birational onto it. The generic Hankel rank is k on Φ_L , two on Σ_ε , and the rank in Table 1 on $\mathcal{T}_{c,k}$. The signed components are rational. In the range $a \leq b$,

$$[\mathcal{D}_L] = c_a \left(\left(\bigoplus_{\nu} \text{Sym}^2 \mathcal{S}_\nu \right)^* \right) \quad \text{in } A^a(X_L). \quad (7.1)$$

The third assertion concerns maximal-dimensional components, not all associated primes in the excess range. The theorem removes the stable restriction from the component and expected-codimension scheme assertions; it does not identify distinct notions of completeness.

7.1. The complete equality analysis.

Lemma 7.2 (All endpoint ties). *Among the rank incidences in the proof of Theorem 1.1, an incidence of dimension at least $M - \min(a, b)$ is either the rank-two incidence with every $s_\nu = 1$, the full-rank incidence with every $s_\nu = c$, or one of the five incidences in Table 1. The dimension in each listed case is exactly $M - \min(a, b)$ when its corresponding endpoint is minimal.*

Proof. Use $D_{r,s}$ and h_r in Lemmas 4.1 and 4.2. For $r < k$ and $s = 0$ the cost is $(Lc - 2)r + 1 > b$. For $1 \leq s \leq c - 2$, minimization in $r \geq 2s$ uses the nonnegative coefficient $L(c - s) - 2$ and gives

$$g(s) = 1 + s(Lc - 4) - Ls(s - 1)/2, \quad g(1) = b, \quad g(c - 2) - b = (c - 3)\{L(c + 2)/2 - 4\}.$$

If $L \geq 2$, the last difference is strictly positive, and strict concavity excludes the intervening integers. The coefficient for $s = 1$ is positive, so equality there requires $r = 2$. For $s = c - 1$, $L \geq 2$, the cost at its minimum exceeds b by $(c - 2)\{L(c + 1)/2 - 4\} > 0$. For $s = c$ and $r < k$, the cost is at least $a + 1$. Thus there are no extra ties for $L \geq 2$.

It remains to take $L = 1$. Saturation gives $k \leq (q + 1)/2$. For $c = 4$, this forces $k = 5$ and feasibility permits only $s \leq 1$ when $r < k$. For $c = 5$, the values at $s = 1, 2, 3$ are $2, 2, 1$. At $s = 2$ equality requires $r = 4$, and feasibility together with the minimal endpoint gives $k = 7$. At $s = 3$ the coefficient of r is zero; feasibility forces $r = 6, k = 8$. For $c = 6$, the only further lower-endpoint tie is $s = 4, r = 8, k = 10$. For $c \geq 7$, strict concavity and the displayed endpoint difference exclude every $2 \leq s \leq c - 2$.

For $s = c - 1$ the cost is decreasing in r and is bounded below by $a + k - 2c + 1$. Feasibility requires $k \geq 2c - 1$. When $k \geq 2c$, this is strictly greater than a . When $k = 2c - 1$, feasibility forces $r = 2c - 2$, and

$$a - b = (c - 7)(c - 2)/2.$$

Saturation permits this case for $c \geq 6$; the minimal-endpoint ties are precisely $c = 6$ and $c = 7$, giving $(6, 11)$ and $(7, 13)$. For $c \geq 8$ the cost a is larger than b . The case $s = c$ was already excluded. This proves the one-contact list.

Finally, if the contacts have different s_ν , write their total cost as the average of $LD_{r,s_\nu} - h_r$. Every summand is at least $\min(a, b)$ by Lemma 4.3; equality of the average requires equality of every summand. The preceding analysis shows that for $L \geq 2$ only the identical endpoint choices remain. The exceptional cases all have $L = 1$. $\square$

Lemma 7.3 (Three-dimensional residual subseries). *For $K \geq 3$ and $L \geq 1$, general three-planes in a K -dimensional binary space have full quadratic product span whenever $6L \geq 2K - 1$, provided either $K = 3$ or $L \geq 2$.*

Proof. For $K = 3$ the plane is the entire binary space. Otherwise stratify the annihilator incidence as before, now with $c = 3$ and $L \geq 2$. For $r < K$, the costs for $s = 0, 1, 2, 3$ are respectively

$$(3L - 2)r + 1, \quad (2L - 2)r - L + 1, \quad (L - 2)r + L + 1, \quad 6L - 2r + 1.$$

On their feasible ranges they are positive: use $r \geq 2$ for $s = 1$, $r \geq 4$ for $s = 2$, and $r \leq K - 1$ for $s = 3$. The full-rank cost, when feasible, is $6L - 2K + 2 \geq 1$. The projective incidence therefore cannot dominate the Grassmannian product. This proves generic surjectivity. $\square$

7.2. Irreducibility and the exceptional geometry.

Lemma 7.4 (Signed incidences in the full range). *For $c \geq 4$, the 2^{L-1} signed secant images are distinct, irreducible and rational of codimension b . A general point of one admits exactly one rank-two projective annihilator.*

Proof. A general secant hyperplane has a unique unordered decomposition into two evaluations: evaluations at four distinct points are independent for $n \geq 3$, and the limiting assertion follows by Hermite interpolation. Its partner hyperplane and its rank-two quadratic annihilator are therefore determined. The secant parameter space is rational, being birational to a projective-line bundle over $\text{Sym}^2 \mathbb{P}^1$. Normalize the first sign to be positive. The incidence is a product of relative Grassmannians of dimension $3 + Lc(k - 1 - c) = M - b$, rational over this rational base.

For a fixed general first hyperplane h , planes admitting a second secant hyperplane have dimension at most $3 + c(k - 2 - c)$ in $\text{Gr}(c, \ker h)$, strictly less than $c(k - 1 - c)$. Thus the

first plane recovers h generically. A general other plane lies in only its chosen partner, proving distinctness of the sign sectors and birationality. Every rank-two isotropic plane is contained in one of its isotropic hyperplanes. The exceptional confluent secants have smaller base dimension. These facts also give uniqueness of the general rank-two annihilator. $\square$

Lemma 7.5 (Orthogonal incidences and their orientation covers). *In the expected-codimension range, the nondegenerate incidence, when feasible, is irreducible of dimension $M - a$ and contains points outside all the signed images. Each exceptional incidence in Table 1 is irreducible and its image also contains points outside the signed images.*

Proof. If $2c < k$, the nondegenerate incidence is smooth over the irreducible open set of nonsingular Hankel forms, with geometrically irreducible fibres $\text{OGr}(c, k)^L$. The dimension of each factor is $c(k - c) - q$. For a fixed nonsingular form, isotropic c -planes in a hyperplane have codimension at least c in this Grassmannian: if the restricted form has a one-dimensional radical, separate planes containing that radical from those avoiding it. Their codimension drops are respectively $k - c - 1 \geq c$ and c . Varying a secant hyperplane contributes three parameters, fewer than c . Hence the first plane can be chosen outside every secant hyperplane.

The inequalities $a \leq b$ and $e \geq p$ imply $2c < k$, except for the four expected-codimension entries of Table 1 and $(c, k, L) = (7, 14, 1)$. This follows at once for $c \geq 8$ from $2k \geq L \binom{c}{2} + 5$; for $4 \leq c \leq 7$ substitution gives exactly these five possibilities. In the four table entries the full-rank incidence is empty. For $(7, 14, 1)$ its maximal orthogonal fibres have two components, but the total incidence is irreducible. Indeed the orientation double cover adjoins a square root of the determinant of the universal Hankel form. This determinant is not a square in the function field: its zero divisor is the irreducible rank-at-most-13 Hankel hypersurface, and it vanishes simply at a general corank-one point. Simplicity also follows by perturbing that point by one further evaluation nonzero on its kernel. The connected orientation cover exchanges the two maximal orthogonal families. Over it each family has geometrically irreducible fibres, which proves irreducibility of the total incidence. In a maximal orthogonal Grassmannian the condition of containing a fixed isotropic line has codimension $c - 1 = 6$; the same three-parameter secant argument gives a nonsecant first plane.

For an exceptional incidence write $r = 2(k - c)$ and $d = 2c - k$. On the dense distinct-support open set,

$$\ell = \sum_{i=1}^r \alpha_i \text{ev}_{x_i}^{(2n)}, \quad E = gV_{d-1}, \quad V_n/E \simeq \mathbb{C}^r,$$

with quotient form $\text{diag}(\alpha_i)$ and all $\alpha_i \neq 0$. The product $\prod_i \alpha_i$ is nonsquare in the function field of the ordered support and projective weight parameter space (normalize the last weight to one). Thus its orientation cover is connected. The same argument for maximal orthogonal fibres proves irreducibility of this incidence. Its closure defines the incidence used here; lower-dimensional support collisions do not add another generic family.

To find a point outside the secant images, first suppose $d \geq 2$. A secant functional annihilating gV_{d-1} must have both its support points among the roots of g , since evaluations of V_{d-1} at two distinct points are independent. A confluent secant contributes only a pure evaluation, because g has simple roots. In the nondegenerate quotient an admissible hyperplane must have isotropic dual vector. Each of the finitely many secant lines through support points contains only two such vectors; a pure evaluation has anisotropic dual. A maximal isotropic s -plane containing any fixed isotropic line is a proper subvariety of codimension $s - 1 > 0$. Avoid this finite union. If $d = 1$, the secant functionals annihilating g form a set of dimension at most two. Here $s = 5$ or 6 , and $s - 1 > 2$, so the same incidence bound again leaves a nonempty complement. This proves the assertion. $\square$

Lemma 7.6 (A corank-one point in the excess exception). *The image $\mathcal{T}_{5,7}$ has dimension $M - b$ and general multiplication corank one.*

Proof. Dehomogenize and choose a monic square-free polynomial g of degree four. In V_6 put

$$U = \text{span}\{1, t, g, tg, t^2g\}.$$

For its four roots x_i , the nonzero barycentric weights $\alpha_i = 1/g'(x_i)$ satisfy $\sum_i \alpha_i x_i^j = 0$ for $j = 0, 1, 2$. Thus $\ell = \sum_i \alpha_i \text{ev}_{x_i}^{(12)}$ has Hankel rank four, radical gV_2 , and annihilates U^2 . The twelve products

$$1, t, t^2, g, tg, t^2g, t^3g, g^2, tg^2, t^2g^2, t^3g^2, t^4g^2$$

have distinct degrees, hence are independent. They span a hyperplane of V_{12} . The annihilator is unique, so the incidence projection is an isomorphism on a neighbourhood in its corank-one open set. Its image has the incidence dimension $M - b$. This example is not in a secant hyperplane. Since it contains gV_2 , any secant annihilator would have both support points among the roots of g , by the argument in Lemma 7.5. Evaluations of $1, t$ at two distinct such roots are independent; a pure evaluation is also excluded. $\square$

7.3. Proof of the component theorem.

Proof of Theorem 7.1. We use Propositions 6.1 and 6.2 for the descent of maximal orthogonal families; in particular a connected orientation cover is not being confused with geometric irreducibility of its fibres.

Assume first $a \leq b$. Proposition 5.6 makes $\mathcal{D}_L$ pure of dimension $M - a$. By Proposition 5.5 and Lemma 7.2, its components must be dominated by the listed equal-dimensional incidences. In the strict case only the nondegenerate incidence, or the specified exceptional replacement, can dominate. It is irreducible, so the nonempty failure scheme has just that support. At the wall the signed images already have the component dimension. The nonsecant point of Lemma 7.5, purity, and the same rank-gap argument force one further component, dominated by the nondegenerate incidence or its exceptional replacement. There can be no others.

For each nondegenerate or exceptional component, its incidence and image have the same dimension. In the nondegenerate case full rank is open in its projective annihilator space. In an exceptional case the rank $r = 2(k - c)$ is the largest rank permitting an isotropic c -plane: $\dim U \leq k - r + \lfloor r/2 \rfloor$ excludes every larger rank. It too is an open rank condition in the annihilator space. A zero-dimensional nonempty open subset of a projective space forces that space to be a point. This proves corank one and birationality.

When $b < a$, the same incidence-to-component argument lists all possible maximal-dimensional components. The signed images have the required dimension, and Lemma 7.6 supplies the additional component at $(5, 7, 1)$. No other rank incidence has sufficient dimension.

We verify generic corank one on the signed components without assuming it in this dimension argument. At a general signed point the rank-gap lemma excludes all strictly smaller incidences. If its annihilator space had positive dimension and generic rank greater than two, its open maximum-rank part would have positive dimension. The only remaining possible incidences have dimension $M - b$, so cannot have a positive-dimensional fibre over a dense subset of a signed component of that same dimension. If the generic rank were two, the entire open part would contradict uniqueness of the rank-two annihilator in Lemma 7.4. These alternatives exhaust a projective space containing its known rank-two point.

The signed residual planes are general by Proposition 5.3. Since $b \leq a$ is exactly

$$L \binom{c}{2} \geq 2k - 5,$$

their multiplication is onto. If $k = c + 1$, each residual plane is the entire residual binary space, whose square spans the target. Otherwise apply the earlier codimension theorem at $(k - 2, c - 1)$ when $c \geq 5$. For $c = 4$ use Lemma 7.3; if $L = 1$, saturation here forces $k = 5$ and the residual space is the entire three-dimensional binary space. Corollary 5.4 therefore proves generic scheme smoothness on the signed components.

For a nondegenerate component all W_ν are zero, so Theorem 5.1 gives tangent dimension $M - a$. For an exceptional component $L = 1$, $W = E = gV_{d-1}$, and $\dim W^2 = 2d - 1$. Its tangent dimension is

$$M - a + \binom{d+1}{2} - (2d - 1) = M - a + (d - 1)(d - 2)/2.$$

For $d = 1, 2$ this is $M - a$. For the excess exception $d = 3$, it is $M - a + 1 = M - b$. In each case it is exactly the local dimension, proving generic scheme smoothness. Proposition 5.7 now proves reducedness, and integrality in the strict case, of the entire expected-codimension scheme.

Finally apply the Thom–Porteous formula to the dual map with source $V_{2n}^* \otimes \mathcal{O}$ of rank p and target $(\bigoplus_\nu \text{Sym}^2 \mathcal{S}_\nu)^*$ of rank e , at rank bound $p - 1$ [16, 11]. Expected codimension has been proved and gives (7.1). Reducedness makes this the sum of the component cycles with multiplicity one. No excess-codimension cycle formula is asserted. $\square$

8. SIGNED INTERSECTIONS AND THE RESIDUAL WALL

This section distinguishes three objects: the incidence over the rank-two base, its reduced normalization, and its image in the full multiplication failure scheme. They agree neither globally nor in all scheme structures.

8.1. Normalization and the sign incidence complex. Let B_2° be the open set of rank-two projective Hankel forms with two distinct support points and nonzero weights. Choosing one of the two isotropic hyperplanes gives a finite etale double cover $\tilde{B}_2^\circ \rightarrow B_2^\circ$. On that cover the hyperplane bundles H_+, H_- and their common radical bundle E are labelled. Denote by $\mathfrak{I}_2^\circ$ the restriction of $\mathfrak{I}$ to this base.

Theorem 8.1 (Normalization and all same-annihilator signed intersections over the split rank-two base). *After this double-cover base change, the normalization of $(\mathfrak{I}_2^\circ)_{\text{red}}$ is the disjoint union of the smooth relative varieties*

$$Y_\varepsilon = \prod_{\nu=1}^L \text{Gr}(c, H_{\varepsilon_\nu}), \quad \varepsilon \in \{+, -\}^L.$$

Descent identifies ε with $-\varepsilon$, giving 2^{L-1} irreducible components before the cover.

For any nonempty collection $\mathcal{A}$ of sign patterns let J be the set of contacts at which those patterns do not all agree. Their scheme-theoretic intersection as reduced components over the labelled base is

$$\bigcap_{\varepsilon \in \mathcal{A}} Y_\varepsilon = \prod_{\nu \in J} \text{Gr}(c, E) \times \prod_{\nu \notin J} \text{Gr}(c, H_{\varepsilon_\nu}). \quad (8.1)$$

When nonempty it is smooth of dimension $M - b - c|J|$ and has codimension $c|J|$ in each Y_ε . These are also the local projected intersections wherever the multiplication map has corank one and its annihilator belongs to B_2° .

Proof. For a split rank-two form its quadratic restriction is a product of two linear functionals. A vector subspace is isotropic exactly when one of these functionals vanishes identically on it: the polynomial ring on that subspace is a domain. The connected cover and its deck action are also established algebraically in Proposition 9.1 with $m = 2$. This identifies the reduced incidence as the union of the displayed Y_ε . Each is smooth and normal. Their disjoint union maps

finitely to the union, is birational on every component, and hence is its normalization. The deck transformation simultaneously interchanges the two hyperplanes; its action on sign patterns has no fixed pattern. The cover is connected, as is also seen from the parameterization by a secant hyperplane and its uniquely recovered partner. Thus its orbits give the stated component count.

The condition $U \subset H_+$ and $U \subset H_-$ is exactly $U \subset E$. On any frame chart these are independent linear containment equations; adding the ideals of the reduced components therefore gives precisely the Grassmannians in (8.1), with their reduced scheme structures. Replacing $\text{Gr}(c, k-1)$ by $\text{Gr}(c, k-2)$ lowers dimension by c at each differing contact. This proves the formula, including emptiness if a required c -plane cannot fit in E . The corank-one incidence isomorphism transfers the local assertion to the projected scheme. $\square$

The restriction on the last statement is essential. Two signed images can also meet through different annihilators at a higher-corank point, or over a support collision outside B_2° . Formula (8.1) does not identify such points with a single labelled annihilator.

Proposition 8.2 (The embedded structure of a rank-two fibre). *Fix a split rank-two form and one contact with $U \subset E$. Up to smooth local factors, its full isotropy fibre has coordinate ring*

$$R/J_c, \quad R = \mathbb{C}[a_1, \dots, a_c, b_1, \dots, b_c], \quad J_c = (a_i b_i, a_i b_j + a_j b_i \ (i < j)). \quad (8.2)$$

Its radical is $(a_1, \dots, a_c) \cap (b_1, \dots, b_c)$. For $c \geq 2$ its nilradical has dimension $\binom{c}{2}$ as a $\mathbb{C}$ -vector space, is annihilated by all a_i, b_i , and squares to zero. Its associated primes are exactly

$$(a_1, \dots, a_c), \quad (b_1, \dots, b_c), \quad (a_1, \dots, a_c, b_1, \dots, b_c).$$

For several contacts the fibre is the product of the corresponding isotropy fibres, with independent contact coordinates.

Proof. Choose quotient coordinates so that the rank-two quadratic form is $2ab$. In a Grassmannian frame chart at $U \subset E$, its two quotient coordinates on the i th frame vector are a_i, b_i . The remaining coordinates are unconstrained radical directions. Restricting the bilinear form to all pairs of frame vectors gives exactly J_c . Its zero set is the union of the two coordinate c -planes, by the factorization argument above, proving the radical assertion.

Modulo J_c , the mixed quadratic monomials satisfy $a_i b_i = 0$ and $a_i b_j = -a_j b_i$. Their classes therefore form $\wedge^2 \mathbb{C}^c$, with no further degree-two relations. Every mixed cubic vanishes: for example

$$a_i a_k b_j = -a_i a_j b_k = a_j a_k b_i = -a_k a_i b_j,$$

so it is zero in characteristic zero; the argument for one a and two b 's is identical. All mixed monomials of higher degree vanish as well. This proves the assertions about the nilradical N . In the exact sequence

$$0 \longrightarrow N \longrightarrow R/J_c \longrightarrow R/\sqrt{J_c} \longrightarrow 0,$$

N has only the origin as associated prime, whereas the reduced quotient has its two minimal primes and no embedded prime. The associated-prime containment for a short exact sequence permits no others, and all three occur when $c \geq 2$. The equations for different contacts use disjoint variables, proving the product presentation. $\square$

Proposition 8.3 (Primary decomposition and Hilbert series of the fibre). *In the notation of (8.2), let $\mathfrak{a} = (a_1, \dots, a_c)$, $\mathfrak{b} = (b_1, \dots, b_c)$, and $\mathfrak{m} = \mathfrak{a} + \mathfrak{b}$. For $c \geq 2$ an irredundant primary decomposition is*

$$J_c = \mathfrak{a} \cap \mathfrak{b} \cap (J_c + \mathfrak{m}^3). \quad (8.3)$$

The last ideal is $\mathfrak{m}$ -primary, and the Hilbert series is

$$\text{Hilb}_{R/J_c}(t) = \frac{2}{(1-t)^c} - 1 + \binom{c}{2} t^2. \quad (8.4)$$

Proof. The intersection $\mathfrak{a} \cap \mathfrak{b}$ is generated by all mixed monomials $a_i b_j$. Every mixed monomial of degree at least three belongs to J_c , by the cubic calculation in the preceding proof. In degree two, intersection with $J_c + \mathfrak{m}^3$ leaves exactly the degree-two part of J_c . Homogeneity proves (8.3) degree by degree. The quotient by $J_c + \mathfrak{m}^3$ is an Artinian local ring with maximal ideal the image of $\mathfrak{m}$; every zero divisor is nilpotent. Thus this ideal is $\mathfrak{m}$ -primary. Neither minimal prime can be omitted: a_1^3 witnesses the need for $\mathfrak{b}$ and b_1^3 for $\mathfrak{a}$. The embedded component cannot be omitted because $a_1 b_2$ survives modulo J_c when $c \geq 2$. The reduced quotient has in each positive degree the direct sum of the pure a - and pure b -monomials, and in degree zero one common constant. The nilradical adds exactly $\binom{c}{2}$ degree-two classes. This proves (8.4) and verifies algebraically that the three primes listed above are all the associated primes. For $c = 1$ the third component is redundant and $J_1 = (a_1 b_1)$ is already reduced. $\square$

This embedded prime is a statement about a fibre with its annihilator fixed. It does not assert an embedded prime of $\mathcal{Z}_L$. In particular a reduced total scheme can have these nonreduced fibres.

8.2. Ordinary double crossings at the residual wall. For the next theorem suppose

$$c \geq 8, \quad k \geq 2c + 1, \quad a = b. \quad (8.5)$$

The parity in $L\binom{c}{2} = 2k - 5$ in fact forces $c \geq 10$ in this range. On the signed open set of Proposition 5.3, the residual map is square:

$$\mu'_K : \bigoplus_{\nu} \mathrm{Sym}^2 K_{\nu} \longrightarrow V_{2n-4}, \quad L\binom{c}{2} = 2k - 5.$$

Let Z_{ε} be the inverse image of its determinant divisor. It is a divisor in the signed incidence, rather than a new arbitrary choice of a discriminant.

8.3. Relative determinant and the normal derivative. In this subsection $B = \widetilde{B}_2^{\circ}$ is the connected base labelling the two isotropic hyperplanes, and $\mathcal{R}$ is their common radical bundle. We impose the wall hypotheses (8.5). These relative statements provide the bundle identifications and the nonemptiness arguments for the node normal form.

Proposition 8.4 (The relative residual divisor). *There is a subbundle $\mathcal{T} \subset V_{2n} \otimes \mathcal{O}_B$ with fibre $g^2 V_{2n-4}$ and rank $h = p - 4$. On*

$$P = \prod_{\nu=1}^L \mathrm{Gr}_B(c-1, \mathcal{R})$$

let $\mathcal{W}_{\nu}$ denote the tautological bundles. The determinant of residual multiplication

$$\tau : \bigoplus_{\nu} \mathrm{Sym}^2 \mathcal{W}_{\nu} \longrightarrow \mathcal{T}$$

is a section of the line bundle

$$\det \mathcal{T} \otimes \left(\det \bigoplus_{\nu} \mathrm{Sym}^2 \mathcal{W}_{\nu} \right)^{-1}. \quad (8.6)$$

Its zero scheme Z_P is an integral reduced Cartier divisor in P . For each sign pattern the signed open incidence $S_{\varepsilon} \rightarrow P$ is a smooth affine-space torsor. Its inverse image $Z_{\varepsilon} = S_{\varepsilon} \times_P Z_P$ is likewise an integral reduced divisor. These assertions hold on the entire distinct-support labelled base, not only after fixing the two points.

Proof. The unordered pair of support points gives a line subbundle $\mathcal{G} \subset V_2 \otimes \mathcal{O}_B$. Multiplication gives isomorphisms

$$\mathcal{R} = \mathcal{G} \otimes V_{n-2}, \quad \mathcal{T} = \mathcal{G}^{\otimes 2} \otimes V_{2n-4} \subset V_{2n} \otimes \mathcal{O}_B.$$

These formulas are homogeneous and so include a support point at infinity. The source and target of τ have the same rank h at the wall, proving (8.6). Trivializing $\mathcal{G}$ identifies P with the product of the base chart and $\text{Gr}(c-1, V_{n-2})^L$, and identifies the determinant zero scheme with the same product for the ordinary residual multiplication divisor. By Theorem 7.1 at $(k', c') = (k-2, c-1)$, this residual divisor is integral and reduced: its expected codimension is one and $L(c-1) - 3 > 1$.

For completeness, these local product descriptions imply the global assertion. Their projections are flat, with geometrically integral fibres. Every irreducible component dominates B , whereas the geometric generic fibre is irreducible, so there is just one component. The product descriptions are reduced. The section is nonzero and is locally a nonzerodivisor, hence defines a Cartier divisor. The fibres of $S_\varepsilon \rightarrow P$ are the affine torsors

$$\prod_{\nu} \text{Hom}(H_{\varepsilon_\nu}/\mathcal{R}, \mathcal{R}/\mathcal{W}_\nu).$$

Local splittings trivialize them. Pulling back Z_P is consequently locally a product with affine space, proving the last assertion by the same generic-fibre and reducedness arguments. $\square$

Proposition 8.5 (Intrinsic normal-derivative factorization). *Let $Y = \mathbb{P}(V_{2n}^*) \times X_L$, let $S = S_\varepsilon$ be the signed open incidence over B , and let $\mathcal{E} = \bigoplus_{\nu} \text{Sym}^2 \mathcal{U}_\nu$. The incidence equations are a section s of $\mathcal{E}^* \otimes \mathcal{L}^*$, where $\mathcal{L}$ is the annihilator line. Their normal derivative is a map of rank- e bundles*

$$d_N s : N_{S/Y} \longrightarrow \mathcal{E}^* \otimes \mathcal{L}^*.$$

After passing to the labelled etale chart where necessary, the normal bundle has an exact sequence

$$0 \longrightarrow \bigoplus_{\nu} \text{Hom}(\mathcal{U}_\nu, V_n/H_{\varepsilon_\nu}) \longrightarrow N_{S/Y} \longrightarrow \mathcal{L}^* \otimes \mathcal{T}^* \longrightarrow 0. \quad (8.7)$$

Put $Q = \bigoplus_{\nu} \text{Sym}^2 \mathcal{W}_\nu$. The first term maps isomorphically onto the kernel of the restriction $\mathcal{E}^ \otimes \mathcal{L}^* \rightarrow Q^* \otimes \mathcal{L}^*$. On the remaining associated quotients, $d_N s$ is $\text{id}_{\mathcal{L}^*} \otimes \tau^*$. Consequently its determinant and the residual determinant are identified by an isomorphism of their determinant line bundles. In any local frames this reads*

$$\det(d_N s) = u_0 \det(\mu'_K), \quad u_0 \in \mathcal{O}_S^*. \quad (8.8)$$

The unit ambiguity is exactly the change of frames; the equality of zero divisors and their multiplicities is intrinsic.

Proof. First restrict Y to the rank-two base. Relative to this base the normal bundle of $\text{Gr}(c, H_{\varepsilon_\nu})$ in $\text{Gr}(c, V_n)$ is $\text{Hom}(\mathcal{U}_\nu, V_n/H_{\varepsilon_\nu})$. The normal sequence for $S \subset Y|_B \subset Y$ gives the first term of (8.7) and the normal bundle of B in $\mathbb{P}(V_{2n}^*)$ as quotient.

The affine tangent space of the rank-two cone at ℓ is spanned by the two evaluations and their first derivatives, with the latter weighted by the nonzero coefficients. A polynomial annihilating this four-dimensional space vanishes to order two at both support points. Its space is precisely $g^2 V_{2n-4}$. Therefore

$$N_{B/\mathbb{P}(V_{2n}^*)} = \text{Hom}(\mathcal{L}, \mathcal{T}^*) = \mathcal{L}^* \otimes \mathcal{T}^*.$$

This identification is independent of coordinates on $\mathbb{P}^1$.

For one contact write $U = W \oplus \mathbb{C}a$ locally. The restriction of a rank-two form has one nonzero quotient functional in the V_n/H_ε direction. Varying each vector of W in this direction prescribes the wa entries, and varying a prescribes the a^2 entry, with its nonzero factor two. The ww' entries remain zero. Thus the plane-normal map is an isomorphism onto the rank- c space of equation forms vanishing on $\text{Sym}^2 W$. Doing this for each contact proves the first block assertion.

Restrict the remaining equations to Q . Variations of the planes contribute zero there. An annihilator-normal variation $\dot{\ell}$ contributes $q \mapsto \dot{\ell}(\mu(q))$, exactly τ^* under the preceding normal-bundle identification. Splittings of the two exact sequences therefore give an upper block triangular matrix with an invertible plane block and the block $\text{id}_{\mathcal{L}^*} \otimes \tau^*$. Taking determinants proves the assertion on determinant lines. A new splitting alters only an off-diagonal block, and new frames change determinants by units. Finally trivializing $\mathcal{G}$ cancels the common g^2 factor and identifies τ with μ'_K , giving (8.8). $\square$

Lemma 8.6 (A simultaneous nonempty corank-one wall open). *The integral divisor Z_ε has a nonempty open subset where the residual map has corank one and nonsingular Hankel annihilator η , $K_1 \rightarrow \mathbb{C}^2$ has full evaluation rank at the support points, and the original multiplication has corank one. This open subset is dense in Z_ε .*

Proof. The first two residual conditions hold on a nonempty open subset of the integral residual divisor by Theorem 7.1. We verify that the evaluation-rank condition meets this subset rather than presuming generality of a residual plane.

Write $c' = c - 1$, $k' = k - 2$; here $k' \geq 2c' + 1$ and $c' \geq 9$. For a nondegenerate form on a k' -space, the locus of isotropic c' -planes contained in a fixed hyperplane has codimension at least c' in its orthogonal Grassmannian. For a nonisotropic normal vector this follows by replacing k' by $k' - 1$ in the dimension formula. For an isotropic normal vector v , planes not containing v are lifts of isotropic c' -planes in $v^\perp/\mathbb{C}v$, with c' affine lift parameters, giving the same bound. Planes containing v are parametrized by $\text{OGr}(c' - 1, k' - 2)$ and have codimension $k' - c' - 1 \geq c'$. This proves the bound in all cases.

Failure of K_1 to map onto the two evaluation coordinates means containment in one member of their hyperplane pencil. The union has codimension at least $c' - 1 > 0$. Over the nondegenerate residual annihilator incidence, its complement is thus nonempty and open. That incidence is irreducible and dominates the residual determinant divisor; its corank-one open is nonempty. Intersecting these two nonempty opens in an irreducible space proves the simultaneous residual assertion, for each fixed pair of support points.

Choose such a tuple and a nonzero $k_0 \in K_1 \cap gV_{n-4}$; this intersection has dimension at least $c - 3 > 0$. Write $k_0 = gk'_0$ and $U_1 = gK_1 + \mathbb{C}a_1$. The already spanned space $g^2 \text{im } \mu'_K$ has one missing direction in g^2V_{2n-4} . The product $(gk_0)a_1 = g^2k'_0a_1$ supplies it if $\eta(k'_0a_1) \neq 0$. Changing the lift by $a_1 \mapsto a_1 + gw$ changes this value by $H_\eta(k_0, w)$. This is a nonzero functional because η is nondegenerate. It vanishes on K_1 , since K_1 is isotropic, so it is a well-defined nonconstant affine function on the actual lift torsor, not merely on a space of redundant frames.

After this choice, products from $(gK_1)a_1$ supply both independent first-derivative coordinates modulo g^2V_{2n-4} , using full evaluation rank and the nonzero values of a_1 at both support points. The square a_1^2 supplies a further, nonzero value coordinate. The product span therefore has dimension at least $(p - 4) + 3 = p - 1$. It is annihilated by ℓ , so its dimension is exactly $p - 1$. All the conditions used are open, and we have exhibited choices meeting them. Integrality of Z_ε from Proposition 8.4 proves density. $\square$

8.4. A simple normal determinant gives an exact double crossing. The passage from a normal derivative to a local equation is an independent analytic statement. We record the precise hypotheses; first-order corank alone would not suffice.

Lemma 8.7 (Simple normal determinant). *Let Y be a smooth complex variety of dimension $d + e$, let $S \subset Y$ be smooth of codimension e , and let s be a section of a rank- e vector bundle $\mathcal{F}$ which vanishes identically on S . At $z \in S$, assume that the normal derivative*

$$d_N s : N_{S/Y} \longrightarrow \mathcal{F}|_S$$

has rank $e - 1$. Suppose its determinant has a simple zero on S at z , meaning that a local representative has nonzero differential in T_z^*S . Then $d \geq 1$ and the zero scheme $Z = V(s)$ has analytic local equation $uv = 0$ in a smooth $(d + 1)$ -dimensional space. More precisely,

$$\widehat{\mathcal{O}}_{Z,z} \simeq \mathbb{C}[[t_1, \dots, t_{d-1}, u, v]]/(uv), \quad (8.9)$$

with S corresponding to the branch $u = 0$. No reducedness or component-count assumption on Z is needed.

Proof. Choose coordinates $(t_1, \dots, t_d, z_1, \dots, z_e)$ in which $S = \{z_1 = \dots = z_e = 0\}$, and trivialize $\mathcal{F}$. After invertible changes of the normal coordinates and of the section's components, the first $e - 1$ components have an invertible Jacobian in $z_1, \dots, z_{e-1}$ at the point. The implicit function theorem solves these components as $z_i = h_i(t, u)$, where $u = z_e$. Since the components vanish on S , uniqueness implies $h_i(t, 0) = 0$. Their common zero scheme is a smooth $(d + 1)$ -dimensional neighbourhood containing $S = \{u = 0\}$. For $e = 1$ this step has no equations to eliminate.

Restrict the last component to this neighbourhood and call it $f(t, u)$. It vanishes on $u = 0$, so in the convergent power-series ring it factors exactly as $f = uG(t, u)$. Along S , the chain rule for the eliminated equations gives

$$G(t, 0) = \partial_u f(t, 0) = \frac{\det(d_N s)(t)}{\det(\partial(s_1, \dots, s_{e-1})/\partial(z_1, \dots, z_{e-1}))(t, 0)}.$$

For $e = 1$ the denominator is 1. This is the elementary Schur complement identity for the normal Jacobian. The denominator is a unit. Since the numerator vanishes simply on S , the differential of $G(t, 0)$ is nonzero. Consequently du and dG are linearly independent at z . Complete $u, v = G(t, u)$ to analytic coordinates $(u, v, t'_1, \dots, t'_{d-1})$ on the neighbourhood. In these coordinates the final equation is exactly $uv = 0$. Completion proves (8.9). Changing any trivialization multiplies the determinants by units, so the simple-zero condition and the conclusion are intrinsic. $\square$

Corollary 8.8 (Normalization and conductor of the local crossing). *In Lemma 8.7, the normalization of the completed local ring is*

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u]] \oplus \mathbb{C}[[t_1, \dots, t_{d-1}, v]],$$

and its conductor in that normalization is $(u) \oplus (v)$. The conductor in the original ring is (u, v) , and the scheme-theoretic intersection of the two branches is reduced and smooth of dimension $d - 1$.

Proof. The ring in (8.9) embeds into the displayed direct sum as the pairs whose restrictions at $u = v = 0$ agree. The sum is finite over it: adjoining the idempotent $(1, 0)$ suffices. Both summands are regular and integrally closed, so this is the normalization. A pair multiplies every pair in the sum back into the original ring precisely when both restrictions at zero vanish. This is $(u) \oplus (v)$, whose inverse image is (u, v) . The quotient by this ideal is $\mathbb{C}[[t_1, \dots, t_{d-1}]]$. $\square$

In the binary wall calculation below, Y is the annihilator ambient space, S the chosen signed incidence, and s its isotropy section. The relative determinant proposition identifies $\det(d_N s)$, up to a unit, with the residual multiplication determinant. The relative reduced divisor and the nonemptiness proposition supply its simple-zero locus while the original multiplication still has corank one. Lemma 8.7 then proves the exact local equation used in Theorem 8.9. Thus neither reducedness inferred from a tangent count nor an unproved removal of higher-order terms enters the crossing argument.

8.5. The local normal form.

Theorem 8.9 (The residual double-crossing divisor). *Under (8.5), Z_ε is irreducible and reduced. It has a nonempty dense open subset on which the following assertions hold. The original multiplication has corank one, and the residual multiplication has corank one with a nonsingular residual Hankel annihilator. Put $d = M - a$. The analytic germ of $\mathcal{D}_L$ at its projected point has equation $uv = 0$ in smooth local coordinates $(t_1, \dots, t_{d-1}, u, v)$, and its completed local ring is*

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u, v]]/(uv). \quad (8.10)$$

The two branches are Σ_ε and Φ_L ; no other signed branch passes through this open subset. Their intersection has codimension one in either branch, and codimension $a + 1$ in X_L . The tangent cone is the union of two distinct hyperplanes in the $(d + 1)$ -dimensional tangent space. The germ is seminormal, its normalization separates its two smooth branches, and its conductor in (8.10) is (u, v) .

Proof. Proposition 8.4 gives the relative integral divisor, Lemma 8.6 supplies a simultaneous dense open, and Proposition 8.5 gives the intrinsic determinant identity. We spell out how they lead to the local equation. At residual dimensions $k' = k - 2$, $c' = c - 1$ the expected codimension is one and the signed codimension is $L(c - 1) - 3 > 1$. Here $c' \geq 9$ and $k' \geq 2c' + 1$. The already proved Theorem 7.1 says that the residual determinant is an integral reduced divisor whose general annihilator η is nonsingular. Pullback by the smooth affine-space bundle in Proposition 5.3 preserves this generic irreducibility and reducedness; bundle trivializations as the support points vary give the same assertion for Z_ε .

We check that the original map has corank one on a nonempty open part of this divisor. A general residual tuple has K_1 mapping onto the two evaluation coordinates at x, y . Indeed failure of this condition places K_1 in one hyperplane of a pencil, and the isotropic Grassmannian incidence count excludes that closed subset. Choose a nonzero

$$k_0 \in K_1 \cap gV_{n-4}, \quad k_0 = gk'_0.$$

Such vectors exist because this intersection has dimension at least $c - 3 > 0$. Write $U_1 = gK_1 + \mathbb{C}a_1$. The product $(gk_0)a_1 = g^2k'_0a_1$ fills the missing direction in g^2V_{2n-4} exactly when $\eta(k'_0a_1) \neq 0$. Varying the lift by $a_1 \mapsto a_1 + gw$, $w \in V_{n-2}$, changes this value by $H_\eta(k_0, w)$, a nonzero functional because H_η is nonsingular. Thus general lifts fill this direction. The products $(gK_1)a_1$ give both first-derivative coordinates modulo g^2V_{2n-4} , since K_1 has full evaluation rank and a_1 has nonzero values at x, y . The product a_1^2 gives one additional value coordinate. Consequently the original product span has dimension $(p - 4) + 3 = p - 1$. All these conditions are nonempty open conditions.

Work on this corank-one open set in the incidence. The signed incidence S is smooth of dimension d in the smooth ambient $Y = \mathbb{P}(V_{2n}^*) \times X_L$. Its codimension is

$$(M + p - 1) - d = p - 1 + b = e,$$

and the full incidence is the zero scheme of e restriction equations. The normal derivative of these equations along S splits into the Lc mixed and square equations involving the nonradical vector a_ν , and the $L\binom{c}{2} = p - 4$ equations on $gK_\nu \times gK_\nu$. The first block is invertible in transverse plane coordinates, by the dual-frame proof of Theorem 5.1. The normal space to the rank-two Hankel base is identified with $(g^2V_{2n-4})^*$: its tangent space is the annihilator of g^2V_{2n-4} modulo $\mathbb{C}\ell$, as is seen by differentiating the two evaluations and their weights. In these normal coordinates the remaining derivative block is precisely $(\mu'_K)^*$. Hence, up to a nowhere-zero factor,

$$\det(\text{normal derivative along } S) = \det \mu'_K. \quad (8.11)$$

At a general point of Z_ϵ this derivative has corank one and its determinant vanishes simply along S .

Eliminate $e-1$ equations and their normal coordinates by the analytic implicit function theorem. The remaining ambient germ is smooth of dimension $d+1$, and S is the smooth hypersurface $u=0$ in it. The last equation vanishes on S , so it is uG . Its normal derivative along S is $G|_S$; by (8.11) this has a simple zero along the residual divisor. Thus dG has a nonzero component tangent to S , and $u, v = G$ are independent analytic coordinates. This proves (8.10), not only an equality of tangent dimensions. The corank-one projection identifies this germ with the one in X_L .

Only one sign choice is possible locally because no U_ν is contained in the radical and the annihilator is unique. The other branch must therefore be Φ_L , by Theorem 7.1. All dimension and tangent-cone assertions follow from the displayed local ring. Its normalization is

$$\mathbb{C}[[t_1, \dots, t_{d-1}, u]] \oplus \mathbb{C}[[t_1, \dots, t_{d-1}, v]],$$

Replacing formal by convergent power series gives the analytic normalization. In either case the original ring consists of pairs with the same restriction at $u=v=0$. This identifies the conductor as (u, v) and proves seminormality: an element of the total fraction ring whose square and cube belong to the original ring lies in each normal branch ring, and its two restrictions agree because their squares and cubes agree. $\square$

The theorem describes a specified irreducible divisor and its dense open double-crossing locus. It does not claim that every point of $\Phi_L \cap \Sigma_\epsilon$ has this type, or that higher-corank and colliding-support intersections are normal crossings.

Corollary 8.10 (A logarithmic conditioning law at a crossing). *At a real point of the double-crossing locus, with its two real branches, there are real analytic coordinates $(z_1, \dots, z_{a-1}, u, v, t)$ in X_L in which*

$$\gamma_{\text{mult}} \asymp \left(\sum_{j=1}^{a-1} z_j^2 + u^2 v^2 \right)^{1/2}. \quad (8.12)$$

On a sufficiently small product box, a probability law whose density is bounded above and below by positive constants satisfies

$$\Pr\{\gamma_{\text{mult}} \leq \epsilon\} \asymp \epsilon^a \log(1/\epsilon), \quad \epsilon \downarrow 0. \quad (8.13)$$

The same law holds in any real loading chart submersing onto this neighbourhood. Constants are local and depend on the chosen chart and density bounds.

Proof. At corank one bounded invertible row and column operations compare γ_{mult} with the norm of the Schur-complement row. The analytic argument in Theorem 8.9, applied in the ambient Grassmannian chart, makes its ideal $(z_1, \dots, z_{a-1}, uv)$. Each finite set of generators is a bounded analytic linear combination of the other on a smaller compact chart, so their norms are comparable. This proves (8.12). For the upper bound integrate first over the $a-1$ normal coordinates, whose volume is $O(\epsilon^{a-1})$. In a fixed square the area of $|uv| \leq C\epsilon$ is $O(\epsilon \log(1/\epsilon))$, by integrating $\min(r, C\epsilon/|u|)$ in u . For the lower bound restrict the normal coordinates to a ball of radius $\epsilon/2$ and take $|uv| \leq \epsilon/2$ in a smaller fixed square. The same elementary integral is bounded below by a positive constant times $\epsilon \log(1/\epsilon)$. Bounded density and Jacobian factors give the assertion. A submersion adds unconstrained smooth coordinates and does not change the exponent or logarithm. $\square$

This is different from the smooth-stratum law ϵ^a . It is a conditioning statement for a specified real design law, not a claim that the exact count-score compression experiment has Fisher

information equal to an unwhitened squared singular value. The covariance of that score compression also depends on the design.

9. HIGHER PRODUCTS AND DIHEDRAL PHASE SECTORS

The general residual theorem of Section 2 already applies to every symmetric power. Here we calculate explicitly one distinguished family of annihilators within that general setting. The two-hyperplane description has a higher-product counterpart, but its monodromy is no longer exhausted by a simultaneous sign change. For $m \geq 2$ consider

$$\mu_m : \bigoplus_{\nu=1}^L \text{Sym}^m U_\nu \longrightarrow V_{mn}, \quad p_m = mn + 1, \quad e_m = L \binom{c+m-1}{m}.$$

Here “two-point annihilator” refers to the family

$$[\ell] = [\text{ev}_x^{(mn)} - t^m \text{ev}_y^{(mn)}], \quad x \neq y, \quad t \neq 0,$$

not to a purported classification of every higher catalecticant rank. Let μ_m be the group of m th roots of unity.

Proposition 9.1 (Algebraic splitting cover and phase descent). *On the distinct-support two-point base for m th multiplication, the irreducible components of the reduced isotropic incidence are exactly the orbits of $\mu_m \rtimes \{1, \sigma\}$ on μ_m^L , where simultaneous multiplication acts on all labels and σ inverts all labels. No further splitting occurs for an inversion-fixed relative pattern.*

Proof. It suffices first to work over the dense chart where both support points are finite. Its labelled splitting cover is the integral variety

$$T = \{(x, y, t) \in \mathbb{A}^2 \times \mathbb{G}_m : x \neq y\}.$$

The transformations

$$\tau_\zeta(x, y, t) = (x, y, \zeta t), \quad \sigma(x, y, t) = (y, x, t^{-1})$$

satisfy $\sigma \tau_\zeta \sigma = \tau_{\zeta^{-1}}$. They form a group of order $2m$, acting freely: a nontrivial rotation cannot fix $t \neq 0$, and a reflection interchanges the distinct points. The quotient is a finite étale Galois cover of the two-point base. To identify the quotient with that base, note that the annihilator $[\text{ev}_x - t^m \text{ev}_y]$ recovers its unordered support and weight ratio. For example the radical of its middle Hankel pairing recovers the degree-two support polynomial, and a two-column Vandermonde system recovers the weights. This also gives algebraic inverses on the usual nonvanishing-minor charts. For the degrees under consideration the pairing has both side dimensions at least two. Thus the displayed transformations are the entire deck group, not just selected monodromies.

Over T the hyperplanes H_ζ are labelled. Their reduced incidences are the integral smooth bundles $\prod_\nu \text{Gr}(c, H_{\zeta_\nu})$; they have geometrically integral fibres. A deck transformation permutes these bundles by the displayed action on the labels (using the inverse rotation gives the same orbit action). The image of any integral bundle under the finite quotient map is irreducible and is exactly the image of its orbit union. Different orbits have distinct generic geometric components. Hence these images are precisely the irreducible components downstairs. An element stabilizing one pattern acts on its integral bundle; its quotient remains integral, since an invariant subring of a domain is a domain. In particular a fixed relative inversion pattern does not split after descent.

Changing affine coordinates on $\mathbb{P}^1$ only changes trivializations of the evaluation lines. Such charts cover the base, and their nonempty overlaps give the same hyperplane-label cover. Alternatively, each component of the full incidence dominates the base, because its split local models are Grassmannian bundles. Thus it is the closure of a component on this dense chart. No additional component is hidden at infinity. This completes the algebraic descent proof. $\square$

Theorem 9.2 (Phase-sector classification for every symmetric power). *Suppose $n \geq 4$, $4 \leq c < n + 1$, $L \geq 1$, and $m \geq 2$. The reduced incidence of two-point annihilators of μ_m has*

$$N_{m,L} = \frac{m^{L-1} + \gcd(m, 2)^{L-1}}{2} \quad (9.1)$$

irreducible components on the distinct-support open base. The closure of the image of each component in X_L is irreducible and rational of codimension $Lc - 3$. Its incidence projects birationally, and its general point has a unique two-point annihilator. These sectors are indexed by relative phase patterns

$$(\xi_2, \dots, \xi_L) \in \boldsymbol{\mu}_m^{L-1}, \quad (\xi_2, \dots, \xi_L) \sim (\xi_2^{-1}, \dots, \xi_L^{-1}).$$

For $m = 2$ this recovers 2^{L-1} . In general the count is not m^{L-1} .

Proof. By polarization in characteristic zero, ℓ annihilates $\text{Sym}^m U$ if and only if the degree- m polynomial $\ell(f^m)$ vanishes identically for $f \in U$. Its factorization is

$$f(x)^m - t^m f(y)^m = \prod_{\zeta \in \boldsymbol{\mu}_m} (f(x) - \zeta t f(y)).$$

Since the polynomial ring on U is a domain, this holds precisely when U lies in one of the m hyperplanes $H_\zeta = \ker(\text{ev}_x - \zeta t \text{ev}_y)$. The splitting cover of the two-point base labels all these hyperplanes. Changing the chosen m th root t acts by simultaneous cyclic translation of phase labels. Interchanging the two support points acts by inversion after that translation. Proposition 9.1 constructs the finite etale cover, identifies its full deck group, and proves the descent assertion, including sectors with stabilizers. Thus the component orbits are those of the dihedral action on $\boldsymbol{\mu}_m^L$.

Quotient first by simultaneous translation, normalizing the first phase to one. The remaining involution inverts every relative phase. Its fixed patterns have each entry of order dividing two; there are $\gcd(m, 2)^{L-1}$ of them. The two-element orbit formula gives (9.1). Over the splitting cover every phase sector is a smooth irreducible product of relative Grassmannians; monodromy orbits are therefore exactly the irreducible components after descent.

Each incidence has dimension $3 + Lc(k - 1 - c) = M - (Lc - 3)$. The first plane determines its general secant hyperplane uniquely by the dimension argument in Lemma 7.4. That hyperplane determines its unordered support points, weight ratio and resulting two-point annihilator, and the other planes determine their phase hyperplanes. This proves generic injectivity and distinctness of the image sectors, hence birationality in characteristic zero.

For rationality, a relative pattern not fixed by inversion chooses an ordering of the support points. Its parameterization uses an open subset of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{G}_m$, with rational Grassmannian fibres trivialized on a dense open set, and is birational. For an inversion-fixed pattern, its first hyperplane instead parameterizes the unordered secant threefold, which is rational; all other hyperplanes are rationally determined from it. The same Grassmannian argument applies. Taking image closures does not change these birational assertions. $\square$

The theorem classifies the two-point incidence and its image sectors. It does not assert that every such sector is an irreducible component of the entire higher-product failure scheme, or that the full higher-product failure codimension has been determined.

Theorem 9.3 (Higher-product residual conormal). *On a phase sector suppose no U_ν is contained in $E = gV_{n-2}$, and write $U_\nu = gK_\nu + \mathbb{C}a_\nu$. Put*

$$R_m = \sum_\nu (gK_\nu)^2 U_\nu^{m-2} = g^2 \sum_\nu K_\nu^2 U_\nu^{m-2} \subset V_{mn}, \quad \delta_m = (p_m - 4) - \dim R_m.$$

If $e_m \geq p_m$ and μ_m has corank one, then the tangent dimension of its maximal-minor scheme is

$$M - (Lc - 3) + \delta_m. \quad (9.2)$$

In particular $\delta_m = 0$ implies scheme smoothness of codimension $Lc - 3$ at this point.

Proof. For a fixed contact split $U = W + \mathbb{C}a$, with $W = gK$. The derivative with respect to the plane vanishes on every monomial having at least two W factors: after varying one factor another still vanishes at both support points. On the quotient spanned by a^m and wa^{m-1} , $w \in W$, it is surjective. Indeed the nonzero quotient linear functional

$$v \mapsto v(x)a(x)^{m-1} - t^m v(y)a(y)^{m-1}$$

allows the coordinates of the variation of each basis vector to solve these c equations independently; the coefficient for a^m is the nonzero integer m . Thus the plane derivative has rank c . The remaining compatibility equations for ℓ are precisely annihilation of R_m . Independent solutions for the contacts give

$$\dim T\mathcal{J}_m = M + (p_m - 1) - Lc - \dim R_m,$$

which is (9.2). The corank-one Schur argument identifies this tangent space with that of the maximal-minor scheme. Multiplication by g^2 embeds V_{mn-4} , of dimension $p_m - 4$, so $\delta_m \geq 0$. If it is zero, the tangent dimension equals the dimension $M - (Lc - 3)$ of the local phase incidence, which is embedded by the corank-one projection. Equality of tangent and local dimensions proves smoothness. $\square$

For $m = 2$ the residual span is independent of the lifts a_ν and reduces to the multiplication of the K_ν . For $m > 2$ it retains U_ν^{m-2} . This difference is why a higher-product phase classification alone does not imply the quadratic saturation formula with q replaced by $\binom{c+m-1}{m}$.

10. A GLOBAL HIGHER-PRODUCT FAMILY WITH EMBEDDED STRUCTURE

We now apply the residual mechanism to a natural family in every symmetric degree, determining the whole reduced failure locus rather than only an obstruction subfamily. The scheme structure contains a further phenomenon: its support is independent of the degree, but its nilpotent structure is not.

Fix $n \geq 4$ and put $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$. A hyperplane $U \subset V_n$ is the kernel of a unique point $[h] \in \mathbb{P}(V_n^*)$. For $m \geq 2$ let

$$\mu_m : \text{Sym}^m \mathcal{U} \longrightarrow V_{mn} \otimes \mathcal{O}_{\mathbb{P}(V_n^*)}, \quad D_m = V(I_{mn+1}(\mu_m)).$$

Its source rank is $e_m = \binom{n+m-1}{m} \geq mn + 1$. Write $C_n \subset \mathbb{P}(V_n^*)$ for the evaluation rational normal curve, and S_n for its secant threefold, including tangent secants. Let $\mathcal{J}_m$ be the ideal sheaf of D_m , and $\mathcal{I}_{S_n}, \mathcal{I}_{C_n}$ the reduced ideal sheaves. Associated points below are those of coherent sheaves on projective space; no assertion about an irrelevant-ideal component of an affine homogeneous presentation is implicit.

Theorem 10.1 (Higher-product hyperplanes: support, ranks and associated points). *For every $n \geq 4$ and $m \geq 2$,*

$$(D_m)_{\text{red}} = S_n, \quad \text{corank } \mu_{m, \ker h} = \begin{cases} 0, & [h] \notin S_n, \\ 1, & [h] \in S_n \setminus C_n, \\ m, & [h] \in C_n. \end{cases} \quad (10.1)$$

The scheme D_m is smooth of dimension three at every point of $S_n \setminus C_n$, including the non-evaluation tangent secants. In particular

$$\mathcal{J}_m : \mathcal{I}_{C_n}^\infty = \mathcal{I}_{S_n}. \quad (10.2)$$

For every $m \geq 3$, the complete set of associated points is

$$\text{Ass}_{\mathcal{O}_{\mathbb{P}(V_n^*)}}(\mathcal{O}_{D_m}) = \{\eta_{S_n}, \eta_{C_n}\}. \quad (10.3)$$

Thus there is exactly one reduced irreducible component, and exactly one embedded associated subvariety, the evaluation curve. Its nilradical $\mathcal{N}_m = \mathcal{I}_{S_n}/\mathcal{J}_m$ satisfies, at every closed point $x \in C_n$,

$$(\mathcal{N}_m^j)_x \neq 0 \quad \text{whenever } 1 \leq j \text{ and } 2j < m. \quad (10.4)$$

Consequently the nilpotency index at x is at least $\lceil m/2 \rceil$. The theorem does not assert that this lower bound is the exact index.

Lemma 10.2 (Propagation from quadratic surjectivity). *Let $U \subset V_n$ be base-point-free. Then $UV_d = V_{n+d}$ for every $d \geq n-1$. In particular, $U^2 = V_{2n}$ implies $U^m = V_{mn}$ for every $m \geq 2$.*

Proof. There are coprime $f, g \in U$: fix a nonzero f and choose g outside the finitely many proper evaluation hyperplanes at its zeros. The kernel of

$$V_d \oplus V_d \longrightarrow V_{n+d}, \quad (A, B) \longmapsto fA + gB$$

consists exactly of $(gh, -fh)$ with $h \in V_{d-n}$. For $d = n-1$ its kernel is zero and both dimensions are $2n$. For $d \geq n$ the rank is $2(d+1) - (d-n+1) = d+n+1$. Thus the map is onto in both cases. If $U^2 = V_{2n}$ then U is automatically base-point-free; apply the first assertion successively with $d = 2n, 3n, \dots$ $\square$

Lemma 10.3 (All higher-product ranks on secants). *For a non-evaluation point $h \in S_n$, the image U^m has codimension one in V_{mn} for every $m \geq 2$. For an evaluation point it has codimension m .*

Proof. First take a distinct-support secant and choose the two points as 0 and ∞ . In the binary monomial basis $z_j = X^j Y^{n-j}$, its hyperplane has a basis

$$z_1, \dots, z_{n-1}, \quad a = z_0 + \lambda z_n, \quad \lambda \neq 0.$$

Products of two of the interior monomials give all exponents $2, \dots, 2n-2$. Multiplication of a by z_1 and z_{n-1} then supplies exponents 1 and $2n-1$, since the other terms are already in that span. Finally a^2 gives the one allowed combination of the endpoint monomials. Thus U^2 contains every interior monomial and one endpoint combination, and has dimension $2n$.

Inductively suppose U^m contains all monomials of exponents $1, \dots, mn-1$, as well as a^m . Multiplying its interior monomials by $z_1, \dots, z_{n-1}$ gives exponents $2, \dots, (m+1)n-2$. Multiplying its two extreme interior monomials by a supplies the two missing interior exponents. The element a^{m+1} supplies an endpoint combination. All products satisfy the single endpoint relation inherited from $f(0) = \lambda^{-1}f(\infty)$ for $f \in U$. Hence there can be no second endpoint direction, and the image has dimension $(m+1)n$.

At a non-evaluation tangent secant, a change of homogeneous coordinates puts the hyperplane in the form with the first derivative at 0 equal to zero. To justify the normalization, in a local coordinate a change fixing 0 changes a first-jet functional by a nonzero scalar and an arbitrary multiple of evaluation; the latter multiple can therefore be removed. The hyperplane is then

$$U = \text{span}\{z_0, z_2, z_3, \dots, z_n\}.$$

Sums of m exponents in $\{0, 2, 3, \dots, n\}$ give exactly $\{0, 2, 3, \dots, mn\}$. This follows by induction, since the intervals obtained by adding $[2, n]$ overlap or are adjacent for $n \geq 3$. Thus precisely exponent 1 is missing.

Finally, for h an evaluation, $U = lV_{n-1}$ for its vanishing linear form l . Since products of the complete binary series span, $U^m = l^m V_{m(n-1)}$, of dimension $m(n-1) + 1$. Its corank is m . $\square$

Lemma 10.4 (Scheme smoothness off the evaluation curve). *At every point of $S_n \setminus C_n$, the plane derivative of the annihilator incidence has rank n , and its polar residual image has dimension $mn - 3$. Consequently the tangent space of D_m has dimension three at each such point.*

Proof. For distinct support, let g be its vanishing quadratic and write $U = W \oplus \mathbb{C}a$, with $W = gV_{n-2}$. The plane derivative vanishes on monomials with at least two factors in W , while it is an isomorphism onto the dual of the n -dimensional quotient spanned by a^m and wa^{m-1} . Indeed the functional on V_n/U obtained by varying one factor and evaluating against the annihilator is nonzero; the images of the n basis vectors of U can be varied independently. This is the computation of Theorem 9.3, now with $c = n$. The residual image is

$$W^2U^{m-2} = g^2V_{2n-4}U^{m-2} = g^2V_{mn-4}.$$

The last equality follows from Lemma 10.2, because U is base-point-free and $2n - 4 \geq n - 1$. Its dimension is $mn - 3$.

For a tangent secant in the normal form of Lemma 10.3, let $W = \text{span}\{z_2, \dots, z_n\}$ and $a = z_0$. The annihilator in V_{mn}^* is the coefficient of exponent 1. Since V_n/U is spanned by z_1 , the derivative independently prescribes the coefficients dual to a^m and wa^{m-1} , and has rank n . Once again its kernel on the dual source consists of monomials with at least two W factors. Their products span precisely the monomials of exponents $4, \dots, mn$: W^2 gives the full interval $[4, 2n]$, and successive factors from U extend it without gaps. This image also has dimension $mn - 3$.

The incidence tangent sequence now gives, in both cases,

$$\dim TD_m = \dim T\mathcal{I} = n + (mn + 1) - 1 - n - (mn - 3) = 3,$$

because the original multiplication has corank one by Lemma 10.3. $\square$

Proof of Theorem 10.1. We first identify the quadratic support. A hyperplane U can be totally isotropic for a nonzero Hankel form H_ℓ only if that form has rank at most two: the maximal isotropic dimension of a rank- r form on an $(n+1)$ -space is $n+1 - \lceil r/2 \rceil$. A rank-one Hankel form gives an evaluation hyperplane. A rank-two Hankel form has radical $H^0(\mathcal{O}(n)(-Z))$ for a length-two divisor Z on $\mathbb{P}^1$. For clarity, this last assertion includes a double point: the proper family of secant lines indexed by length-two divisors covers the rank-at-most-two Hankel variety of Lemma 4.1; on a length-two quotient the rank-two pairing is nondegenerate, so its kernel is exactly the indicated space. Since an isotropic hyperplane contains this radical, its annihilating functional lies on the corresponding secant or tangent line to C_n . Hence $|D_2| \subset S_n$. Conversely a distinct-support secant imposes a nonzero quadratic endpoint relation. Closedness supplies the whole S_n , proving $|D_2| = S_n$.

Outside S_n , quadratic multiplication is onto; the propagation lemma proves surjectivity in every higher degree. On S_n the rank lemma gives the two remaining cases of (10.1). Thus $|D_m| = S_n$ globally. Since S_n is an irreducible threefold, Lemma 10.4 equates local Krull and tangent dimensions at every point off C_n . The local rings there are regular. In particular $\mathcal{J}_m = \mathcal{I}_{S_n}$ away from C_n .

The coherent quotient $\mathcal{N}_m = \mathcal{I}_{S_n}/\mathcal{J}_m$ is therefore supported on C_n , and some power of $\mathcal{I}_{C_n}$ kills it. This gives the inclusion of $\mathcal{I}_{S_n}$ in the left side of (10.2). The reverse inclusion holds because $\mathcal{I}_{S_n}$ is prime and S_n is not contained in C_n . This proves the saturation identity.

At an evaluation point x , the original corank is m . Schur elimination consequently expresses $\mathcal{J}_{m,x}$ as the ideal of m -minors of a matrix whose entries vanish at x . In the regular local ring $\mathcal{O}_{\mathbb{P}(V_n^*),x}$, with maximal ideal $\mathfrak{m}_x$, one has

$$\mathcal{J}_{m,x} \subset \mathfrak{m}_x^m. \tag{10.5}$$

Choose coordinates $h_0, \dots, h_n$ dual to the binary monomials so that $x = [1 : 0 : \dots : 0]$, and work on $h_0 = 1$. The function

$$F = \det \begin{pmatrix} 1 & h_1 & h_2 \\ h_1 & h_2 & h_3 \\ h_2 & h_3 & h_4 \end{pmatrix}$$

vanishes on S_n , since every two-evaluation moment sequence has Hankel rank at most two. Its initial form at x is $h_2h_4 - h_3^2$, a nonzero quadratic. Therefore F^j has order $2j$, and (10.5) implies that its class modulo $\mathcal{J}_m$ is nonzero whenever $2j < m$. Transitivity of $\mathrm{SL}_2(\mathbb{C})$ on C_n proves (10.4) at every evaluation point. For $m \geq 3$ this shows in particular that $\mathcal{N}_m$ is nonzero with support all of C_n .

It remains to exclude any additional associated points. The multiplication and its maximal-minor ideal are $\mathrm{SL}_2(\mathbb{C})$ -equivariant. The associated points of a coherent sheaf form a finite set, stable under automorphisms. Off C_n the scheme is the smooth irreducible open part of S_n , so its only associated point there is η_{S_n} . An associated point whose closure is contained in C_n is either η_{C_n} or a closed point of C_n . The latter is impossible: transitivity would force every closed point of the curve into this finite set. Thus the only possible embedded associated point is η_{C_n} . It actually occurs because the nonzero submodule $\mathcal{N}_m$ has support C_n , whose generic point is a minimal point of its support and hence an associated point. Associated points of a submodule are associated points of the ambient module. This proves (10.3), and completes the proof. $\square$

The expected determinantal codimension here is $\binom{n+m-1}{m} - mn$, while the actual codimension is $n - 3$. For $m \geq 3$ this is an excess family with a complete associated-point classification. The nilpotency bound is unbounded as m grows even though the reduced threefold and its unique embedded support do not change. This conclusion is stronger than the two-point-sector calculation: it excludes every other reduced failure component and every other embedded associated subvariety in this entire family. It does not identify the full embedded primary ideal, or assert an all-contact, all- c higher-product classification.

11. COMPARISON WITH PRIOR MULTIPLICATION AND FAILURE-LOCUS RESULTS

The use of secants to account for failure of normal generation is classical. It is important here to distinguish the fixed complete series, its varying proper subseries, the Hankel annihilator space, and the scheme on the parameter space of subseries. The following comparisons refer to specific results, not to an inference from the titles of articles.

Result compared	Parameter space and conclusion	Use and distinction here
[7, Section 1; Theorem 2.1]	Hankel matrices of indeterminates; determinantal dimensions, normality and rational singularities.	These supply the annihilator-rank geometry. They do not identify the singularities of its varying isotropic-plane image. Our residual Fitting formulas concern that image.
[8], [19, Section 2.9]	A matrix satisfying the expected-grade condition has its maximal-minor resolution.	We use this resolution after proving the height. The rank competition, exceptional incidences and residual normal derivative are separate arguments.
[5, Theorem 0.2; Proposition 2.2]	A fixed completely embedded surface; numerical cohomology conditions force failure on finite linear sections, including a higher-product version.	Here the subseries vary, without cross-products. These cited statements do not give the component or residual Fitting formulas on $\text{Gr}(c, V_n)^L$.
[14, Theorems 1 and 3]	Complete very ample curve series; criteria and secant explanations for failure of normal generation.	The ambient complete series on $\mathbb{P}^1$ here is normally generated. The failure comes from varying proper subseries.

For the component theorem the additional assertions are the full rank-incidence optimization, the five exceptional isotropic incidences, and descent of the signed families. For the global singularity theorem they are the identification of the polar residual space and the equality of the restricted cotangent Fitting ideals with its multiplication minors. The local node theorem further uses a relative determinant divisor and its normal derivative, not merely the dimensions of rank strata. The cotangent presentation, Schur elimination and determinantal resolution themselves are standard algebraic tools and are not claimed as new constructions.

The closely related paper [4] requires a distinct qualification. Its publisher's first page has been inspected: it introduces higher-order embedding conditions and varying restriction maps to finite schemes. The remaining pages, containing the theorem statements needed for a complete comparison, were not available in the primary text retrieved for this revision. Accordingly no assertion that its theorems exclude or contain the results here is made. The comparison with this particular paper remains unverified; the table above is not a substitute for it. In particular a difference of parameter-space descriptions on the first page would not by itself establish a priority claim.

The local presentations in Section 3 use classical Schur elimination and the analytic implicit function theorem. The generic determinantal singularity, the quadratic cone $xz - y^2 = 0$, and the cubic cusp are not claimed as new singularities. The assertion specific to multiplication is the formula for the joint polar relations, its single-annihilator specialization, and the explicit residual obstruction and realization maps. A theorem-level comparison of these formulations with the failure-locus literature must distinguish an exact nonlinear presentation from the restricted Fitting identities; neither is a substitute for the unresolved comparison with [4].

11.1. The intrinsic derivative and classical determinantal geometry. The intrinsic map to $\text{Hom}(\ker \mu, \text{coker } \mu)$ is the normal derivative to a matrix rank stratum. This interpretation, and the tangent condition $\psi(\ker \varphi) \subset \text{im } \varphi$, are classical; see [10, Remark 1.18]. Invertible row and column operations preserve minor ideals [10, Lemma 1.3], and the generic rank-stratum germ is a smooth factor times a smaller determinantal germ [10, Section 1.3.2, equation (13)]. Essential nonsingularity is transversality to that stratum [10, Definition 2.11 and Proposition 2.12]. Fitting ideals are independent of the chosen presentation and commute with base change [25, Tags 07Z8 and 07ZA].

Accordingly, neither the existence of an intrinsic normal derivative nor the generic determinantal normal form is claimed as a new general theorem. Section 3 gives a self-contained application of these tools, identifies their dual obstruction through the actual multiplication polars, and separates the resulting exact nonlinear equations from their first-order Fitting data. The structural hypotheses are explicitly tested there; the generic realization example is not used as a substitute for a natural higher-product classification.

The latter role belongs to Theorem 10.1. It proves a global statement on the natural parameter space $\mathbb{P}(V_n^*)$, for every $n \geq 4$ and $m \geq 2$, rather than inferring a classification from a general transversality criterion. The base-point-free pencil argument and rank-two Hankel description are classical ingredients. The additional scheme conclusions proved here are smoothness at every non-evaluation secant, saturation off the evaluation curve, the complete associated-point set for $m \geq 3$, and a degree-dependent lower bound on nilpotency. They follow from the polar rank computation, the order of the Schur minors at an evaluation, and equivariance. They are not consequences of the generic matrix normal form alone. This comparison identifies the content requiring a priority assessment; it does not assert that an exhaustive search has established its historical originality.

12. AN APPLICATION TO CALIBRATED INFORMATION RECOVERY

Let J be fixed and invertible, let $H(\theta)_{ij} = \theta_{i+j}$, and set $R_\theta = JH(\theta)J^\top$ in its positive moment cone. For a prescribed smooth contact, a normal frame Z determines $U = J^\top \text{ran } Z$. The normal Hessian of the squared R_θ^{-1} -distance is the inverse of $Z^\top R_\theta Z$ (Lemma B.1). Hence this contact gives exactly the restriction of the moment functional to U^2 . Several independent contacts give the dual of (1.1). The information-loss locus is therefore the real positive realization of the failure geometry studied above, subject to the exact contact hypotheses.

Theorem D.1 supplies the positive native realization and global separation conditions. At a real crossing with two real branches, Corollary 8.10 gives the local small-singular-value law $\varepsilon^a \log(1/\varepsilon)$. These are applications of the geometry, not assumptions in its proofs. The appendices give the complete likelihood, boundary, score-compression, finite-offset and estimation arguments. In particular they do not identify a computed feature with an independent noisy distance observation.

APPENDIX A. CALIBRATED INFORMATION METRICS AND THE GEOMETRIC REDUCTION

Let $M \subset \mathbb{R}^k$ be a known residual image and $R > 0$ an inverse information matrix. Consider

$$C_R(b) = \inf_{x \in M} (x - b)^\top R^{-1} (x - b). \quad (\text{A.1})$$

At a smooth point, its full second jet sees only the normal compression of R . A statistical model may constrain R enough to remove this ambiguity. The inverse problem is therefore governed both by the geometry of M and by the linear space of admissible information matrices.

We study the calibrated family

$$R_\theta = JH(\theta)J^\top, \quad H(\theta)_{ab} = \theta_{a+b}, \quad 0 \leq a, b \leq k-1,$$

where J is fixed and invertible, and θ lies in an open positive moment cone of dimension $p = 2k-1$. Set $n = k-1$ and $V_n = H^0(\mathbb{P}^1, \mathcal{O}(n))$. Dehomogenization identifies real coefficient vectors with $\mathcal{P}_n$. A normal space N corresponds to $U = J^\top N$. The observations specify the moment functional on $U^2 = \text{span}\{fg : f, g \in U\}$. Multiple contacts specify a sum of such spaces, without cross-contact products.

The central geometric issue is not only whether a general product map is onto, but how large its entire failure locus is. Put $q = \binom{c+1}{2}$ and

$$X_L = \text{Gr}(c, V_n)^L, \quad \Delta^{(L)} = \{(U_1, \dots, U_L) : \sum_{\nu} U_{\nu}^2 \neq V_{2n}\}.$$

Here codimension refers to complex codimension of the reduced closed set. Theorem 1.1 gives the precise formula. The common-secant condition means that all U_{ν} lie in a hyperplane represented by a point of the secant variety of the evaluation rational normal curve in $\mathbb{P}(V_n^*)$. Thus the second term of (1.2) is a concrete geometric obstruction, not a formal expected codimension. For one contact, the common-base-point locus lies properly inside this larger irreducible family and is not itself a component of the failure locus.

The numerical formula has a structural refinement. Write $a = Lq - p + 1$ and $b = Lc - 3$. In the stable range $c \geq 8$, $k \geq 2c + 1$, Theorem G.1 proves that the maximal-minor scheme is integral and Cohen–Macaulay when $a < b$. Its generic annihilator is a nondegenerate Hankel form. At $a = b$ the scheme is reduced and Cohen–Macaulay, with precisely one nondegenerate component and 2^{L-1} signed secant components. If $b < a$, the latter are exactly the maximal-dimensional components. They are rational and generically smooth in the determinantal scheme. The signs record independent choices of the two isotropic hyperplanes of a rank-two form, modulo their simultaneous interchange; the common-secant family above is only the all-equal choice.

The change of regime has an intrinsic residual description. At two distinct secant points, write $U_{\nu} \cap \text{rad } H_{\ell} = gK_{\nu}$, where g is their vanishing quadratic. The wall is

$$L \binom{c}{2} = 2k - 5,$$

exactly the saturation threshold for multiplication of the $(c - 1)$ -planes K_{ν} in the $(k - 2)$ -dimensional residual space. This identity determines the conormal dimension of a general signed component. The expected-codimension cycle and local conditioning exponents follow from the structural theorem. The earlier all-dimension codimension formula is not restricted by the additional hypotheses of this refinement.

For the synchronized residual image

$$q_A(\eta) = ((a_i^{\top} \eta)^2)_{i=1}^k, \quad A \in \mathbb{R}^{k \times d},$$

the normal dimension is $c = k - d$. Native realizability is an open positive-vector condition. Global separation at a prescribed normal space requires, in addition, that the corresponding tangent plane not split across a nontrivial coordinate partition. This exact criterion is proved in Theorem D.1.

Theorem A.1 (Sharp native single-contact threshold). *Fix $d \geq 2$ and any separated root and clock configuration of Section C. For $k > d$, there exist full-rank loadings A with $q_A(e_1) > 0$ for which the single full second jet $D^2 C_{R_{\theta}}(q_A(e_1))$ identifies every $\theta \in \Omega$ if and only if*

$$\binom{k - d + 1}{2} \geq 2k - 1.$$

In the sufficient range the assertion holds on a nonempty Zariski-open loading set, for the global distance function. In particular

$$k_{\min}(d) = d + \left\lceil \frac{3 + \sqrt{16d + 1}}{2} \right\rceil = d + 2\sqrt{d} + O(1). \quad (\text{A.2})$$

Unrestricted positive matrices at the same contact retain an invisible space of dimension $d(2k - d + 1)/2$.

A contact here is a matrix-valued observation with q entries, not a scalar datum. The comparison in (A.2) is across calibrated families with varying k , fixed d , and arbitrary separated configurations at each k ; it is not a universal threshold for every information model. Independently designed contacts can use different loadings with the same J . Several points of one fixed loading are not asserted to vary independently.

Method and relation to earlier theory. Hankel duality with polynomial multiplication is classical [21]. The saturated one-space maximal-rank statement is also a consequence of classical postulation of general rational curves [2, 3, 20]. Our proof of Theorem 1.1 instead combines the classical dimensions of Hankel rank strata [7, Sections 1–2] with an isotropic-incidence count, an optimization over all ranks, and an explicit common-secant family. This proves the full codimension formula and the coupled multi-space assertion, not just generic surjectivity. The classical maximal-minor height bound supplies the other matching upper bound [9, Introduction]. These classical inputs are distinguished from the incidence optimization and native fixed-space realization proved here.

Failure of quadratic normality has a substantial antecedent literature. Green–Lazarsfeld [14, Theorems 1 and 3] study complete very ample linear series on curves, giving normal-generation criteria and finite secant configurations explaining failure under degree and cohomology hypotheses. Ballico [5, Theorem 0.2] studies failure cycles for a fixed embedded surface and detects failure on finite linear sections, using a Grassmannian incidence and cohomology. The principle that secants can explain quadratic-normality failure is therefore classical. Here the complete series on $\mathbb{P}^1$ is fixed and normally generated; the varying objects are its proper c -dimensional subseries, and for $L > 1$ the observation uses a direct sum of within-subseries products, without cross-products. Our structural assertion concerns the components and generic scheme structure over this product of Grassmannians. Neither the complete-series degree criterion nor the surface-section failure-cycle theorem supplies this signed component classification or the residual wall. We distinguish these particular conclusions from the classical inputs on Hankel ranks, determinantal resolutions, and cycles [7, 8, 19, 16, 11].

Polynomial product spaces also occur in the study of Gram spectrahedra [23]; the present observation is the dual moment functional on within-contact products. The monomial specialization is a restricted additive-basis problem [27, 17, 18], and can have a strictly worse threshold. Quadratic sensing and phase retrieval concern reconstruction of the input to a quadratic measurement map [1]; our unknown is instead an information metric, and injectivity on all of Sym_d is not assumed. Complete-conormal projective duality [12] addresses a different, larger observation category.

The statistical statements likewise keep their experiments separate. The coalesced-root model has a smooth positive extension, but its physical local parameter set is a cone. Its efficient Gaussian quotient and $N^{-1/4}$ root-amplitude scale are derived explicitly in Section H. For observed Poisson exposures we form normal-compression score statistics directly from the counts. Their local information loss is exactly the loss of rank of the product map. This is a local likelihood statement in a specified target subexperiment, not a claim of global equivalence of all raw data and finitely many distance readings. The finite-sample estimators and computed finite-offset features in the appendices are retained as distinct procedures. Standard constrained and regular asymptotic principles are attributed to [13, 26].

APPENDIX B. EXACT FIBRES OF PRESCRIBED CONTACT DATA

For $\theta \in \mathbb{R}^p$ write

$$\ell_\theta \left(\sum_{a=0}^{2n} f_a t^a \right) = \sum_a f_a \theta_a.$$

Let $\Omega \subset \mathbb{R}^p$ be open with $R_\theta > 0$. At each prescribed point x_ν , assume that the global image has one embedded smooth germ and that its local minimizing projection computes (A.1). Parameter differentiation below uses analytic germs and a common projection tube on compact metric sets. These conditions are verified for the native loading family in Section D.

Let Z_ν have orthonormal columns spanning the Euclidean normal space. Put

$$B_\nu = Z_\nu^\top R_\theta Z_\nu, \quad G_\nu = \frac{1}{2} Z_\nu^\top D^2 C_{R_\theta}(x_\nu) Z_\nu.$$

Lemma B.1 (Normal block). *One has $G_\nu = B_\nu^{-1}$. The full Hessian has tangent kernel and is determined by this normal block.*

Proof. The quadratic contact with a manifold equals that with its tangent space T . At the minimizer of $(u - v)^\top R^{-1}(u - v)$ over $v \in T$, the vector $R^{-1}(u - v)$ is normal. Hence $u - v = RZn$, and normal projection gives $Bn = Z^\top u$. The minimum is $(Z^\top u)^\top B^{-1} Z^\top u$. A local parametrization and its positive tangent Gram matrix justify the minimizing projection by the implicit function theorem and yield this quadratic term. $\square$

Set $U_\nu = J^\top \text{ran } Z_\nu$ in polynomial coordinates, and define

$$\mathcal{U} = \text{im} \left(\bigoplus_{\nu=1}^L \text{Sym}^2 U_\nu \xrightarrow{\text{multiplication}} \mathcal{P}_{2n} \right), \quad \mathcal{L}\theta = (B_\nu(\theta))_\nu.$$

The codomain of $\mathcal{L}$ has the direct-sum Frobenius norm.

Theorem B.2 (Exact contact fibre). *The exact second-jet fibre through θ is*

$$(\theta + \mathcal{U}^\perp) \cap \Omega.$$

Its local dimension is $p - \dim \mathcal{U}$. Recovery on Ω is equivalent to $\mathcal{U} = \mathcal{P}_{2n}$, or to injectivity of $\mathcal{L}$.

Proof. For normal vectors v, w with coefficient polynomials f_v, f_w ,

$$v^\top (R_{\theta'} - R_\theta) w = \ell_{\theta' - \theta}(f_v f_w).$$

Equality of the jets is equality of the positive normal blocks, by Lemma B.1. It is therefore precisely annihilation of $\mathcal{U}$. Inversion is a diffeomorphism on positive blocks; openness of Ω gives the asserted fibre dimension. $\square$

For one d -dimensional germ there are at most $q = \binom{k-d+1}{2}$ independent products. For unrestricted matrices, fixing the normal-normal block leaves tangent-tangent and mixed blocks free, with total dimension

$$\frac{k(k+1) - (k-d)(k-d+1)}{2} = \frac{d(2k-d+1)}{2}. \quad (\text{B.1})$$

Small perturbations in every such direction preserve positivity.

APPENDIX C. THE NATIVE INFORMATION FAMILY

Fix distinct strictly positive rank-one four-cell probability tables M_1, M_2 , roots $0 < r_1 < \dots < r_k < D$, and $m = 2k + 1$ clocks $D < T_1 < \dots < T_m$. Use

$$f_1(T) = f_{1,0}(T) \prod_i ((T - r_i - s_i/2)^2 - v_i), \quad p_j = \pi_j M_1 + (1 - \pi_j) M_2,$$

$$\frac{\pi_j}{1 - \pi_j} = \frac{\alpha f_1(T_j)}{(1 - \alpha) f_2(T_j)}.$$

The fixed factors are positive at the clocks. At $s = v = 0$, the free coordinates are log mixing odds and the root-sum displacements; the physical retained coordinates satisfy $v_i \geq 0$. The likelihood

is positive and analytic on a two-sided neighbourhood of this point. Its restriction to $v \geq 0$, not that extension's unrestricted estimator, is the physical experiment.

Let $\lambda_j > 0$ be information weights and set

$$\kappa_j = \pi_j^2 (1 - \pi_j)^2 \sum_{a=1}^4 \frac{(M_{1a} - M_{2a})^2}{p_{ja}}, \quad V_j = (1, -((T_j - r_i)^{-1})_i, -((T_j - r_i)^{-2})_i).$$

The k simple poles precede the k double poles. Write

$$\begin{aligned} h(t) &= \prod_i (t - r_i)^2, \quad p_*(t) = \prod_i (t - r_i), \quad P(t) = \prod_j (t - T_j), \\ d_i &= \frac{P(r_i)}{\prod_{a \neq i} (r_i - r_a)^2}, \quad \xi_j = \frac{h(T_j)}{P'(T_j)}, \\ J_{ia} &= [t^a] d_i \frac{p_*(t)}{t - r_i}, \quad \rho_j = \frac{\xi_j^2}{p_*(T_j)^2}. \end{aligned} \tag{C.1}$$

Proposition C.1 (Profiled quadratic information). *The retained block of the inverse full information, equivalently the inverse nuisance Schur complement, is*

$$R = JH(\theta)J^\top, \quad \theta = \sum_{j=1}^m \frac{\rho_j}{\lambda_j \kappa_j} w_{2n}(T_j), \quad w_s(t) = (1, t, \dots, t^s)^\top. \tag{C.2}$$

The matrix J is invertible. Independent positive information weights give an open moment cone Ω of dimension $p = 2k - 1$.

Proof. Multiplying a linear combination of the columns of V by h gives a polynomial of degree at most $2k$. Vanishing at the $2k + 1$ clocks forces it to be zero; principal parts at the roots and the constant at infinity then force every coefficient to vanish. Thus V is invertible and the full information is $V^\top \text{diag}(\lambda_j \kappa_j) V$.

For arbitrary log-odds increments y_j , interpolate

$$g(t) = \sum_j h(T_j) y_j \frac{P(t)}{(t - T_j) P'(T_j)}.$$

The retained coefficient at r_i is $-g(r_i) / \prod_{a \neq i} (r_i - r_a)^2$. Thus the retained rows Z_0 of V^{-1} are

$$(Z_0)_{ij} = \frac{\xi_j d_i}{T_j - r_i} = \frac{\xi_j}{p_*(T_j)} (Jw_n(T_j))_i.$$

Their weighted Gram matrix gives (C.2). Evaluation of the polynomials defining J at the distinct roots is diagonal with nonzero entries, proving invertibility. The positive moment weights range over an open orthant. The moment matrix has rank p by Vandermonde independence, so its image is open. Positive atomic weights at distinct clocks give positive definite $H(\theta)$. $\square$

The weights in this proposition need not be unknown likelihood parameters. For a realized fixed allocation they are n_j/N ; for a limiting allocation they are its limits; a different prespecified λ defines a target design metric. Section I uses a further protocol in which exposures are unknown but their counts are observed. Section H identifies (C.2) as the covariance of the efficient Gaussian quotient of the physical boundary experiment.

APPENDIX D. NATIVE REALIZATION AND GLOBAL SEPARATION

All spaces in this section are real. The following result holds for any fixed $J \in \mathrm{GL}_k(\mathbb{R})$, not just the interpolation matrix.

Theorem D.1 (Exact fixed-space realization). *Let $1 \leq d < k$, $c = k - d$, $U \in \mathrm{Gr}(c, k)$, and $T = JU^\perp$. The space U is the normal-polynomial space of a local immersed image q_A at $q_A(e_1) > 0$ if and only if*

$$T \cap \mathbb{R}_{>0}^k \neq \emptyset. \quad (\text{D.1})$$

There is such a realization with

$$q_A^{-1}(q_A(e_1)) = \{e_1, -e_1\} \quad (\text{D.2})$$

if and only if, in addition,

$$D_s T \neq T \quad \text{for every } s \in \{\pm 1\}^k \setminus \{\mathbf{1}, -\mathbf{1}\}. \quad (\text{D.3})$$

Here $D_s = \mathrm{diag}(s)$. On each sign chamber of the full-rank loading domain with Ae_1 nowhere zero, the map

$$A \mapsto U(A) = J^\top (\mathrm{ran}(\mathrm{diag}(Ae_1)A))^\perp$$

is a submersion onto its positive realizable image. That image is nonempty, Euclidean open, and Zariski dense.

Proof. Put $z = Ae_1$. The tangent space is $T = \mathrm{ran}(\mathrm{diag}(z)A)$ and contains $t = z^2 > 0$. Taking orthogonal complements in $U = J^\top T^\perp$ gives $T = JU^\perp$ and proves necessity of (D.1). Conversely choose $t > 0$ in T , extend it to a basis $t, t_2, \dots, t_d$, and set

$$z_i = \sqrt{t_i}, \quad A = [z, \mathrm{diag}(z)^{-1}t_2, \dots, \mathrm{diag}(z)^{-1}t_d]. \quad (\text{D.4})$$

Then $\mathrm{diag}(z)A$ has this prescribed basis, and $q_A(e_1) = t$.

Equality of coordinatewise squares is equivalent to $A\eta = D_s z$ for a sign vector s . Because D_s commutes with $\mathrm{diag}(z)$,

$$D_s z \in \mathrm{ran} A \iff D_s t \in T. \quad (\text{D.5})$$

If $D_s T = T$ for a nonconstant s , the right side holds for every $t \in T$; positivity ensures $D_s t \neq \pm t$, giving another preimage. Conversely, under (D.3), each $T \cap D_s T$ is a proper linear subspace of T . Their finite union cannot cover the relatively open positive cone in (D.1). Choose t outside that union and use (D.4). Equation (D.5) leaves only the constant signs, hence exactly the two preimages in (D.2).

For the submersion statement, on a fixed sign chamber the change of frames $A \mapsto B = \mathrm{diag}(z)A$ is a diffeomorphism onto full-rank frames whose first column is strictly positive. Its inverse has $z_i = \pm \sqrt{B_{i1}}$ with the fixed signs and $A_{ij} = B_{ij}/z_i$. The ordinary frame-to-plane map is a submersion. Orthogonal complement and multiplication by fixed J preserve that property. Positivity persists under small changes of a frame, and planes containing $\mathbf{1}$ show nonemptiness. A polynomial vanishing on a nonempty real open subset of an affine Grassmannian chart vanishes identically, proving Zariski density. $\square$

The obstruction in (D.3) has a concrete interpretation. If I is the set of positive signs of s , then $D_s T = T$ precisely when

$$T = (T \cap \mathbb{R}^I) \oplus (T \cap \mathbb{R}^{I^c}).$$

Thus a prescribed normal space may be locally realizable yet admit no globally separated realization at that same space. Theorem D.1 classifies exactly this distinction.

Lemma D.2 (Generic global distance). *A nonempty Zariski-open subset of the full-rank loading domain with Ae_1 nowhere zero satisfies (D.2). For each such fixed A , the global image has one embedded analytic germ at $b_0 = q_A(e_1)$. For every compact set $\mathcal{R} \subset \text{Sym}_k^{>0}$ there is a neighbourhood of b_0 on which the distance to that germ equals the global distance, and the minimizing projection is analytic with uniform bounds for $R \in \mathcal{R}$.*

Proof. For a nonconstant sign vector, the condition $D_s Ae_1 \in \text{ran } A$ is algebraic, defined by augmented $(d+1)$ -minors. It is proper: choose a vector z with no zero entries and a d -plane containing z but not $D_s z$, then take a frame with first column z . Removing the finitely many such proper sets proves the first assertion. The ambient loading space is the irreducible affine space $\mathbb{R}^{kd}$; full rank and nonzero first coordinates specify a Zariski-open domain, so these finite open intersections remain dense.

Full rank gives

$$\|q_A(\eta)\|_2 \geq k^{-1/2} \|A\eta\|_2^2 \geq k^{-1/2} \sigma_{\min}(A)^2 \|\eta\|_2^2,$$

so q_A is proper. Its derivative is injective at both preimages, and their local image germs coincide by evenness. Properness excludes a remote sheet accumulating at b_0 . The trial image b_0 bounds the distance by $C\|b - b_0\|^2$. Uniform positive metric bounds on $\mathcal{R}$ therefore force all minimizing images into this germ when b is close enough to b_0 . The positive tangent Hessian and the implicit function theorem give the projection tube. Its radius and derivative constants depend on A , the local germ and $\mathcal{R}$; no uniformity over all generic loadings is claimed. $\square$

Proof of Theorem A.1. Necessity is the dimension bound in Section B. If it holds and $d \geq 2$, then $c = k - d \geq 5$. Theorem 1.1 with $L = 1$ gives a nonempty complex Zariski-open full-product locus. Its defining minors have real coefficients and some minor is a nonzero polynomial on a real affine chart. A nonzero real polynomial cannot vanish on any nonempty real open set. Thus the full-product locus meets the positive realizable image of Theorem D.1.

On pivot charts normal frames are rational functions of A , so full product rank pulls back to a nonempty Zariski-open subset of the irreducible loading domain. Intersect it with Lemma D.2's dense open set. Theorem B.2 gives recovery from the global distance for every $\theta \in \Omega$ with the same J . Writing $k = d + c$ gives $c^2 - 3c - 4d + 2 \geq 0$, whose positive root yields (A.2). Formula (B.1) gives the unrestricted ambiguity. $\square$

For $d = 2, 3, 4, 5, 6$, the optimal k are 7, 8, 10, 11, 13. At the optimum the full symmetric measurement $S \mapsto (a_i^T S a_i)_i$ is necessarily noninjective for $d \geq 5$: check $d = 5, 6$ directly, and use $k_{\min} \leq 2d < d(d+1)/2$ for $d \geq 7$. This is not full-symmetric phase retrieval in disguise.

Corollary D.3 (Independent native designs and the real secant family). *Fix $4 \leq c < k$ and a native interpolation matrix J . For L independently chosen loading matrices of size $k \times (k - c)$, generic globally separated contacts recover the common native moment parameter if and only if $Lq \geq p$. The common-secant family Σ_L contains real, positive, globally separated native realizations. At its general such points its loading pullback has real local codimension $Lc - 3$.*

Proof. Apply the product of the loading submersions and generic global-separation opens to Theorem 1.1, using real nonvanishing of minors as above. For the second assertion choose distinct $x, y > D$ and $h = \text{ev}_x + \varepsilon \text{ev}_y$ with $\varepsilon > 0$. By (C.1), every coordinate of $Jw_n(t)$ is negative for $t > D$. Thus $t = -Jh > 0$. A general d -plane T containing t avoids each of the finitely many vectors $D_s t$ for nonconstant s , since none is proportional to t and $d < k$. Such a $T = JU^\perp$ has $U \subset \ker h$ and, by (D.4)–(D.5), a globally separated realization. Choose the T_ν independently. These choices form real open subsets in the secant incidence. Its generically one-to-one projection is birational in characteristic zero and is an isomorphism on dense open subsets. The loading submersion consequently preserves its codimension there. $\square$

The exact formula (1.2) is an algebraic statement over $\mathbb{C}$. The corollary supplies a real native component of that dimension when the secant term is minimal. It does not infer real points on every complex component in the other regime. It also does not replace the independent-design assumption by multiple arbitrary points on one fixed quadratic image.

APPENDIX E. RANK STRATA AND QUANTITATIVE INVERSION

Let Δ_r be the determinantal scheme on $\text{Gr}(c, V_n)$ where multiplication has rank at most r . At a point of rank *exactly* r , put $K_U = \ker \mu_U$ and $Q_U = V_{2n}/U^2$.

Proposition E.1 (Infinitesimal rank map). *The tangent space to Δ_r is the kernel of*

$$\begin{aligned} \beta_U : \text{Hom}(U, V_n/U) &\longrightarrow \text{Hom}(K_U, Q_U), \\ \beta_U(\phi) \left(\sum_i f_i \odot g_i \right) &= \sum_i \{ \phi(f_i)g_i + f_i\phi(g_i) \} \pmod{U^2}. \end{aligned} \quad (\text{E.1})$$

If β_U is onto, the scheme is smooth there of codimension $(q-r)(p-r)$. The corresponding native contact fibre has dimension $p-r$.

Proof. Express nearby planes as graphs and differentiate products. A different lift of ϕ changes the result only by an element of U^2 . Near a nonzero r -minor, invertible block operations reduce multiplication to an invertible r -block and a Schur complement. Rank at most r is equivalent to the vanishing of that complement, whose differential is (E.1). Its kernel is the scheme tangent space, and surjectivity gives smoothness with the displayed codimension. The fibre assertion is Theorem B.2. $\square$

The same construction on X_L uses the direct sum of the graph derivatives. The assertion is about the scheme defined by minors, and does not presuppose its reducedness.

Use coefficient Euclidean norms, the induced symmetric tensor norm on multiplication, direct-sum Frobenius norms on blocks, and the standard Grassmannian Riemannian distance. Let $\Delta_{\mathbb{R}}$ denote the real failure set. Let γ be the least singular value of an injective $\mathcal{L}$.

Theorem E.2 (Stability and transverse degeneracy). *Suppose $m_0 I \leq B_\nu \leq M_0 I$ and $\hat{G}_\nu = G_\nu + E_\nu$, with $\max_\nu \|E_\nu\|_{\text{op}} \leq (2M_0)^{-1}$. Then*

$$\left\| \mathcal{L}^\dagger(\hat{G}_\nu^{-1})_\nu - \theta \right\|_2 \leq \frac{2M_0^2}{\gamma} \left(\sum_\nu \|E_\nu\|_F^2 \right)^{1/2}. \quad (\text{E.2})$$

The least multiplication singular value is bounded above by a constant times distance to $\Delta_{\mathbb{R}}$. Near a real corank-one point at which the corresponding kernel-to-cokernel differential is onto, it is bounded above and below by positive local constants times that distance. The same local comparison holds in loading space for distance to the pulled-back failure set.

Proof. The inverse identity gives $\hat{G}_\nu^{-1} - B_\nu = -\hat{G}_\nu^{-1} E_\nu B_\nu$. The assumed bounds make its Frobenius norm at most $2M_0^2 \|E_\nu\|_F$. Apply $\mathcal{L}^\dagger$, whose norm is γ^{-1} .

In finitely many compact orthonormal frame charts, multiplication is Lipschitz in the plane coordinates. Singular values are Lipschitz in matrix entries and vanish on the real failure set, giving the upper estimate. At corank one, the Schur complement is a row of length $Lq - p + 1$. Bounded invertible block operations compare the least singular value with the norm of this row. Surjectivity of its differential gives local coordinates in which the row is a coordinate projection, proving the lower estimate as well. For fixed J , passage from an orthonormal polynomial frame to $J^\top Z$ changes norms by bounded positive factors. The loading submersion gives local product coordinates and transfers the distance comparison. $\square$

No linear distance law is asserted at nontransverse or singular points. Compact design sets disjoint from failure have a positive lower stability constant by continuity; this alone is not a dimension-uniform numerical estimate or a random-loading tail bound.

APPENDIX F. STRUCTURED COMPRESSION BEYOND HANKEL MATRICES

Let $\mathcal{K} \subset \text{Sym}_k$ have dimension s and contain a positive matrix. Write $\rho(\mathcal{K}, c)$ for the maximum rank of $X \mapsto X|_{N \times N}$ over $N \in \text{Gr}(c, k)$.

Proposition F.1 (Generic transfer). *For generic full-rank native loadings with Ae_1 nowhere zero, compression has rank $\rho(\mathcal{K}, c)$ and the global contact fibre in an open positive subset of $\mathcal{K}$ has dimension $s - \rho(\mathcal{K}, c)$.*

Proof. Maximal compression rank is a nonempty Zariski-open condition. Apply Theorem D.1 with $J = I$ and Lemma D.2. Positive block inversion preserves fibres, and rank-nullity gives the dimension. $\square$

This is a transfer principle, not a computation of ρ for an unspecified family. The following computation includes multiplicities in a fixed spectral decomposition.

Theorem F.2 (Block-spectral rank). *Let $P_1, \dots, P_s$ be nonzero mutually orthogonal real symmetric projections with sum I_k , and let $\mathcal{K} = \text{span}\{P_1, \dots, P_s\}$. Then*

$$\rho(\mathcal{K}, c) = \min\{s, \binom{c+1}{2}\}.$$

For the diagonal family the sharp threshold is $\binom{c+1}{2} \geq k$, with least ambient dimension $d + \lceil (1 + \sqrt{1 + 8d})/2 \rceil$ at fixed $d = k - c$. For the family of real symmetric circulant matrices it is $\binom{c+1}{2} \geq \lfloor k/2 \rfloor + 1$.

Proof. The upper bound is dimensional. Suppose $c \leq s$. Choose one unit vector in each $\text{ran } P_i$. In their span, a normal frame with rows $u_i^\top$ gives compressed matrices $u_i u_i^\top$. The c vectors e_a followed by the $c(c-1)/2$ vectors $e_a + e_b$ have linearly independent outer products: first inspect off-diagonal entries, then diagonal entries. Select the first s when $s \leq q$, or retain all q and add rows when $s > q$. Starting with the e_a gives full column rank. Column-orthonormalization transforms every compressed matrix by the same invertible congruence and therefore preserves linear independence. If $c > s$, take a c -plane containing one unit vector from every $\text{ran } P_i$ and extend their span. Evaluation on these distinguished vectors separates the s coefficients. This proves the rank formula in all cases.

For diagonal matrices, $s = k$ and substitution $k = d + c$ gives $c^2 - c - 2d \geq 0$. A real symmetric circulant matrix is diagonal in the real Fourier decomposition: the zero frequency, the Nyquist frequency when present, and the paired sine–cosine frequencies give $\lfloor k/2 \rfloor + 1$ orthogonal spectral projections. Its eigenvalue is arbitrary on each such block, so the preceding formula applies. $\square$

Independent Bernoulli log-odds coordinates with information weights λ_i have inverse information $\text{diag}((\lambda_i p_i (1 - p_i))^{-1})$, providing a genuine positive diagonal information family. The block-spectral theorem concerns the stated matrix families and does not claim that every such family is generated by the calibrated polynomial experiment, or compute ranks for all Toeplitz or graph-sparse families.

APPENDIX G. COMPONENTS, SCHEME STRUCTURE, AND THE RESIDUAL WALL

Let $\mathcal{D}_L$ denote the scheme of maximal minors of μ , and write

$$e = Lq, \quad a = e - p + 1, \quad b = Lc - 3, \quad m_X = Lc(k - c), \quad \mathcal{E} = \bigoplus_{\nu=1}^L \text{Sym}^2 \mathcal{S}_\nu.$$

Thus $|\mathcal{D}_L| = \Delta^{(L)}$. The letters a, b in this section are codimensions, not loading vectors. Theorem 1.1 holds throughout its stated range. The structural results below assume the additional stable-range conditions

$$c \geq 8, \quad k \geq 2c + 1, \quad e \geq p. \quad (\text{G.1})$$

For the classical input in Lemma 4.1, the precise affine dimension statement is $\dim R = 2t - 2$ on page 2 of [7]'s arXiv version; set $t = r + 1$. The secant identification is its equation (2.0.1). Exact-rank strata are open in their rank-at-most closures; confluent forms remain included. These inputs do not supply the component classification below.

These conditions exclude the small equality cases in the rank optimization and the two families of maximal even-dimensional orthogonal Grassmannians.

For distinct $x, y \in \mathbb{P}^1$, choose nonzero representatives of their evaluation functionals and let $t \neq 0$. Set

$$h_+ = \text{ev}_x + t \text{ev}_y, \quad h_- = \text{ev}_x - t \text{ev}_y, \quad \ell = \text{ev}_x^{(2n)} - t^2 \text{ev}_y^{(2n)}. \quad (\text{G.2})$$

Rescaling the representatives only reparametrizes this family. For $\varepsilon \in \{+, -\}^L / \{\varepsilon \sim -\varepsilon\}$, let Σ_ε be the closure of the tuples satisfying $U_\nu \subset \ker h_{\varepsilon_\nu}$. The all-equal pattern is the common-secant variety Σ_L of Proposition 4.4. Define also

$$\mathfrak{F}_L = \{([\ell], (U_\nu)) : \text{rank } H_\ell = k, H_\ell|_{U_\nu \times U_\nu} = 0\}, \quad \Phi_L = \overline{\text{pr}_{X_L} \mathfrak{F}_L}. \quad (\text{G.3})$$

The closure in this definition is essential: the generic annihilator of Φ_L will be nondegenerate, but its specializations need not be.

Theorem G.1 (Component classification and determinantal scheme). *Assume (G.1).*

(i) *If $a < b$, the scheme $\mathcal{D}_L$ is integral and Cohen–Macaulay of codimension a , with support Φ_L . Its general point has multiplication corank one, a unique projective annihilator of Hankel rank k , and a smooth determinantal neighbourhood. The projection from $\mathfrak{F}_L$ is birational onto Φ_L .*

(ii) *If $a = b$, the scheme $\mathcal{D}_L$ is reduced and Cohen–Macaulay of pure codimension a . Its irreducible components are exactly*

$$\Phi_L \quad \text{and} \quad \Sigma_\varepsilon \quad (\varepsilon \in \{+, -\}^L / \{\pm 1\}).$$

There are $1 + 2^{L-1}$ components. The general point of each has multiplication corank one and is smooth in the determinantal scheme. The generic annihilator has rank k on Φ_L and rank two on every Σ_ε .

(iii) *If $b < a$, the maximal-dimensional irreducible components of $\Delta^{(L)}$ are exactly the 2^{L-1} varieties Σ_ε , of codimension b . Their general points have multiplication corank one and are smooth in $\mathcal{D}_L$. Every Σ_ε is rational.*

(iv) *In the expected-codimension range $a \leq b$, the fundamental cycle is*

$$[\mathcal{D}_L] = c_a(\mathcal{E}^*) = \left[\prod_{\nu=1}^L \prod_{1 \leq i \leq j \leq c} (1 + x_{\nu i} + x_{\nu j}) \right]_a \quad \text{in } A^a(X_L), \quad (\text{G.4})$$

where $x_{\nu i}$ are the formal Chern roots of $\mathcal{S}_\nu^$ and the brackets select codimension a .*

Part (iii) classifies maximal-dimensional components, not possible smaller components or embedded structure in the excess range. Parts (i) and (ii), by contrast, concern the entire determinantal scheme. Formula (G.4) uses the classical determinantal cycle formula after the expected codimension and reducedness have been established; the cycle formula itself is not a new general intersection-theoretic principle.

G.1. The two extremal incidences.

Lemma G.2 (Strict rank gap). *Under (G.1), every annihilator incidence stratum has dimension strictly less than $m_X - \min(a, b)$, except possibly the strata*

$$r = 2, \quad s_1 = \cdots = s_L = 1; \quad r = k, \quad s_1 = \cdots = s_L = c.$$

Their dimensions are respectively $m_X - b$ and $m_X - a$.

Proof. Use $D_{r,s}$ and h_r from Lemmas 4.1 and 4.2. If $r = k$, the radical is zero and $s = c$. Suppose $r < k$. For $s = 0$, $LD_{r,0} - h_r = (Lc - 2)r + 1 > b$. For $1 \leq s \leq c - 2$, the lower endpoint $r = 2s$ in the feasible interval gives

$$g(s) = 1 + s(Lc - 4) - Ls(s - 1)/2.$$

Its coefficient before minimizing in r is $L(c - s) - 2 \geq 0$. Moreover,

$$g(1) = b, \quad g(c - 2) - b = (c - 3)\{L(c + 2)/2 - 4\} > 0.$$

Concavity places every integer $2 \leq s \leq c - 2$ strictly above b . At $s = 1$ the coefficient $L(c - 1) - 2$ is strictly positive, so equality requires $r = 2$.

For $s = c - 1$ and $L = 1$, the calculation in Lemma 4.3 gives at least $a + k - 2c + 1 \geq a + 2$. For $s = c - 1$ and $L \geq 2$, it gives at least $b + (c - 2)\{L(c + 1)/2 - 4\} > b$. For $s = c$, $r < k$, it gives at least $a + 1$. Thus, for each fixed nonexceptional r and every feasible s , $LD_{r,s} - h_r > \min(a, b)$. Averaging this inequality over the L choices s_ν gives $\sum_\nu D_{r,s_\nu} - h_r > \min(a, b)$. At $r = 2$, replacing any $s_\nu = 1$ by zero increases this sum. The two displayed dimensions follow from $D_{2,1} = c$ and $D_{k,c} = q$. $\square$

The proof explains the two ends of the codimension formula: an arbitrary nondegenerate Hankel form imposes q isotropy equations per contact, whereas a rank-two form leaves a choice between two hyperplanes. Intermediate ranks incur a strict dimension loss in (G.1).

Lemma G.3 (Signed secant incidences). *The 2^{L-1} varieties Σ_ε are distinct irreducible rational subvarieties of $\Delta^{(L)}$, each of codimension b . Their incidence parametrizations are birational. A general point of any one of them admits exactly one rank-two projective annihilator.*

Proof. A general point $[h]$ of the secant threefold S_n has a unique unordered secant decomposition. Indeed, evaluations at four distinct points are linearly independent for $n \geq 3$, by interpolation; the corresponding confluent statement follows by Hermite interpolation. Thus two different secant lines cannot meet away from their endpoints, and tangent or limiting cases form a proper closed subset. The parameter space is birational to a projective-line bundle over $\text{Sym}^2 \mathbb{P}^1 = \mathbb{P}^2$, hence is rational. Sending $[h_+]$ to $[h_-]$ in (G.2) is a rational involution, independent of the ordering of the two evaluation points.

Normalize the sign pattern by $\varepsilon_1 = +$. Over the dense open set of distinct secants with nonzero coefficients, the signed incidence is a product of relative Grassmannians, with dimension $3 + Lc(k - 1 - c) = m_X - b$. It is irreducible and rational: its underlying hyperplane bundles are trivial on a dense open subset of the rational base. The functional ℓ in (G.2) annihilates every product, for either sign, so its image closure lies in $\Delta^{(L)}$.

Here is the projection calculation explicitly. Fix $h \in S_n$ in the above open set. In $\text{Gr}(c, \ker h)$ the exceptional planes admitting another secant hyperplane are the projection of

$$\{(h', U) : h' \in S_n \setminus \{h\}, U \subset \ker h \cap \ker h'\}.$$

Its dimension is at most $3 + c(k - 2 - c)$, strictly smaller than $c(k - 1 - c)$ because $c > 3$; if $c > k - 2$ it is empty. Therefore a general first plane determines h uniquely. This proves generic injectivity and, in characteristic zero, birationality of the signed projection. The recovered h also determines its partner and the rank-two annihilator. For a general tuple no U_ν lies in both

partner hyperplanes. Consequently two different normalized sign patterns have different images. The exact-rank-two stratum not represented by two distinct evaluations has smaller dimension; it cannot affect these generic assertions. $\square$

Lemma G.4 (The nondegenerate incidence). *The variety $\mathfrak{F}_L$ is smooth and irreducible of dimension $m_X - a$. Its image contains a point whose first plane lies in no secant hyperplane.*

Proof. Over the open set of nonsingular Hankel forms, the fibres are products of $\text{OGr}(c, k)$. The restriction equations have independent differentials, as in Lemma 4.2. Since $2c < k$, the special orthogonal group acts transitively on these isotropic Grassmannians; a Witt basis proves transitivity, and the special orthogonal group over $\mathbb{C}$ is connected. Thus the fibres are smooth and geometrically irreducible of dimension $c(k - c) - q$. The base is a smooth irreducible open subset of $\mathbb{P}^{p-1}$, so the relative incidence is smooth and irreducible, with the stated dimension.

Fix one nonsingular form H and put $d_H = c(k - c) - q$. In any hyperplane K the restricted form has radical of dimension zero or one. In the nondegenerate case its isotropic c -planes have dimension $d_H - c$. In the one-dimensional radical case, the strata avoiding and containing the radical have dimensions respectively

$$d_H - c, \quad d_H - (k - c - 1) \leq d_H - c,$$

when nonempty. These formulas follow by the same choice-of-radical-intersection and lift calculation as Lemma 4.2. Varying K among secant hyperplanes contributes only three parameters. Their union in $\text{OGr}(c, k)$ therefore has dimension at most $d_H - c + 3 < d_H$. Choose U_1 outside it and choose the other isotropic planes arbitrarily. This proves the last assertion. $\square$

G.2. Corank one and the residual multiplication map. Let $\mathfrak{J} \subset \mathbb{P}(V_{2n}^*) \times X_L$ be the scheme defined by $\ell\mu = 0$.

Lemma G.5 (Corank-one chart). *Above the open subset on which μ has rank at least $p - 1$, the projection from $\mathfrak{J}$ to $\mathcal{D}_L$ is an isomorphism. At a point of corank one, infinitesimal pairs satisfy*

$$\dot{\ell}(uv) + H_\ell(\dot{u}, v) + H_\ell(u, \dot{v}) = 0 \quad (u, v \in U_\nu, \ 1 \leq \nu \leq L). \quad (\text{G.5})$$

Here $\dot{U}_\nu \in \text{Hom}(U_\nu, V_n/U_\nu)$ and $\dot{\ell}$ is taken modulo $\mathbb{C}\ell$.

Proof. On a nonzero $(p - 1)$ -minor, invertible row and column operations reduce μ to an identity block of size $p - 1$ and a remaining row of length $e - p + 1$. Its entries generate the maximal-minor ideal. The normalized last coordinate of a projective left kernel vector is one; the other coordinates are uniquely eliminated by the identity block. The remaining equations are exactly that row. This proves the scheme isomorphism, not only a bijection of points. Differentiating the product equations gives (G.5). $\square$

Lemma G.6 (Generic smoothness of the signed components). *If $b \leq a$, a general point of each Σ_ε has corank one and the scheme $\mathcal{D}_L$ is smooth there of codimension b .*

Proof. The strict rank gap excludes every intermediate-rank annihilator at a general point of Σ_ε . If $b < a$, the full-rank incidence has smaller dimension and is excluded as well. If $a = b$ and full-rank annihilators occurred over a dense subset of Σ_ε , their fibres would have positive dimension: the projective annihilator space already contains its rank-two point, and its full-rank part is open. This would force $\dim \mathfrak{F}_L > m_X - a$, a contradiction. Lemma G.3 then implies corank one. Indeed, any further annihilator would give a projective line of annihilators, and hence either a forbidden rank or more than one rank-two point.

For the tangent calculation take the distinct evaluation points in (G.2). Let g be the binary quadratic vanishing at them. Then

$$E = \text{rad } H_\ell = gV_{n-2}, \quad U_\nu = W_\nu \oplus \mathbb{C}a_\nu, \quad W_\nu = U_\nu \cap E = gK_\nu,$$

where the K_ν are independent general $(c-1)$ -planes in V_{n-2} . The inequality determining this regime is precisely

$$a - b = L \binom{c}{2} - (2k - 5) \geq 0. \quad (\text{G.6})$$

Apply Theorem 1.1, already proved in Section 4, with $k' = k - 2$ and $c' = c - 1$. Its hypotheses hold and (G.6) is its saturation condition. Consequently, on a nonempty open subset of the signed incidence,

$$\sum_{\nu} W_{\nu}^2 = g^2 V_{2n-4}, \quad \dim \sum_{\nu} W_{\nu}^2 = p - 4. \quad (\text{G.7})$$

There is no inductive use of the present structural theorem.

Equation (G.5) on $W_{\nu} \times W_{\nu}$ first requires $\dot{\ell}$ to annihilate (G.7). Thus $\dot{\ell}$ has dimension $(p-1) - (p-4) = 3$. For each such $\dot{\ell}$, the remaining equations consist of the $c-1$ pairs (w, a_{ν}) and the pair (a_{ν}, a_{ν}) . Choose $b_{\nu} \in V_n$ with $H_{\dot{\ell}}(a_{\nu}, b_{\nu}) = 1$. The b_{ν} -coordinates of $\dot{U}_{\nu}(w)$ and $\dot{U}_{\nu}(a_{\nu})$ solve these c equations independently; the last coefficient is two, which is nonzero. Thus the tangent dimension of $\mathfrak{J}$, and by Lemma G.5 of $\mathcal{D}_L$, is

$$3 + \sum_{\nu} \{c(k-c) - c\} = m_X - b.$$

The local dimension is at least this number because the point lies on Σ_{ε} . Equality of dimension and tangent dimension proves regularity of the local ring and the stated smoothness. $\square$

Identity (G.6) is a deformation-theoretic interpretation of the change of regime. Saturation of the multiplication map after removing the two evaluation points is exactly the condition under which the rank-two component acquires its predicted smooth normal space. This is stronger than locating a minimum in an incidence count.

G.3. The whole scheme in expected codimension.

Proof of Theorem G.1. Suppose first that $a \leq b$. On a product of standard affine Grassmannian charts the coordinate ring is a polynomial ring in m_X variables over $\mathbb{C}$, the matrix of μ has size $p \times e$, and $I = I_p(\mu)$ is its maximal-minor ideal. On every chart meeting the failure scheme, Theorem 1.1 gives height $I \geq a$ and the maximal-minor height inequality gives height $I \leq e - p + 1 = a$. Since the ring is regular, $\text{grade } I = a$. The Eagon–Northcott grade criterion therefore provides a free resolution of R/I of length a ; in particular it is Cohen–Macaulay and unmixed of codimension a [8, 19]. This application uses the grade criterion, not a claim that the entries of μ are independent variables.

Every component consequently has dimension $m_X - a$. By Lemma G.2, its generic annihilator must have rank k , or, when $a = b$, rank two. In the first case it is dominated by the irreducible incidence $\mathfrak{F}_L$, so there is at most one such component. Lemma G.4 supplies a point outside every signed secant variety; purity and the strict rank gap force a full-rank component through the resulting non-secant part of the scheme. Hence this component exists and is Φ_L , with dimension $m_X - a$.

The full-rank incidence has this same dimension. Its generic fibre is therefore zero-dimensional, but this fibre is the full-rank open subset of the projective annihilator space. It follows that the generic multiplication corank is one. The generic fibre is a single point, and Lemma G.5 identifies a dense open subset of Φ_L with an open subset of the smooth variety $\mathfrak{F}_L$. This proves birationality and generic scheme smoothness there.

If $a < b$, the rank-two incidence has smaller dimension, leaving exactly Φ_L . If $a = b$, every distinct Σ_{ε} has the dimension of a component, and Lemma G.3 identifies all the rank-two components; the confluent rank-two base is a proper lower-dimensional subset. There are no others by the strict rank gap. Lemma G.6 proves generic scheme smoothness on them. A

Cohen–Macaulay scheme has no embedded associated primes; if it is generically reduced at every minimal prime, it is reduced. Applying this fact proves (i) and (ii), including integrality in (i).

If $b < a$, all maximal-dimensional components have dimension $m_X - b$. The full-rank incidence and all the intermediate-rank incidences are too small to dominate one. Lemma G.3 therefore gives exactly the listed components, their codimension and rationality, and Lemma G.6 gives their generic corank and smoothness. This proves (iii).

Finally apply the Thom–Porteous formula to the dual map $V_{2n}^* \otimes \mathcal{O} \rightarrow \mathcal{E}^*$, with source rank p , target rank e , and rank bound $p - 1$ [16, 11]. The expected codimension is a , as verified above, so the formula is $c_a(\mathcal{E}^*)$. The roots of $\text{Sym}^2 \mathcal{S}_\nu^*$ are $x_{\nu i} + x_{\nu j}$, $i \leq j$, which gives (G.4). Reducedness shows that in the equality case this is the sum of the component cycles with coefficient one. No excess-range use of the expected-codimension formula is made. $\square$

G.4. Real native strata and local conditioning.

Corollary G.7 (All signed components have positive native points). *For a native interpolation matrix J , every Σ_ε has a Zariski-dense set of real points admitting positive, globally separated native realizations. If (G.1) holds and $b \leq a$, some such points are smooth points of $\mathcal{D}_L$ of real local codimension b . Their pullbacks to loading space have the same local codimension.*

Proof. Choose real $x, y > D$ and sufficiently small $t > 0$. Every coordinate of $-J \text{ev}_x$ is positive, so both $-Jh_+$ and $-Jh_-$ are strictly positive. Independently for each contact, choose a general $(k - c)$ -plane T_ν containing $-Jh_{\varepsilon_\nu}$ and avoiding the vectors obtained from it by nonconstant coordinate sign changes. The fixed-space construction in Theorem D.1 then gives the required globally separated positive loading and $U_\nu \subset \ker h_{\varepsilon_\nu}$. These choices contain a real open subset of the signed incidence. Its birational parametrization is defined over $\mathbb{R}$ and is an isomorphism on dense open subsets. A nonzero real polynomial cannot vanish on this real open subset, so it meets the generic corank-one and residual-spanning opens of Lemma G.6. The real smooth codimension equals the complex codimension there. The loading submersions of Theorem D.1 transfer the assertion. In the finite sign test, s and $-s$ define the same unordered coordinate partition. Positivity specifies a Euclidean-open, Zariski-dense real image, not a positivity-defined Zariski-open set. $\square$

Corollary G.8 (Local small-singular-value exponent). *At any real smooth corank-one point furnished by Corollary G.7, there is a loading coordinate neighbourhood, disjoint from the other components, on which the least multiplication singular value γ_{mult} satisfies*

$$C_1 \text{dist}(A, |\mathcal{D}_L|_{\text{load}}) \leq \gamma_{\text{mult}}(A) \leq C_2 \text{dist}(A, |\mathcal{D}_L|_{\text{load}}). \quad (\text{G.8})$$

Here the neighbourhood has compact closure in a fixed frame chart and in a fixed full-rank loading sign chamber. On a sufficiently small coordinate product box, a probability law with density bounded above and below by positive constants satisfies

$$c_1 \epsilon^b \leq \Pr\{\gamma_{\text{mult}} \leq \epsilon\} \leq c_2 \epsilon^b \quad (0 < \epsilon < \epsilon_0). \quad (\text{G.9})$$

At a real smooth corank-one point of Φ_L in the range $a \leq b$, the same assertion holds with a in place of b .

Proof. Use the Schur-complement row of Lemma G.5. Bounded invertible matrix operations compare its Euclidean norm with γ_{mult} . At a smooth scheme point of codimension b , the differentials of its entries span a conormal space of dimension b . Select b independent entries as normal coordinates. The whole row vanishes on the failure submanifold; bounded derivatives give the upper distance estimate, and the selected coordinates give the lower estimate. The loading submersion and compact chart bounds preserve these comparisons. Integration in the b normal coordinates, with bounded Jacobian and density on the product box, gives (G.9). The full-rank case is identical with codimension a . $\square$

These are local probability laws near specified smooth strata. They do not assert a global tail for arbitrary random loadings or constants uniform in c, k, L . In particular, singular intersections and smaller components in the excess range are not covered by this local tube calculation.

Residual structure and the scope of the classification. The codimension formula leads to a description beyond the stable component theorem retained in the appendices. Theorem 7.1 removes its dimension restriction: in the original range $4 \leq c < k$ there are exactly five nonendpoint equality incidences. In each the plane contains the entire radical and is maximal isotropic in the quotient. Orientation monodromy, rather than a choice of one connected orthogonal family, determines their irreducibility.

The common mechanism is Theorem 5.1. For an annihilator ℓ , put $W_\nu = U_\nu \cap \text{rad } H_\ell$. The obstruction to solving the infinitesimal plane equations is exactly the restriction of ℓ to $\sum_\nu W_\nu^2$. This statement holds at every annihilator rank. At rank two it turns singularity of a corank-one point into failure of a smaller multiplication map. At the wall a general point of the residual determinant divisor is an ordinary double crossing between the nondegenerate and signed components (Theorem 8.9), with a logarithmic conditioning law (Corollary 8.10).

The reduced rank-two incidence has a normalization indexed by sign patterns. Its intersections with a common annihilator have a complete relative-Grassmannian description, while its full isotropy fibres carry an explicit square-zero nilradical (Theorem 8.1 and Proposition 8.2). These are different scheme statements. A reduced global scheme need not have reduced fibres. For higher symmetric powers the analogous labels form dihedral phase orbits, rather than a simple power of the number of hyperplanes (Theorems 9.2 and 9.3).

The all-dimension component theorem is proved from the codimension formula and the residual tangent sequence; it does not rely on the stable theorem retained in the appendices. The full boundary, count-comparison and finite-sample arguments are also retained there, so that their statistical assumptions do not enter the geometric proof.

Our use of secant obstructions is situated in the classical failure-locus literature, including Ballico's higher-order failure-locus paper [4], the complete-series work of Green–Lazarsfeld [14], and the failure-cycle study [5]. The statements proved here specify the varying subseries, the absence of cross-contact products, the exceptional incidences, and the local scheme strata. The accompanying literature audit records which primary texts were checked and which theorem-level comparisons remain unverified. For [4], the first page has now been inspected in addition to the bibliographic record; this is not a completed theorem-by-theorem comparison. Section 11 specifies this distinction.

For clarity, the Hankel shape identification used below is an equality of ideals in the same polynomial ring, not just a dimension analogy. Writing $S = \mathbb{C}[z_0, \dots, z_{2k-2}]$ and $H_{u,v} = (z_{i+j})_{0 \leq i < u, 0 \leq j < v}$ for $u + v = 2k$, the classical identity recalled in [7, Section 1] is

$$I_{r+1}(H_{k,k}) = I_{r+1}(H_{r+1, 2k-r-1}) \subset S \quad (1 \leq r < k).$$

The latter is a maximal-minor Hankel ideal with quotient of affine dimension $2r$. Projectivization gives $2r - 1$; removing the smaller rank locus gives the exact-rank stratum used in the incidence proof.

APPENDIX H. THE PHYSICAL BOUNDARY EXPERIMENT

The algebraic extension through $v = 0$ in Section C is useful for differentiation, but it does not remove the physical restriction $v \geq 0$. We now identify the actual local experiment. The argument is the finite-cell likelihood expansion underlying constrained Gaussian limits [13, 26]; all normalizations and the synchronized profile are specified here.

Let $u \in \mathbb{R}^{k+1}$ denote the nuisance log-odds and root-sum coordinates. Observe independent categorical samples with $n_j/N \rightarrow \lambda_j > 0$, and let $\Delta_N = (\Delta_{N,u}, \Delta_{N,v})$ be the score at $(u, v) = (0, 0)$ divided by $\sqrt{N}$. Partition

$$I = V^\top \text{diag}(\lambda_j \kappa_j) V = \begin{pmatrix} I_{uu} & I_{uv} \\ I_{vu} & I_{vv} \end{pmatrix}, \quad Q = I_{vv} - I_{vu} I_{uu}^{-1} I_{uv} = R^{-1}.$$

Theorem H.1 (Cone likelihood and synchronized profile). *Fix a compact set $K \subset \mathbb{R}^{k+1} \times \mathbb{R}_{\geq 0}^k$, with the Euclidean product norm. For $(a, h) \in K$, the physical local alternatives $u = a/\sqrt{N}$, $v = h/\sqrt{N}$ satisfy, uniformly on compact index sets,*

$$\log \frac{dP_{N,a,h}}{dP_{N,0,0}} = (a, h)^\top \Delta_N - \frac{1}{2}(a, h)^\top I(a, h) + o_P(1). \quad (\text{H.1})$$

Here uniformity means that the supremum over K of the absolute remainder converges to zero in $P_{N,0,0}$ -probability. Under the corresponding alternatives,

$$Y_N = R(\Delta_{N,v} - I_{vu} I_{uu}^{-1} \Delta_{N,u}) \implies N(h, R). \quad (\text{H.2})$$

After quotienting out unrestricted nuisance translation, this is a Gaussian shift with parameter restricted to the physical cone. For synchronized alternatives $h = q_A(\zeta)$, the profiled limiting log likelihood, relative to its value at $h = 0$, is

$$\frac{1}{2} Y^\top R^{-1} Y - \frac{1}{2} C_R(Y), \quad C_R(Y) = \inf_{\zeta} (Y - q_A(\zeta))^\top R^{-1} (Y - q_A(\zeta)). \quad (\text{H.3})$$

The common physical root displacement has scale $N^{-1/4} \zeta$.

Proof. The finite set of clock probabilities is strictly positive on a two-sided neighbourhood of the baseline. Its derivatives through order three are bounded on a smaller neighbourhood. At $u = a/\sqrt{N}$, $v = h/\sqrt{N}$, expand each cell log probability. Normalization makes the mean linear term vanish, the expectation of the Hessian is minus the score covariance, and the summed cubic remainder is $O(N^{-1/2})$ on compact index sets. The centred Hessian fluctuation is $o_P(1)$. The finite-dimensional independent-group central limit theorem gives $\Delta_N \Rightarrow N(0, I)$ at the baseline and proves (H.1). Under the local alternatives its mean tends to $I(a, h)$ and its covariance to I , either by the same cell calculation or the likelihood change of measure. The efficient score in (H.2) has mean Qh and covariance Q , proving the assertion after multiplication by R .

In the limiting score experiment the nuisance block is independent of the efficient block. Its mean is $I_{uu}a + I_{uv}h$, which is an unrestricted translation as a varies. Discarding this nuisance translation yields (H.2); this is a quotient, not a claim that an unknown nuisance parameter can be simulated without information. Profiling the efficient Gaussian likelihood over $h = q_A(\zeta)$ and completing the square gives (H.3). The map q_A is proper for full-rank A , so the infimum is attained and its optimizing ζ are bounded whenever Y is bounded. Uniform LAN therefore also gives the iterated compact-localization limit of profiled likelihoods: first restrict (a, ζ) to a fixed compact set, pass to the limit, and then exhaust the limiting experiment by such sets. No assertion about remote maximizers of an unrestricted finite-sample likelihood is required. Finally $q_A(N^{-1/4} \zeta) = N^{-1/2} q_A(\zeta)$, proving the displacement scale. $\square$

The unrestricted vector Y is a statistic in the limiting experiment, not the constrained maximum likelihood estimator. The latter is a metric projection onto $\mathbb{R}_{\geq 0}^k$, or onto $q_A(\mathbb{R}^d)$ in the synchronized experiment. At the boundary it need not be normal; for the nonconvex synchronized image the projection can be set-valued. The distance value in (H.3) is nevertheless unambiguous.

Corollary H.2 (Sharp root-amplitude scale on a physical ray). *Fix $a_0 \in \mathbb{R}^d \setminus \{0\}$ and $v_0 = q_A(a_0) \neq 0$. On the one-sided submodel with physical displacement ta_0 , $0 \leq t \leq TN^{-1/4}$, and*

nuisance $u = a/\sqrt{N}$ in a fixed bounded set, the minimax squared risk for estimating t is of order $N^{-1/2}$, for each fixed $T > 0$. Thus the amplitude scale is $N^{-1/4}$, whereas its squared amplitude is on the $N^{-1/2}$ parameter scale.

Proof. The variance parameter is $v = t^2 v_0$. Set $J_0 = v_0^\top Q v_0 > 0$ and use the baseline efficient score to define

$$\hat{b}_N = \frac{v_0^\top Q Y_N}{\sqrt{N} J_0}, \quad \hat{t}_N = \sqrt{\max\{\hat{b}_N, 0\}}.$$

For completeness, let s_{je} be the baseline score of cell e in group j . There are finitely many such bounded vectors. Taylor expansion of the cell probabilities, rather than convergence in distribution, gives

$$\sup_{(a,h) \in K} \|\mathbb{E}_{a,h} \Delta_N - I_N(a,h)\| \leq C_K N^{-1/2}, \quad \sup_{(a,h) \in K} \|\text{Var}_{a,h} \Delta_N\|_{\text{op}} \leq C_K,$$

where I_N uses the actual allocations n_j/N and $I_N \rightarrow I$. The first bound sums the bounded second-order probability remainders; the second sums variances of independent bounded scores. The efficient linear map consequently has mean $h + o(1)$ and bounded covariance, uniformly on K . With $h = \sqrt{N} t^2 v_0$, the definition of $\hat{b}_N$ now gives bias $o(N^{-1/2})$ and variance $O(N^{-1})$, and hence $\mathbb{E}(\hat{b}_N - t^2)^2 \leq C/N$. This argument supplies the moment control separately from LAN. For $b \geq 0$, $(\sqrt{\max\{x, 0\}} - \sqrt{b})^2 \leq |x - b|$. Cauchy–Schwarz therefore gives the upper bound $C'/\sqrt{N}$.

For the lower bound take the two physical alternatives $t = 0$ and $t = \tau N^{-1/4}$ with the same nuisance, where $0 < \tau \leq T$ is sufficiently small. Their likelihood divergence is bounded by a constant times τ^4 , by (H.1) or its finite-cell Taylor proof. Their total variation is consequently bounded away from one. The two-point testing bound for squared loss gives a positive constant times the squared separation $\tau^2 N^{-1/2}$. The claim concerns this labelled nonnegative ray; it does not assert identifiability of the sign of a general loading vector. $\square$

APPENDIX I. DIRECT COUNT-TO-CONTACT EXPERIMENT COMPARISON

Here the observations differ from fixed-count categorical marks. Let the mark laws, hence κ_j , be known, and observe independent exposures

$$N_j \sim \text{Pois}(N\lambda_j), \quad j = 1, \dots, m.$$

This is the known-mark-law subexperiment of an observed-exposure protocol. Fix an interior centre $\lambda^0 > 0$. Write D for the derivative of the native moment functional at that centre:

$$D_{\cdot j} = -\frac{\rho_j}{(\lambda_j^0)^2 \kappa_j} w_{2n}(T_j), \quad \Lambda = \text{diag}(\lambda_j^0), \quad \Sigma = D\Lambda D^\top > 0.$$

The positivity follows from Vandermonde rank. For a local moment index $h \in \mathbb{R}^p$ in a fixed compact ball, define the least-favourable target submodel

$$\lambda_N(h) = \lambda^0 + N^{-1/2} B_* h, \quad B_* = \Lambda D^\top \Sigma^{-1}. \quad (\text{I.1})$$

It has $\theta(\lambda_N(h)) = \theta(\lambda^0) + h/\sqrt{N} + O(N^{-1})$. Its local information is Σ^{-1} , since $DB_* = I_p$ and $B_*^\top \Lambda^{-1} B_* = \Sigma^{-1}$.

Let $\mathcal{L}$ be the known normal-compression operator, possibly rank deficient, and put $r = \text{rank } \mathcal{L}$. Choose an orthonormal coordinate frame F for its range. Generate independent $U_j \sim \text{Unif}(-1/2, 1/2)$, independently of the counts, and retain only

$$T_N = F^\top \mathcal{L} D \frac{N - N\lambda^0 + U}{\sqrt{N}}. \quad (\text{I.2})$$

The randomization law is part of the protocol; its realized auxiliary values are not retained. This statistic is formed directly by fixed linear weights on the counts. No estimator of θ is

constructed first. The small randomization prevents lattice encoding in this protocol: without such randomization or a genuine finite-precision coarsening, an exact real-valued linear encoding of an integer vector can be injective even when its real matrix has deficient rank. Weak convergence alone would not exclude that lattice encoding.

For experiments $\mathcal{A} = (P_h)$ and $\mathcal{B} = (Q_h)$ on the same parameter set, use

$$\delta(\mathcal{A}, \mathcal{B}) = \inf_K \sup_h \|KP_h - Q_h\|_{\text{TV}}, \quad \Delta(\mathcal{A}, \mathcal{B}) = \max\{\delta(\mathcal{A}, \mathcal{B}), \delta(\mathcal{B}, \mathcal{A})\}.$$

The infimum is over parameter-independent Markov kernels and $\|P - Q\|_{\text{TV}} = \sup_A |P(A) - Q(A)|$, equal to one half the L^1 distance for densities. All comparisons below use a fixed compact local parameter set.

Lemma I.1 (Quantitative jittered Poisson comparison). *Uniformly for h in a fixed compact set, the distribution of*

$$X_N^\circ = (N - N\lambda^0 + U)/\sqrt{N}$$

*under (I.1) is within $CN^{-1/5}$ in total variation of $N(B_*h, \Lambda)$. The constant depends on the fixed dimension, positive exposure bounds, and the compact index set. The full jittered vector and the raw counts are exactly equivalent experiments, by independent jittering and rounding. The raw local target experiment is therefore asymptotically equivalent to $Z \sim N(h, \Sigma)$.*

Proof. We give the density estimate, including the lattice interpolation and tails. In one coordinate write $K \sim \text{Pois}(N\lambda)$, with $0 < \lambda_- \leq \lambda \leq \lambda_+$, and centre first at its own mean. The density $f_{N,\lambda}$ of $(K - N\lambda + U)/\sqrt{N}$ is constant, equal to $\sqrt{N} \Pr(K = j)$, on each cell of width $N^{-1/2}$. Put $z_j = (j - N\lambda)/\sqrt{N}$ and let φ_λ be the centred normal density with variance λ . Stirling's formula with remainder

$$\log j! = (j + \tfrac{1}{2}) \log j - j + \tfrac{1}{2} \log(2\pi) + r_j, \quad 0 < r_j < 1/(12j),$$

and the Taylor formula for $\log(1 + u)$ give, uniformly for $|z_j| \leq 2N^{1/10}$ and all sufficiently large N ,

$$\left| \log \frac{\sqrt{N} \Pr(K = j)}{\varphi_\lambda(z_j)} \right| \leq C\{N^{-1/2}(1 + |z_j|^3) + N^{-1}(1 + |z_j|^4)\}.$$

Indeed $j = N\lambda(1 + z_j/(\sqrt{N}\lambda))$, the quadratic term is $-z_j^2/(2\lambda)$, and the remaining logarithmic, cubic, and Stirling terms have the displayed bounds. Within a cell, $|\log \varphi_\lambda(z) - \log \varphi_\lambda(z_j)| \leq CN^{-1/2}(1 + |z_j|)$. Consequently on $|z| \leq R_N = N^{1/10}$ the density ratio differs from one by at most $CN^{-1/5}$. Integrating this relative bound costs at most $CN^{-1/5}$. Outside this region,

$$\mathbb{E}\{(K - N\lambda + U)/\sqrt{N}\}^2 = \lambda + (12N)^{-1},$$

so Chebyshev's inequality bounds the step-density tail by CR_N^{-2} ; the same bound holds for the Gaussian tail. Thus

$$\|f_{N,\lambda} - \varphi_\lambda\|_{L^1} \leq CN^{-1/5}. \quad (\text{I.3})$$

Translation to the centre $N\lambda^0$ changes neither the bound nor its uniformity. The finitely many small N are absorbed by increasing C .

For m independent coordinates, the total variation distance between products is at most the sum of the coordinate distances. This proves the same order for covariance $\text{diag } \lambda_N(h)$. To replace it by Λ , differentiate the normal density with respect to each variance on the fixed positive compact interval. The L^1 norms of these derivatives are uniformly bounded (they are the normal expectations of the absolute variance scores). Since $\|\lambda_N(h) - \lambda^0\| = O(N^{-1/2})$ uniformly, covariance replacement costs $O(N^{-1/2})$. This proves the asserted quantitative comparison.

Rounding $\sqrt{N}X_N^\circ + N\lambda^0$ to the nearest integer recovers each count almost surely, while independent jittering gives the reverse Markov kernel. On arbitrary Gaussian inputs the rounding

kernel clips negative integers to zero, so it is defined on the entire observation space. In the limiting normal vector $X \sim N(B_*h, \Lambda)$, the statistic $Z = DX$ has law $N(h, \Sigma)$ and is sufficient: conditional simulation is

$$X = B_*Z + E, \quad E \sim N(0, \Lambda - \Lambda D^\top \Sigma^{-1} D \Lambda),$$

with E independent and its law independent of h . These kernels, the total variation approximation, and contraction of total variation under kernels prove the experiment comparison uniformly on the compact parameter set. In particular this proves the asserted local Le Cam comparison, not merely a central limit theorem. $\square$

Theorem I.2 (Known-mark local information retained by contact scores). *With known mark laws and the preassigned centre and target submodel (I.1), the experiment retaining (I.2) is, uniformly on compact local parameter balls, asymptotically equivalent to*

$$T \sim N(F^\top \mathcal{L}h, F^\top \mathcal{L} \Sigma \mathcal{L}^\top F). \quad (\text{I.4})$$

Its limiting information matrix for h is

$$I_{\text{contact}} = \mathcal{L}^\top (\mathcal{L} \Sigma \mathcal{L}^\top)^\dagger \mathcal{L} = \Sigma^{-1/2} P_{\text{ran}(\Sigma^{1/2} \mathcal{L}^\top)} \Sigma^{-1/2}. \quad (\text{I.5})$$

It has rank r and kernel $\ker \mathcal{L}$. The contact-score reduction is locally asymptotically equivalent to the raw target subexperiment if and only if $\mathcal{L}$ is injective. Hence it retains all local moment information exactly when the within-contact polynomial products span V_{2n} .

Proof. Apply the fixed linear map $F^\top \mathcal{L} D$ to the total variation approximation in Lemma I.1. Contraction gives convergence in total variation to (I.4), whose covariance is positive definite in its r coordinates. Identity kernels therefore compare these experiments in both directions with error tending to zero. The usual normal-shift likelihood gives the first expression in (I.5). For $M = \mathcal{L} \Sigma^{1/2}$, the identity $M^\top (M M^\top)^\dagger M = P_{\text{ran } M^\top}$ gives the second. Its rank and kernel follow immediately.

If $\mathcal{L}$ is injective, Z is recovered from $\mathcal{L}Z$ by a left inverse. Thus (I.4) is equivalent to $N(h, \Sigma)$, and Lemma I.1 gives equivalence to the raw target subexperiment. Conversely choose distinct h_0, h_1 in a local parameter ball with $h_1 - h_0 \in \ker \mathcal{L}$. Their limiting contact distributions coincide, whereas $N(h_0, \Sigma)$ and $N(h_1, \Sigma)$ have strictly positive total variation distance. By the triangle inequality, any kernel purporting to reconstruct both raw limit laws has error at least half that distance. The same positive lower obstruction holds asymptotically by the uniform comparisons. The polynomial criterion is Theorem B.2. $\square$

The uncentred contact block is $B_\nu(\theta) = \mathcal{L}_\nu \theta$. Its local fluctuation is precisely the linear type appearing in (I.2). Passing to inverse-Hessian contact scores applies the fixed invertible block transformation $E_\nu \mapsto -B_{\nu,0}^{-1} E_\nu B_{\nu,0}^{-1}$, so it preserves the information statement. These are correlated raw-data score statistics, not independently noisy observations of distance values.

The centre λ^0 and the local submodel (I.1) are specified before observing the data. The theorem does not claim global sufficiency when both exposure and mark laws are unknown, or that a plug-in estimate of the centre automatically gives the same comparison. It supplies a genuine direct reduction and a complete local information-loss calculation in the stated full-dimensional target experiment. When $\mathcal{L}$ is injective, exact range-compatible score encoding does not introduce a γ^{-2} statistical variance penalty; independent measurement errors and numerical perturbations have the distinct stability behaviour of Theorem E.2 and Appendix M.

I.1. Deterministic finite precision and the full exposure tangent model. The auxiliary randomization above prevents an exact real-valued encoding of the integer count vector. A finite-precision observation provides another, deterministic, way to remove that encoding.

Theorem I.3 (Deterministic quantized score reduction). *Keep the known-mark target submodel and put $C = F^\top \mathcal{L}D$. Suppose $r \geq 1$. For a deterministic mesh $\eta_N > 0$, retain only*

$$Q_N = \left\lfloor \eta_N^{-1} C(N - N\lambda^0) / \sqrt{N} \right\rfloor \in \mathbb{Z}^r,$$

with the floor taken coordinatewise. If $\eta_N \rightarrow 0$ and $\sqrt{N}\eta_N \rightarrow \infty$, this experiment is asymptotically equivalent to (I.4). More precisely, for $0 < \eta_N \leq 1$,

$$\Delta(Q_N, \mathcal{T}) \leq C_0 \{N^{-1/5} + (\sqrt{N}\eta_N)^{-1} + \eta_N\}. \quad (\text{I.6})$$

In particular $\eta_N = N^{-1/4}$ is admissible. The information and equivalence criterion of Theorem I.2 therefore hold for this deterministic protocol as well.

Proof. Couple the unjittered score $S_N = C(N - N\lambda^0) / \sqrt{N}$ to $S_N^\circ = CX_N^\circ$ with the same counts. Their coordinatewise difference is at most $d_N = C_1 / \sqrt{N}$, deterministically. The two quantizers can differ only if a coordinate of S_N° lies within d_N of the grid $\eta_N \mathbb{Z}$. By Lemma I.1, its joint law is within $CN^{-1/5}$ of the nonsingular normal law in (I.4).

For a univariate normal density f with variance in a fixed positive interval, the grid-boundary probability is at most Cd_N / η_N if $d_N \leq \eta_N / 2$. To see this uniformly in its mean, sum the supremum of f on each grid interval and use

$$\eta_N \sum_{j \in \mathbb{Z}} \sup_{[j\eta_N, (j+1)\eta_N]} f \leq 1 + \eta_N \int_{\mathbb{R}} |f'(x)| dx.$$

There are at most two boundary pieces per interval. A union bound over the r coordinates proves $\Pr\{\lfloor S_N / \eta_N \rfloor \neq \lfloor S_N^\circ / \eta_N \rfloor\} \leq C\{N^{-1/5} + d_N / \eta_N\}$. When $d_N > \eta_N / 2$ the asserted bound follows by increasing its constant. Quantization contracts total variation, so Q_N is within the same bound of a quantized Gaussian, uniformly in the local index.

The kernel adding an independent uniform point in the reported grid cell reconstructs the cell-average Gaussian density. Its L^1 distance from the Gaussian is at most $C_r \eta_N \int \|\nabla f(x)\| dx \leq C\eta_N$, by the integral mean-value inequality on cubes. The reverse kernel is deterministic quantization. Combining these kernels proves both deficiencies in (I.6). Randomization is used only by this comparison kernel, not by the observed statistic Q_N . The case $r = 0$ is the one-point experiment and needs no mesh. $\square$

An actual finite alphabet may be obtained by an overflow symbol outside a ball of radius $N^{1/10}$. Uniform second-moment bounds add only $O(N^{-1/5})$ to the comparison. Thus infinite digital precision is not hidden in the conclusion.

The least-favourable submodel has a precise relationship to the full exposure tangent experiment, rather than being an intrinsic physical distance observation.

Proposition I.4 (Full exposure model and nuisance quotient). *For known marks and arbitrary bounded exposure perturbations $g \in \mathbb{R}^m$, set $\lambda_N(g) = \lambda^0 + g / \sqrt{N}$. The raw count experiment converges in Le Cam distance, uniformly on compact sets, to $X \sim N(g, \Lambda)$. In this Gaussian limit the transformation*

$$Z = DX, \quad E = (I_m - B_* D)X$$

has independent components, with means $h = Dg$ and $\xi = (I_m - B_ D)g \in \ker D$. Quotienting the limit experiment by translations of this unrestricted nuisance gives exactly $Z \sim N(h, \Sigma)$; its contact compression has information (I.5). The least-favourable submodel is the section $\xi = 0$.*

Proof. The density proof of Lemma I.1 applies unchanged to g in a compact set; jittering and rounding give both kernels. Since $DB_* = I_p$, $g = B_* h + \xi$ is a bijective linear parametrization by $h \in \mathbb{R}^p$ and $\xi \in \ker D$. The cross-covariance is

$$D\Lambda(I_m - B_* D)^\top = D\Lambda - \Sigma B_*^\top = 0.$$

The Gaussian components are therefore independent, the covariance of Z is Σ , and the law of $E - \xi$ does not depend on either parameter. Translation by $\ker D$ acts transitively on its nuisance means and leaves Z fixed. This proves the quotient assertion. It does not supply a kernel simulating an unknown ξ from Z , nor assert equivalence to the full experiment with that nuisance retained. The compact-set comparisons are made before this unrestricted Gaussian translation quotient; a compact nuisance set is not silently replaced by a translation group. $\square$

The centre remains an experimental input. A simple sufficient condition for replacing it by an independent pilot estimate is $\sqrt{N}\|\hat{\lambda}^0 - \lambda^0\| \rightarrow 0$ in probability, together with a fixed positive compact range for the pilot and continuity of the weights. On events where the scaled error is small, the centring shift is $o(1)$, the weights and Gaussian covariances converge, and the same uniform density bounds and mixture kernels apply. A root- M pilot gives this sufficient condition when $M/N \rightarrow \infty$. This is a sufficient, not an optimal, pilot regime. An ordinary pilot of size comparable to N does not meet it: its random centring error must then be included in the limit experiment. Nothing here asserts automatic plug-in sufficiency for unknown marks or an unrecorded pilot.

APPENDIX J. UNIFORM CONSTANTS AND THE NUISANCE INVARIANT

We record explicit constants for the Poisson step-density estimate in Lemma I.1. They concern the same experiment and do not replace its finite-precision protocol.

Proposition J.1 (An explicit central-region threshold). *Let $K \sim \text{Pois}(N\lambda)$, let $U \sim \text{Unif}(-1/2, 1/2)$ independently, and suppose $0 < \lambda_- \leq \lambda \leq \lambda_+$. Let $f_{N,\lambda}$ be the density of $(K - N\lambda + U)/\sqrt{N}$, and let φ_λ be the centred Gaussian density of variance λ . Set*

$$C_0 = 10(1 + \lambda_-^{-1} + \lambda_-^{-2}), \quad N_0 = \left\lceil \max\{1, 2/\lambda_-, (4/\lambda_-)^{5/2}, C_0^5\} \right\rceil.$$

For every integer $N \geq N_0$,

$$\|f_{N,\lambda} - \varphi_\lambda\|_1 \leq (3C_0 + 2\lambda_+ + 1)N^{-1/5}. \quad (\text{J.1})$$

Proof. Write $z_j = (j - N\lambda)/\sqrt{N}$ and $u_j = z_j/(\sqrt{N}\lambda)$. If $|z_j| \leq 2N^{1/10}$ and $N \geq N_0$, then $|u_j| \leq 1/2$ and $j \geq N\lambda_-/2 \geq 1$. For $F(u) = (1+u)\log(1+u) - u$, Taylor's theorem gives $|F(u) - u^2/2| \leq (2/3)|u|^3$ on this interval. Stirling's remainder bound $0 < r_j < 1/(12j)$ therefore yields

$$\left| \log \frac{\sqrt{N} \Pr(K = j)}{\varphi_\lambda(z_j)} \right| \leq \frac{2|z_j|^3}{3\sqrt{N}\lambda_-^2} + \frac{|z_j|}{\sqrt{N}\lambda_-} + \frac{1}{6N\lambda_-}.$$

Within a cell of half-width $1/(2\sqrt{N})$ the change of the Gaussian log density is at most $|z_j|/(2\sqrt{N}\lambda_-) + 1/(8N\lambda_-)$. On $|z| \leq N^{1/10}$ its cell centre satisfies $|z_j| \leq 2N^{1/10}$; the sum of these bounds is at most $C_0N^{-1/5} \leq 1$. Thus the density ratio differs from one by at most $3C_0N^{-1/5}$, which bounds the central integral. The second moments of the step density and the Gaussian are $\lambda + 1/(12N)$ and λ . Chebyshev's inequality bounds the sum of their outside integrals by $(2\lambda_+ + 1)N^{-1/5}$. Adding proves the result. $\square$

For the local exposure model, let $H = \sup_h \|B_*h\|_\infty$ on the prescribed compact index set. Taking also $N \geq (2H/\min_j \lambda_j^0)^2$ puts every exposure in $[\min_j \lambda_j^0/2, \max_j \lambda_j^0 + \min_j \lambda_j^0/2]$. Equation (J.1) then supplies the uniform coordinate constants; product total variation is at most the sum of the coordinate total variations. The covariance replacement and quantization bounds in the preceding appendix remain unchanged.

Definition J.2 (Nuisance translation invariant). In the limiting Gaussian experiment of Theorem H.1, use the independent nuisance score and efficient observation as coordinates. Let $\mathbb{R}^{k+1}$ act by translation on the nuisance score coordinate and trivially on the efficient observation Y . The invariant observation is the measurable maximal invariant of this action, namely Y .

Thus the phrase “quotienting out nuisance translation” in that theorem means passage to this explicitly defined invariant observation. Independence of the Gaussian blocks follows from their zero cross covariance, and arbitrary nuisance translation removes any information in the first block about the retained parameter for invariant procedures. It is not an equivalence of the full experiment, in which the nuisance itself remains an unknown parameter, to Y alone.

APPENDIX K. CLASSICAL MAXIMAL RANK AND MONOMIAL COMPARISON

The proof of the global codimension theorem does not invoke curve postulation. For comparison, the complete classical statement used for a single general space includes the injective range as well.

Proposition K.1 (Classical rational maximal rank). *For $4 \leq c \leq n+1$, a general c -plane $U \subset H^0(\mathbb{P}^1, \mathcal{O}(n))$ satisfies*

$$\text{rank}(\text{Sym}^2 U \longrightarrow H^0(\mathbb{P}^1, \mathcal{O}(2n))) = \min\left\{\binom{c+1}{2}, 2n+1\right\}.$$

The full-rank locus is a nonempty Zariski-open set defined over $\mathbb{R}$.

Precise classical reduction. Apply the maximal-rank theorem in [20, Theorem 1.2] to a general Brill–Noether curve with genus $g = 0$, degree n , ambient projective dimension $r = c - 1 \geq 3$, and restriction degree two. Its Brill–Noether number is $(r+1)(n-r) \geq 0$. A general c -dimensional subseries of $H^0(\mathbb{P}^1, \mathcal{O}(n))$ is base-point-free and embeds $\mathbb{P}^1$ in $\mathbb{P}^r$: its projection centre avoids the secant and tangent locus of the rational normal curve because $r \geq 3$. The space of such subseries is irreducible and parametrizes the general rational Brill–Noether map, up to the choice of a basis and source coordinates. Restriction of quadrics has source $\text{Sym}^2 U$ and target $H^0(\mathbb{P}^1, \mathcal{O}(2n))$, exactly the multiplication map above. This specializes the general theorem to the classical nonspecial rational range of [2, 3]. The complete series is immediate from monomials. Nonvanishing maximal minors in real matrix charts gives the real assertion. This proof attributes, rather than reproves, the classical injective-range theorem. $\square$

For a monomial space $U_S = \text{span}\{t^a : a \in S\}$,

$$U_S^2 = \mathcal{P}_{2n} \iff S + S = \{0, \dots, 2n\}. \quad (\text{K.1})$$

This is the restricted additive 2-basis problem, distinct from an unrestricted polynomial subspace [27, 17, 18].

Proposition K.2 (Strict monomial obstruction and constructive bases). *If $n \geq 3$ and (K.1) holds, then $\binom{|S|+1}{2} \geq 2n+2$. Thus no monomial normal space attains the native optimum at $(d, k) = (3, 8)$ or $(5, 11)$.*

For $a \geq 2$, $n \geq 2a$, write $n = ba + r$, $0 \leq r < a$. The set

$$S = [0, a-1] \cup \{a, 2a, \dots, ba\} \cup [n-a+1, n]$$

has $S+S = [0, 2n]$ and $|S| \leq 2a + \lfloor n/a \rfloor - 1$. In particular such bases exist with $|S| \leq 2\sqrt{2n} + O(1)$.

Proof. Representing $0, 1, 2n-1, 2n$ forces $0, 1, n-1, n \in S$. The distinct unordered pairs $(0, n)$ and $(1, n-1)$ coincide in sum, so at least one of the $\binom{|S|+1}{2}$ products is redundant. At the displayed native optima the pair count is exactly $2n+1$, respectively 15 and 21, making monomial recovery impossible.

For the construction, adding the multiples $0, a, \dots, ba$ to $[0, a-1]$ covers $[0, ba+a-1] \supset [0, n]$. Adding them to $[n-a+1, n]$ covers $[n-a+1, 2n-r]$. Adding the upper interval to itself covers $[2n-2a+2, 2n]$, which overlaps because $r \leq a-1$. The last multiple belongs to the upper interval, giving the cardinality bound. Choose an integer nearest $\sqrt{n/2}$ for the asymptotic bound, handling the finitely many smaller values by the full interval. $\square$

For $k = 3d+2$ one has the explicit family $S = [0, d] \cup [2d+1, 3d+1]$, with complementary exponents $E = [d+1, 2d]$. The three sum intervals are $[0, 2d]$, $[2d+1, 4d+1]$ and $[4d+2, 6d+2]$. Each column of the interpolation matrix J has one strict coordinatewise sign, as follows by expanding $d_i \prod_{a \neq i} (t - r_a)$ with positive roots. In particular a signed Je_{d+1} is positive. Taking $T = J \text{span}\{e_a : a \in E\}$ realizes the monomial complement locally by Theorem D.1; generic small loading perturbations give globally separated full-product realizations. The unrestricted fibre dimension for this family is $5d(d+1)/2$. The smaller sets $\{0, 1, 3, 5, 6\}$ and $\{0, 1, 2, 5, 8, 11, 12, 13\}$ also satisfy (K.1); no optimal restricted-basis constant is claimed.

For a nonorthonormal normal frame $W = ZO$, set $K = W^\top W$ and $G = \frac{1}{2}W^\top D^2 C_R W$. The exact dual readout is

$$KG^{-1}K = W^\top RW. \quad (\text{K.2})$$

Indeed $K = O^\top O$ and $G = O^\top B^{-1}O$, and direct inversion proves the identity. If W has columns $J^{-\top}e_a$ for $a \in S$, its readout entries are θ_{a+b} . Dropping the Gram factors in (K.2) would give an incorrect normalization.

APPENDIX L. FINITE CALIBRATED SAMPLES AND OBSERVED EXPOSURES

Keep the roots, clocks and calibrated tables fixed and separated. Put $\delta = M_1 - M_2 \neq 0$, assume $\pi_j \in [\varepsilon, 1-\varepsilon]$ with $0 < \varepsilon < 1/2$, and define

$$\kappa(u) = u^2(1-u)^2 \sum_{a=1}^4 \frac{\delta_a^2}{M_{2a} + u\delta_a}, \quad g(u) = 1/\kappa(u).$$

On this interval g is Lipschitz with a finite constant L_g . We distinguish a chosen target-design weight λ from the realized sample allocation n_j/N . The theorem below estimates the metric indexed by the chosen known λ . It estimates the realized normalized sample information only when those weights are chosen as n_j/N , and the limiting information when $n_j/N \rightarrow \lambda_j$.

Theorem L.1 (Fixed-design categorical recovery). *Observe $n_j \geq 1$ independent marks at clock j with law p_j , independently across groups. For empirical cell frequencies $\bar{p}_j$, set*

$$\hat{\pi}_j = \text{proj}_{[\varepsilon, 1-\varepsilon]} \frac{\langle \bar{p}_j - M_2, \delta \rangle}{\|\delta\|_2^2}, \quad W_{\cdot j} = \frac{\rho_j}{\lambda_j} w_{2n}(T_j), \quad \hat{\theta} = W(g(\hat{\pi}_j))_j.$$

Then $\hat{\theta} \in \Omega$ and

$$\mathbb{E}\|\hat{\theta} - \theta\|_2^2 \leq \frac{\|W\|_{\text{op}}^2 L_g^2}{\|\delta\|_2^2} \sum_j n_j^{-1}. \quad (\text{L.1})$$

On compact interior design sets, the induced estimates of R and the inverse normal blocks have squared risk $O(\sum_j n_j^{-1})$. If $\mathcal{L}$ is injective, inversion of those induced jets returns $\hat{\theta}$ exactly.

Proof. Before clipping, the estimate of π_j is unbiased. For a one-hot cell vector X , $\text{Var}\langle X, \delta \rangle \leq \|\delta\|_2^2$. Projection cannot increase distance to the true point in the interval, so $\mathbb{E}|\hat{\pi}_j - \pi_j|^2 \leq (n_j \|\delta\|_2^2)^{-1}$. Apply the Lipschitz bound for g and the operator norm of W . All estimated moment weights are positive. On compact design sets they lie in a fixed positive box; the moment image is a compact convex subset of Ω . The map to R and positive block inversion are Lipschitz there. Finally the induced inverse jets satisfy $\hat{G}_\nu^{-1} = \mathcal{L}_\nu \hat{\theta}$, proving the last assertion. $\square$

The bound allows the π_j to be coupled by the polynomial model and does not assert a full-dimensional fixed-design likelihood. Its constants need not stay bounded when roots or clocks collide, or probabilities approach the boundary.

Theorem L.2 (An additional observed-exposure protocol). *Observe independent counts $N_j \sim \text{Pois}(N\lambda_j)$ and, conditionally, N_j categorical marks of law p_j . Suppose $0 < \lambda_- \leq \lambda_j \leq \lambda_+$. Clip N_j/N to this interval, use the preceding clipped mark estimate, and use any value in $[\varepsilon, 1 - \varepsilon]$ when $N_j = 0$. Substitution into (C.2) gives uniform squared risk at most K/N on fixed compact designs.*

An open exposure box has an open image in the moment cone. If the mark laws are known and exposures are interior, the plug-in moment estimate has limiting covariance

$$\Sigma_\theta = \sum_j \frac{\rho_j^2}{\lambda_j^3 \kappa_j^2} w_{2n}(T_j) w_{2n}(T_j)^\top > 0 \quad (\text{L.2})$$

after multiplication of its error by $\sqrt{N}$. This is the efficient functional covariance in that known-mark-law subexperiment; its squared-risk order N^{-1} is locally unimprovable.

Proof. Clipping gives $\mathbb{E}|\hat{\lambda}_j - \lambda_j|^2 \leq \lambda_+/N$. For $X \sim \text{Pois}(a)$,

$$\mathbb{E}[X^{-1}; X > 0] \leq 2\mathbb{E}[(X+1)^{-1}] = 2(1 - e^{-a})/a, \quad \Pr(X=0) = e^{-a} \leq 1/a.$$

Conditioning on the counts therefore gives mark-estimation risk $O(N^{-1})$. The weight functions $\rho_j/(\lambda\kappa(u))$ are uniformly Lipschitz on the fixed positive compact rectangle, proving the risk bound.

The exposure-to-moment Jacobian has columns $-\rho_j w_{2n}(T_j)/(\lambda_j^2 \kappa_j)$ and full row rank. This gives the open-image assertion. The Poisson central limit theorem and the delta method give (L.2). The exposure information per unit N is $\text{diag}(1/\lambda_j)$; its inverse pushed forward by this Jacobian is exactly the displayed covariance. The regular local information bound for smooth functionals, or the normal target experiment in Lemma I.1, gives efficiency and the lower risk order [26]. $\square$

Counts are genuinely observed in this additional protocol. Unknown exposures cannot be inferred from fixed-count marks alone. Jointly unknown marks and exposures need not have the known-mark-law efficient covariance (L.2).

APPENDIX M. COMPUTED FINITE-OFFSET FEATURES AND INDEPENDENT NOISE

At a globally separated informative point b_0 , choose the normal frame Z and the q vectors v_a consisting of its columns and their pairwise sums. Quadratic evaluation is an isomorphism $\mathcal{Q} : \text{Sym}_c \rightarrow \mathbb{R}^q$. Define

$$\mu_h(\theta)_a = \frac{1}{2} \{C_{R_\theta}(b_0 + hv_a) + C_{R_\theta}(b_0 - hv_a)\}.$$

Fix a compact convex $\Theta \subset \Omega$ containing the true parameter and the clipped plug-in estimates. For sufficiently small $h > 0$ all targets are positive and lie in the common analytic projection tube. Symmetric Taylor expansion, also differentiated in θ , gives

$$h^{-2} \mu_h(\vartheta) = \mathcal{Q}((\mathcal{L}\vartheta)^{-1}) + h^2 r_h(\vartheta), \quad \sup_{0 < h \leq h_0} \|r_h\|_{C^1(\Theta)} < \infty. \quad (\text{M.1})$$

Theorem M.1 (Computed finite-offset encoding and inversion). *For a raw estimator with squared risk at most K/N , compute $\hat{\mu}_h = \mu_h(\hat{\theta})$ and set*

$$\tilde{\theta}_h = \mathcal{L}^\dagger [\mathcal{Q}^{-1}(h^{-2} \hat{\mu}_h)]^{-1}.$$

For small enough h all inverses exist and

$$\mathbb{E}\|\tilde{\theta}_h - \theta\|^2 \leq 2K/N + 2C^2h^4/\gamma^2.$$

For a fixed design, $h_N = o(N^{-1/4})$ makes this output equivalent to the raw estimator at scale $\sqrt{N}$. With uniform tube and metric bounds along varying designs, it suffices that $\sqrt{N}h_N^2/\gamma_N \rightarrow 0$. A deterministic contact-computation error at most τ_N adds parameter error at most $C'\tau_N/(\gamma_N h_N^2)$, provided positive inversion remains valid.

Proof. Smooth positive inversion in (M.1) gives

$$F_h(\vartheta) := [\mathcal{Q}^{-1}(h^{-2}\mu_h(\vartheta))]^{-1} = \mathcal{L}\vartheta + h^2E_h(\vartheta), \quad \|E_h\|_{C^1} \leq C.$$

Thus $\tilde{\theta}_h = \hat{\theta} + h^2\mathcal{L}^\dagger E_h(\hat{\theta})$ and its extra error is at most Ch^2/γ . Squaring proves the risk and scaling claims. An error τ_N perturbs the argument of inversion by at most $\|\mathcal{Q}^{-1}\|\tau_N/h_N^2$; use the inverse identity in Theorem E.2. $\square$

This theorem concerns deterministic re-encoding of an already available estimator. It is not an independent observational route or a new minimax rate. Its offset condition controls approximation bias; the leading statistical risk is inherited from the raw estimate. The direct score reduction of Section I does not use this construction.

For comparison, consider the different experiment of independent noisy readings

$$Y_i = \mu_h(\theta) + \sigma\xi_i, \quad \xi_i \sim N(0, I_q), \quad i = 1, \dots, N.$$

For compact convex Θ and sufficiently small fixed h , the exact finite-offset mean satisfies

$$ah^2\|\theta - \theta'\| \leq \|\mu_h(\theta) - \mu_h(\theta')\| \leq bh^2\|\theta - \theta'\|$$

for $a, b > 0$. Indeed $F_h = \mathcal{L}\theta + h^2E_h$ and $h^2\|DE_h\| \leq \gamma/2$ give a lower bound by the triangle inequality and the mean-value bound for E_h ; positive inversion and $\mathcal{Q}$ transfer it to μ_h . The upper bound follows directly from (M.1). Fitting the exact mean by least squares consequently gives

$$\sup_{\theta \in \Theta} \mathbb{E}\|\hat{\theta}_N - \theta\|^2 \leq \frac{4q\sigma^2}{Na^2h^4},$$

since the fitted mean is within twice the sample-mean error of the true mean. At an interior parameter the information is $I_h = \sigma^{-2}D\mu_h^\top D\mu_h > 0$, the asymptotic covariance is I_h^{-1} , and the local asymptotic minimax risk is $\text{tr}(I_h^{-1})/N$ [26]. The least eigenvalue of I_h/h^4 is comparable to γ^2/σ^2 under the fixed metric bounds, by differentiating (M.1). Gaussian tails and the lower Lipschitz bound justify convergence of local risks. A quadratic-truncated fit at fixed h would instead generally retain an $O(h^2)$ bias and is not used. The h^{-4} variance in this independent-noise experiment must not be transferred to the correlated computed or raw-score statistics.

APPENDIX N. COMPLETE CONTACT, DESIGN FIBRES, AND ENDPOINT REPRESENTATIONS

These statements concern different observation categories from a prescribed normal block. We record their mathematical interfaces and proofs so that the one-contact and multi-contact results do not erase the complementary conclusions.

N.1. Complete conormal ambiguity. Suppose $\mathcal{A}_A : \text{Sym}_d \rightarrow \mathbb{R}^k$, $S \mapsto (a_i^\top S a_i)_i$, is injective. The complete conormal union of the rank-one image is dense in the inverse image, under $\mathcal{A}_A^*$, of the singular symmetric matrices: a conormal n at $q_A(\eta)$ satisfies $(\sum_i n_i a_i a_i^\top)\eta = 0$. Conversely a real singular matrix has a nonzero real kernel vector. For $d \geq 3$, no nonzero quadratic form vanishes on all singular symmetric matrices. To see this, restrict a quadratic polynomial to matrices diagonal in any fixed orthogonal basis. Every monomial of degree at most two omits at least one of the d diagonal variables, so vanishing on every coordinate hyperplane forces the

restriction to vanish identically. Every real symmetric matrix is orthogonally diagonalizable, so the polynomial is zero. For $d = 2$, direct substitution of rank-one matrices leaves exactly the determinant line. This is the elementary coordinate consequence of classical Veronese duality [12].

Under the surjective map $\mathcal{A}_A^*$, pure kernel and mixed quadratic terms must also vanish, since singular symmetric matrices span Sym_d . Thus the complete-conormal ambiguity is zero for $d \geq 3$ and one-dimensional for $d = 2$. In the latter case it is generated by $B_{ii} = 0$, $B_{ij} = \det(a_i, a_j)^2$ and is intersected with the admissible native positive cone. In root coordinates the native linear span consists exactly of symmetric matrices satisfying

$$\frac{(r_j - r_i)X_{ij}}{d_i d_j} - \frac{(r_l - r_i)X_{il}}{d_i d_l} + \frac{(r_l - r_j)X_{jl}}{d_j d_l} = 0 \quad (i < j < l). \quad (\text{N.1})$$

The divided-difference identity for $(t - r_i)^{-1}(t - r_j)^{-1}$ proves these relations for $JH(\theta)J^\top$. Conversely they leave k free diagonal entries and $k - 1$ off-diagonal difference coordinates, hence dimension $2k - 1$, equal to the Hankel span. Testing B in (N.1) gives the exact two-parameter transversality criterion. Equality of second jets here is not equality of all higher contact terms.

N.2. Exact dual-jet query count. In a deterministic exact-real oracle whose scalar answers are values of the dual normal quadratic forms, full product rank requires exactly $p = 2k - 1$ scalar queries. Squares span within-contact products by polarization, so select p independent square functionals for sufficiency. For necessity, follow any adaptive transcript at an interior parameter. Fewer than p scalar linear answers have a nonzero common kernel. A sufficiently small perturbation in that kernel remains admissible and reproduces the same answers, hence the same adaptive choices, by induction. This is a query statement about the specified dual oracle, not a statistical sample bound or a count of ordinary scalar distance readings.

N.3. Reciprocal design fibres. Put $a_j = \rho_j / \kappa_j > 0$, let V_m be the $p \times m$ moment matrix, and use reciprocal weights $\lambda_j = a_j / \omega_j$. Define on the open moment cone

$$\chi(\theta) = \min_{V_m \omega = \theta, \omega > 0} \sum_j a_j / \omega_j.$$

The zeroth moment bounds every fibre. The objective tends to infinity at its boundary and is strictly convex, so there is a unique minimizer. Its diagonal Hessian $D_\omega > 0$ and differentiation of the constrained first-order equations give

$$D^2 \chi = (V_m D_\omega^{-1} V_m^\top)^{-1} > 0.$$

Indeed $D_\omega d\omega = V_m^\top d\nu$ and $V_m d\omega = d\theta$, while ν is the gradient of the value. The implicit function theorem also gives smoothness. Here $m = p + 2$, so the affine fibre has dimension two. Every ray from its minimizer meets a higher level once, because the restriction is strictly convex and diverges at the positive-fibre boundary. Its higher levels are smooth circles. Consequently the unit-exposure-budget image is $\{\theta : \chi(\theta) \leq 1\}$, with singleton boundary fibres and circular interior fibres. Homogeneity $\chi(t\theta) = t^{-1}\chi(\theta)$ makes the boundary regular relative to the open cone. Injective contact coordinates transport this geometry without changing the fibres.

N.4. Endpoint value-graph representations. For a labelled semialgebraic value graph specified over algebraic coefficients, fix a hidden dimension and a specified normalization of a positive block matrix. Every representation

$$F(x) = \min_{z \geq 0} (x, z)^\top \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} (x, z)$$

is characterized exactly by positivity, that normalization, and the quantified relations

$$z \geq 0, \quad Cz + B^\top x \geq 0, \quad z_i(Cz + B^\top x)_i = 0,$$

together with the value equality for every labelled x . Strict convexity in z makes these Karush–Kuhn–Tucker conditions necessary and sufficient. Real quantifier elimination therefore computes the representation fibre at the fixed hidden dimension [6]. Enumeration over dimensions terminates under a finite-representability promise, yielding the first nonempty normalized fibre; no unconditional finite hidden-dimension bound is implied. This input is a scalar value graph, not an optimizer map as in inverse parametric programming [15, 22].

N.5. Deterministic local root costs. The native expansion $p(u, v) = p_0 + L_p u + D_p v + O((\|u\| + \|v\|)^2)$ gives the information in Proposition C.1. With squared Hellinger distance $h^2(p, q) = \sum_e (\sqrt{p_e} - \sqrt{q_e})^2$, profiling the free coordinates yields the quadratic efficient cost after multiplying by $4/t^2$. Comparing a physical target variance tb , $b \geq 0$, with synchronized $v = tq_A(\eta)$ gives (A.1). Full rank of A gives coercivity and bounded scaled minimizing sequences, making the Taylor remainder uniform on the relevant local chart. Physical splitting has size $\sqrt{t}$. For C^2 nonlinear loadings g_i with $g_i(0) = 0$ and $Dg_i(0) = a_i^\top$,

$$g_i(\sqrt{t}\eta)^2 = t(a_i^\top \eta)^2 + O(t^{3/2})$$

uniformly on bounded η . These local statements do not require injectivity of $\mathcal{A}_A$ on all of Sym_d ; global chart selection uses the separated-fibre hypotheses. At $t = N^{-1/2}$ the same cost is the physical local Gaussian profile of Theorem H.1.

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